

First Draft

Chapter: Cell_Type | T Cells

T cells are a cornerstone of adaptive immunity, playing crucial roles in both cell-mediated and humoral immune responses. They are primarily divided into two major subsets: cytotoxic T cells (CD8⁺ T cells) and helper T cells (CD4⁺ T cells). CD8⁺ T cells, also known as "killer T cells," directly kill virus-infected cells and cancer cells by releasing perforin and granzymes, while CD4⁺ T cells differentiate into multiple subsets such as Th1, Th2, Th17, etc., each with distinct roles in immune responses.

Upon antigen stimulation, naïve T cells differentiate into CD4⁺ helper and CD8⁺ cytotoxic effector and memory cells, mediating direct killing, diverse immune regulatory functions, and long-term protection. The development and function of T cells are tightly regulated by metabolic pathways, with CD8⁺ T cells primarily relying on glycolysis for energy production during activation. This metabolic shift is crucial for their effector functions, although they still require intact oxidative phosphorylation (OXPHOS) for optimal function.

Metabolic reprogramming in T cells is a critical aspect of their activation and differentiation. For instance, the induction of high glycolytic activity in CD8⁺ T cells favors their differentiation toward effector cells but impairs the survival of long-lived memory cells. Additionally, the presence of branched-chain amino acids (BCAAs) can enhance the anti-tumor immune function of CD8⁺ T cells, while a low-glucose microenvironment can reduce their glycolysis function and induce apoptosis.

T cell function is also influenced by the microenvironment, including hypoxic conditions, which can inhibit the anti-tumor abilities of cytotoxic T cells and natural killer cells. Metabolic checkpoints such as the mammalian target of rapamycin (mTOR) play a key role in cell fate decisions by regulating T cell metabolic responses. The use of metabolic inhibitors and activators can modulate T cell function and improve the efficacy of cancer immunotherapies.

In the context of cancer immunotherapy, T cells can be genetically engineered to express chimeric antigen receptors (CARs) or T cell receptors (TCRs) targeting specific neoantigens. This approach has shown promising results in clinical trials, with CAR-T cells demonstrating significant efficacy in treating certain types of leukemia and lymphoma. However, T cell exhaustion, characterized by chronic antigen exposure, can impair their function. Strategies to overcome this include the use of checkpoint inhibitors such as anti-PD-1 and anti-PD-L1 antibodies, which have shown success in clinical settings.

The role of T cells in autoimmune diseases is complex. Anergic T cells, which are metabolically anergic, can contribute to the pathogenesis of autoimmune conditions. Understanding the metabolic and signaling pathways that regulate T cell function is crucial for developing targeted therapies to modulate their activity.

In summary, T cells exhibit profound metabolic reprogramming that supports their differentiation, activation, and effector functions. Metabolic checkpoints and signaling pathways play critical roles in T cell function, and targeting these pathways can enhance the efficacy of cancer

immunotherapies and improve the treatment of various diseases.

Chapter: Cell_Type | B Cells

B cells are a crucial component of the adaptive immune system, playing a pivotal role in the production of antibodies and the development of immunological memory. These lymphocytes are primarily found in secondary lymphoid organs such as the spleen and lymph nodes, where they encounter antigens and undergo activation, proliferation, and differentiation.

Development and Function

B cells develop in the bone marrow, where they undergo gene rearrangement to generate diverse B cell receptors (BCRs) capable of recognizing a wide array of antigens. In birds, B cells mature in the bursa of Fabricius, a lymphoid organ. Mature B cells express MHC class II molecules to present antigens to T cells, but upon BCR engagement, they secrete soluble immunoglobulins, mediating humoral immunity. B cells can differentiate into plasma cells, which immediately begin producing antibodies, or memory B cells, which confer lasting protection from the antigen.

Activation and Differentiation

B cell activation occurs in secondary lymphoid organs following engagement of the BCR with an antigen. This interaction causes the B cell to divide and differentiate into an antibody-producing cell, known as a plasma cell. Activated B cells form a germinal center within the lymphoid follicle, where they proliferate and are selected with T-helper cell support. During this process, the antibody isotype can change in response to cytokines released by helper T cells, and the antigen-binding properties of the antibody can be altered through somatic hypermutation of V-region genes.

Mechanisms and Regulation

The molecular biology of B cells involves intricate mechanisms such as class switch recombination and somatic hypermutation. Class switch recombination allows B cells to switch from producing IgM to producing other isotypes like IgG, IgA, or IgE. Somatic hypermutation introduces point mutations in the variable regions of the immunoglobulin genes, enhancing the affinity of the BCR for the antigen. NF- κ B activation, induced by BCR engagement and CD40 signaling, supports B cell survival, proliferation, and antibody production.

T Cell Interactions and Immune Regulation

B cells and T cells are not independent; they interact in a reciprocal manner. B cells can activate antigen-specific T cells for a cellular immune response, while an efficient B-cell antibody response depends on T-cell activation of B lymphocytes. B cells also regulate autoimmunity by providing IL-10, which can promote recovery from autoimmune encephalomyelitis (EAE) and restrain T cell responses during viral and bacterial infections.

Autoimmunity and Tolerance

To avoid the development of autoreactive B cells, BCR ligation of self-antigens during early B cell development in the bone marrow promotes negative selection through NF- κ B-induced receptor editing of the self-reactive BCR or apoptosis. B cells can also contribute to T cell negative selection for central T cell tolerance by expressing autoimmune regulator (Aire), which increases MHC class

II and CD80 levels.

Genetic and Environmental Influences

Genetic factors, such as defects in Jak3, can impair B lymphocyte maturation and T lymphocyte activation. Environmental factors, including microbiota exposures, shape the B cell repertoire. The presence of certain antigens, like PEGylated lipids, can induce T cell-independent B cell responses, leading to the production of specific antibodies.

Conclusion

B cells are essential for the immune system, playing critical roles in antibody production, isotype switching, and the development of immunological memory. Their activation and differentiation are tightly regulated through complex molecular mechanisms and interactions with T cells. Understanding these processes is crucial for developing novel therapeutic strategies targeting B cells in various diseases, including autoimmune disorders and cancer.

Chapter: Cell_Type | Macrophages

Macrophages are a heterogeneous and plastic population of cells that play a crucial role in the immune system. They are present in various tissues and are involved in both innate and adaptive immunity. Macrophages are known for their ability to polarize into two distinct phenotypes: M1 (classically activated macrophages) and M2 (alternatively activated macrophages). M1 macrophages are primarily activated by T cell response, specifically CD4⁺ Th1 cells, and exhibit robust antigen-presenting and pro-inflammatory functions. They are involved in pathogen elimination and produce proinflammatory cytokines with strong killing effects on pathogens. In contrast, M2 macrophages are involved in tissue repair and contribute to tissue homeostasis by secreting anti-inflammatory and pro-repair factors such as IL-10 and TGF- β . They are expanded during helminth infections and play an important immune regulation and tissue repair role by competing for L-arginine and generating proline.

Macrophages are also involved in the EMT process via the secretion of various soluble factors such as IL-1 β , IL-8, TNF- α , and TGF- β . They are polarized toward the M1 profile during the early phase of *Mycobacterium tuberculosis* infection, as well as other mycobacterial species such as *Mycobacterium ulcerans* and *Mycobacterium avium*. Macrophages can also undergo apoptosis or polarize to an M2 phenotype to protect the host from excessive injury. The ability of macrophages to polarize in this highly plastic manner is crucial in allowing the immune system to adapt to varying threat levels from microbes and the resulting tissue damage, thus allowing infections to fully resolve.

Macrophages are also involved in the regulation of fibrosis, where they orchestrate all phases of tissue injury and repair. They are present in adipose tissue and liver in obesity, where their metabolism is regulated by adaptations to the lipid-rich tissue environment. Macrophages are also involved in the tumor microenvironment, where they play a central role in inflammation and immune regulation. They are essential immune cell components in the tumor microenvironment (TME) and their distribution is closely associated with cancer progression, therapeutic response, and prognosis.

Macrophages can be activated by various stimuli, including type-1 or type-2 cytokines, and their

polarization state is dynamic. They can sense danger signals over pattern recognition receptors (PRRs), such as TLRs and C-type lectin receptors (CLRs). The polarization of macrophages is not fixed and they can integrate multiple signals from microbes, damaged tissues, and the normal tissue environment. Macrophages are also involved in the regulation of inflammation and immune responses, and their function can be modulated by various factors, including metabolic reprogramming and the expression of specific genes.

In summary, macrophages are a diverse and plastic population of cells that play a crucial role in the immune system. Their ability to polarize into M1 and M2 phenotypes allows them to adapt to different threats and contribute to both pathogen elimination and tissue repair. Their function is regulated by various factors, including metabolic reprogramming and the expression of specific genes, and they play a central role in the regulation of inflammation and immune responses.

Chapter: Cell_Type | Dendritic Cells

Dendritic cells (DCs) are the most potent and versatile professional antigen-presenting cells (APCs), playing a pivotal role in the initiation and regulation of adaptive and innate immune responses. They were first described by Paul Langerhans in the late nineteenth century, and their development from hematopoietic stem cells (HSCs) is a multistep process that begins in the bone marrow. DCs can induce T-cell tolerance, and in vitro, their behavior and capabilities may differ from those isolated ex vivo. Dendritic cells are key in vaccine development by presenting antigens to T cells, initiating and regulating immune responses. They are also involved in the recognition, capture, processing, and presentation of antigens to T and B cells, bridging innate and adaptive immune responses.

Immature dendritic cells constantly sample the environment for pathogens such as viruses and bacteria, phagocytosing them and degrading their proteins into small pieces, which are then presented at the cell surface using major histocompatibility complex (MHC) molecules. Upon maturation, DCs stimulate naïve T cells through increased surface expression of peptide-loaded MHC antigens, costimulatory receptors, and ligands such as CD80 and CD86, and the release of cytokines like IL-6, IL-12, and interferons (IFNs). Mature DCs also produce high levels of IL-10, IL-2, and TGF- β , promoting the proliferation and activation of inducible regulatory T cells (iTreg), which decrease autoimmune reactions. However, if tolerance cannot be restored, CD220+ B cells can modulate Th17 responses, maintaining a proinflammatory state.

Dendritic cells can be classified into several types, including conventional dendritic cells (cDCs), plasmacytoid dendritic cells (pDCs), and follicular dendritic cells (FDCs). Conventional DCs are further divided into cDC1s and cDC2s, each with distinct functions. cDC1s are potent stimulators of T cells and are involved in the induction of Th1 responses, while cDC2s are more involved in Th2 responses. pDCs are specialized for producing significant quantities of type I interferons, especially in response to viral infections. FDCs are located in lymphoid follicles and have long "dendritic" processes, but they are not of hematopoietic origin and do not express MHC class II.

The migration of dendritic cells from tissues to lymph nodes is a complex process influenced by chemokines and cytokines, which guide their movement through the body. Dendritic cells are heterogenous and include precursors of dendritic cells, monocytes, and neutrophils. They are essential for mounting a precise defense against pathogens and play a crucial role in the innate and adaptive immune responses. Dendritic cells are also involved in the regulation of immune responses

in various diseases, including cancer, diabetes, and infectious diseases. They can be used in cancer immunotherapy, such as dendritic cell-based vaccines, tolerogenic DCs, and combined nanotechnology/checkpoint inhibitors.

Chapter: Cell_Type | Natural Killer Cells

Natural killer (NK) cells are a crucial component of the innate immune system, playing a pivotal role in the rapid recognition and elimination of abnormal cells, including virus-infected cells and tumor cells. These cells are characterized by their ability to induce cytolytic activity on target cells without prior sensitization or activation, a process that depends on a balance of signals from activating and inhibitory receptors on the NK cell surface. NK cells express high levels of Toll-like receptors (TLRs) such as TLR-1, TLR-2, TLR-3, TLR-5, and TLR-6, which contribute to their antiviral responses and innate immunity. Additionally, NK cells are equipped with scavenger receptors that can activate tumor cell killing by NK cells, highlighting their versatility in immune defense mechanisms.

The function of NK cells is tightly regulated by two complementary inhibitory receptor families that bind to classical and non-classical HLA class I molecules: the CD94/NKG2A receptors and the killer cell immunoglobulin-like receptors (KIRs). This regulation ensures that NK cells are tolerant against healthy tissue while being able to attack infected cells. The activity of NK cells is further modulated by signaling adaptors such as Vav2 and Vav3 isoforms, SLP-76, Shc, and LAT, which are involved in downstream signaling pathways initiated by activating receptors. NK cells also express the Fc receptor (FcR) molecule, specifically CD16, which plays a role in antibody-dependent cell-mediated cytotoxicity by binding the Fc portion of IgG antibodies.

NK cells are involved in the regulation of autoimmune responses, where they can either suppress or augment autoimmunity. This dual role is influenced by the balance of activating and inhibitory signals received by NK cells. Recent studies have shown that NK cells can display features typically attributed to adaptive immune cells, such as dynamic expansion and contraction of subsets, increased longevity, and a form of immunological memory. This suggests that NK cells can respond more potently upon secondary challenges with the same antigen, indicating a more adaptive-like behavior.

NK cells are also crucial in the context of cancer immunotherapy. They can be engineered to express chimeric antigen receptors (CARs) for targeted therapy of specific cancers, such as glioblastoma. Additionally, nanoparticles loaded with RNA effectors, such as siRNA, microRNA (miRNA), and short hairpin RNA (shRNA), can enhance antitumor activity by silencing specific genes to alter genomic function, thereby enhancing NK cell activity. The therapeutic success of haploidentical hematopoietic stem cell transplantation (HSCT) to cure high-risk leukemias further underscores the importance of NK cells in clinical settings.

NK cells play a significant role in the immune response against viral infections, including HIV. They are activated in response to HIV infection, contributing to the antiviral immune response. However, viruses like HIV have evolved strategies to evade NK cell responses, such as shedding decoy NKG2D soluble ligands. Similarly, tumor cells can evade NK cell recognition by downregulating ligands for activating receptors or by expressing inhibitory receptors that bind to MHC class I molecules.

The education of NK cells, a process that determines the threshold for activation in mature NK

cells, is influenced by interactions with neighboring cells. This process is critical for the development and maturation of NK cells in the bone marrow, where interactions between inhibitory receptors and MHC-I molecules are necessary. IL-15 is essential for the maturation of NK cells at all stages of development, and STAT5b-deficient NK cells exhibit decreased proliferation and toxicity after stimulation by IL-2 and IL-15. These findings highlight the importance of cytokine signaling in the regulation of NK cell function.

In conclusion, NK cells are a heterogeneous population of cells that exhibit cytotoxic, immunomodulatory, and tumor cell-killing properties. They play a crucial role in the immune response against viral infections and cancer, and their function is tightly regulated by a complex interplay of activating and inhibitory signals. Understanding the mechanisms underlying NK cell activation and regulation is essential for the development of novel immunotherapeutic strategies.

Chapter: Cell_Type | Hematopoietic Stem Cells

Hematopoietic stem cells (HSCs) are primitive pluripotent cells capable of self-renewal and differentiation into any cells of the hematopoietic lineage, including myeloid (monocytes and macrophages, neutrophils, basophils, eosinophils, erythrocytes, megakaryocytes/platelets, dendritic cells) and lymphoid lineages (T-cells, B-cells, NK-cells). HSCs are utilized to treat a diverse array of hematologic and non-hematologic diseases and conditions, and they can be sourced from marrow, mobilized peripheral blood, or umbilical cord blood (UCB). Previously described with inducible lineage tracing mice, HSCs in bone marrow are partially derived from PDGFR α ⁺ mesenchymal progenitors transiently present during embryonic day E7.5-E8.5 (PDGFR α -lineage; Ding et al., 2013). Hematopoietic stem cells (HSCs) are being clinically tested for their use in non-hematopoietic tissue regeneration, and in ex vivo approaches, HSCs are harvested, edited using CRISPR-Cas9, and reinfused following conditioning, akin to hematopoietic stem cell transplantation (HSCT).

HSCs are primarily derived from embryonic yolk sac-derived erythromyeloid progenitors (EMPs) and fetal liver monocytes, with a minor contribution from HSCs, and are maintained in adults through local self-renewal. The first definitive HSCs arise from the ventral endothelial wall of the embryonic aorta within the aorta-gonad-mesonephros region, through a process known as endothelial-to-hematopoietic transition. Slightly later, HSCs are also found in the placenta, yolk sac, embryonic head, and fetal liver. HSCs can be identified or isolated by the use of flow cytometry where the combination of several different cell surface markers (particularly CD34) are used to separate the rare hematopoietic stem cells from the surrounding blood cells. Lack of expression of mature blood cell markers is used in combination with detection of several positive cell-surface markers to isolate hematopoietic stem cells. Hematopoietic stem cells (HSCs) constitute 1:10,000 of cells in myeloid tissue and resemble lymphocytes in shape. A small number of HSCs can expand to generate a very large number of daughter HSCs, and when plated onto a confluent culture of stromal feeder layer, a fraction of HSCs creep between the gaps and eventually settle between the stromal cells and the substratum or trapped in the cellular processes between the stromal cells.

HSCs are quiescent and mostly exist in a state of reversible growth arrest, but when provoked by cell death or damage, they exit quiescence and begin actively dividing again. Quiescent HSCs accumulate DNA damage during aging that is repaired upon entry into the cell cycle. Senescence and apoptosis block hematopoietic activation of quiescent HSCs with short telomeres. The altered metabolism of quiescent HSCs helps the cells survive for extended periods of time in the hypoxic

bone marrow environment. Balanced (Bala) HSCs repopulate peripheral white blood cells in the same ratio of myeloid to lymphoid cells as seen in unmanipulated mice (on average about 15% myeloid and 85% lymphoid cells, or $3 \leq \rho \leq 10$). The sensitivity of HSCs to Lig4, DNA polymerase mu, and NHEJ1 deficiency suggests that NHEJ is a key determinant of the ability of stem cells to maintain themselves against physiological stress over time.

HSCs may also settle in the liver or spleen and develop, and the hematopoietic stem cell compartment can be split into three categories of HSC using the ratio $\rho = L / M$ of lymphoid (L) to myeloid (M) cells in blood as a quantitative marker. Repopulation kinetics of HSCs cannot be easily observed directly, and their behaviors need to be inferred indirectly. The definition of hematopoietic stem cell has developed since they were first discovered in 1961. CD47 is upregulated on circulating hematopoietic stem cells and leukemia cells to avoid phagocytosis. Rossi et al. found that endogenous DNA damage accumulates with age even in wild type HSCs, and suggested that DNA damage accrual may be an important physiological mechanism of stem cell aging.

Chapter: Cell_Type | Monocytes and Granulocytes

Monocytes and granulocytes play crucial roles in immune responses and tissue repair. Monocytes, upon activation, can differentiate into dendritic cells (DCs) or macrophages, depending on the cytokine milieu. For instance, treatment with interleukin 4 (IL-4) and granulocyte-macrophage colony stimulating factor (GM-CSF) leads to the differentiation of monocytes into immature dendritic cells (iDCs) within about a week. Macrophages derived from either tissue residence or extravasated monocytes are pivotal in the resolution process, with any functional influence by extravasation occurring at the level of the monocyte, during macrophage differentiation, or at the level of the differentiated macrophage. Monocytes and T cells enhance their secretion of inflammatory cytokines in a stiff environment, a characteristic consistent with the perivascular macrophages abundantly seen. However, it is unclear whether this differentiation boost is due to monocytes being in contact with the basement membrane or their natural tendency to migrate towards blood vessels.

Granulocytes, a collection of phagocytes, include neutrophils, eosinophils, and basophils. Neutrophils, the most abundant white blood cells, are essential in the initial response to infection and inflammation. They can respond to microorganisms via phagocytosis, generation of reactive oxygen species (ROS), and degranulation or the release of neutrophil extracellular traps (NETs). Neutrophils can also release mitochondrial DNA to form neutrophil extracellular traps, which help in immobilizing and killing invading pathogens. Neutrophils have been shown to facilitate ovarian cancer premetastatic niche formation and to promote Alzheimer's disease-like pathology via LFA-1 integrin. Additionally, neutrophils play a role in autoimmune diseases, such as systemic lupus erythematosus, by releasing self-DNA-peptide complexes.

Eosinophils, another granulocyte, are increased in the airways and sputum of individuals with severe asthma and are involved in the killing of helminth parasites. They are also important mediators of allergic responses and asthma pathogenesis, with increased eosinophil numbers correlating with greater asthma severity. Eosinophils can prime a long-lived effector macrophage phenotype that mediates accelerated helminth expulsion. They promote humoral immunity by priming B cells and play a central role in type 2 immunity, including antigen presentation to CD4+ T cells and secretion of granular contents containing type 2 mediators such as IL-4, IL-5, and IL-13.

In summary, monocytes and granulocytes, particularly neutrophils and eosinophils, are critical in immune responses and inflammation. Their functions are context-dependent and can vary from pathogen defense to contributing to disease progression.

Chapter: Cell_Type | Lymphocytes

Lymphocytes are a diverse group of white blood cells that play crucial roles in the immune system. They include T cells, B cells, and natural killer (NK) cells, each with distinct functions in immune responses. Cytotoxic T lymphocytes (CTLs) are a subset of T cells that directly kill infected or cancerous cells by inducing apoptosis. They achieve this through the release of perforin and granzymes, which are delivered by a process of directed exocytosis. CTLs are key players in the elimination of tumor or pathogen-infected cells via the phagocytic granule (lysosome) and Fas ligand (FasL) pathways. These cells are essential for cancer immunosurveillance, as proposed by Burnet and Thomas in 1957, where they act as sentinels recognizing and eliminating nascent transformed cells.

Innate lymphoid cells (ILCs) represent an emerging family of innate immune cells, described about 10 years ago. ILCs are divided into four main groups based on their transcription factors and cytokine production profiles: NK cells, group 1 ILCs (expressing T-bet and producing IFN- γ), group 2 ILCs (expressing ROR α and GATA3 and producing type 2 cytokines), and group 3 ILCs (expressing ROR γ t and producing IL-17A and IL-22). These cells play crucial roles in immune responses, particularly at mucosal surfaces, where they regulate innate immune responses. For instance, group 2 ILCs (ILC2s) are involved in helminth immunity, while group 1 ILCs are essential for type 1 immunity.

The process of lymphocyte isolation from peripheral blood mononuclear cells (PBMCs) involves density gradient centrifugation to remove granulocytes, red blood cells, and platelets. PBMCs are then separated and collected, and the lymphocyte population is gated using forward scatter (FSC) and side scatter (SSC) parameters, followed by negative gating for CD3 to isolate T cells. The reprogramming of lymphocytes can enhance their anti-tumor responses in a CXCL9-dependent manner. Additionally, the metabolism, migration, and memory of lymphocytes are critical for their function. For example, the major metabolic changes in activated lymphocytes involve a switch to glycolytic metabolism, while reducing oxidative metabolic pathways.

Lymphocytes, including T cells and NK cells, can escape from cytotoxic T cells by ceasing to express essential co-stimulatory molecules such as CD80 or CD86. The expression of these molecules is crucial for the activation of cytotoxic T cells. The interaction of lymphocytes with high endothelial venules (HEVs) is essential for controlling lymphocyte traffic. In the absence of extrinsic signals, lymphocytes rely on nutrient utilization for cell size and viability. Enhanced mRNA delivery into lymphocytes has been achieved using lipid-varied libraries of charge-altering releasable transporters, as demonstrated by McKinlay et al.

In the context of autoimmune diseases, lymphocytes can be targeted by specific antibodies, such as anti-T3 linked to anti-target cell antibodies, to modulate their activity. The signaling pathway involving Janus kinase 3 (JAK3) is particularly relevant to lymphocytes, as its function is mainly restricted to these cells. The role of lymphocytes in various diseases, including asthma, cancer, and autoimmune disorders, underscores their importance in immune responses.

Chapter: Cell_Type | Fibroblasts and Stem Cells

Fibroblasts play a crucial role in the healing process after tissue damage, where they proliferate and remodel the extracellular matrix (ECM) to restore tissue integrity. These cells are also recognized as non-professional phagocytes, capable of phagocytosis with low efficiency. In the tumor microenvironment (TME), fibroblasts can differentiate into cancer-associated fibroblasts (CAFs), which are implicated in cancer progression and are potential targets for therapy and diagnosis. For instance, Cai et al. (2020) used single-cell RNA sequencing to reveal pro-invasive CAFs in hypopharyngeal squamous cell carcinoma, highlighting the heterogeneity of CAFs.

Pericytes, defined as mesenchymal cells at vascular proximity, are also significant in the TME. They express markers such as PDGFR β , CD146, NG2, and α -SMA and are involved in the regulation of endothelial junctions through the release of Ang1 and activation of endothelial Tie2 signaling. Proebstl et al. (2012) demonstrated that pericytes may stretch low expression regions in the basement membrane to enlarge gaps between them, contributing to the organization of the perivascular space and subcellular junctional architecture of endothelial cells (ECs).

In the context of the TME, fibroblasts and CAFs contribute to the network by promoting the recruitment of tumor-associated macrophages (TAMs) through various axes, such as the CXCL12/CXCR4 axis in breast cancer and the IL-8/CXCR2 axis in colorectal cancer. Additionally, cancer-associated fibroblasts can induce antigen-specific deletion of CD8⁺ T cells, thereby protecting tumor cells from immune attack. For example, Monteran and Erez (2020) highlighted the dark side of fibroblasts, emphasizing their role in immunosuppression within the TME.

Mesenchymal stem cells (MSCs) are another important component of the TME, modulating immune responses by balancing pro- and anti-inflammatory cytokines and promoting tissue repair. However, these cells can also contribute to tumor progression by enhancing the growth and metastasis of colon cancer, as demonstrated by Shinagawa et al. (2018). Furthermore, CAFs and other stromal cells can differentiate into various cell types, including cancer-associated fibroblasts, as noted by Baghban et al. (2020).

The TME is a complex system involving a variety of cell types, including fibroblasts, epithelial cells, endothelial cells, neutrophils, mesenchymal stem cells, myeloid cell-derived suppressor cells, and mast cells. These cells interact in intricate ways, with TAMs playing a central role in promoting the formation, survival, and proliferation of cancer stem cells (CSCs) and other stem cell subtypes. The interplay between these cells underscores the complexity of the TME and the multifaceted roles of fibroblasts and CAFs in cancer progression.

Chapter: Cell_Type | Epithelial and Other Cells

Epithelial cells form the mucosal barrier, supporting M cells and secreting mucus to trap antigens. Mucus production from goblet cells is induced by type 2 cytokines, with IL-13 and IL-4 playing dominant roles. FAE (fermentation-associated epithelium) differs from typical small intestinal villus epithelium in having fewer goblet cells, resulting in a thinner mucus layer, and is characterized by the presence of specialized M cells or microfold cells, which provide uptake and transport of antigens from the lumen. These cells are critical in the immune surveillance process, with M cells in the follicle-associated epithelium sampling and transporting antigens from the intestinal lumen.

In the context of helminth infections, the host type 2 immune response is regulated by a complex interplay between epithelial cells, eosinophils, basophils, mast cells, ILC2s, T helper type 2 cells (T_H 2), and effector cytokines such as IL-3, IL-4, IL-5, IL-9, IL-13, and IgE antibodies. Activated ILC2 cells secrete IL-13, which increases mucus secretion in goblet cells. Goblet cells are the main source of mucins, the major family of glycoproteins that comprise the mucus barrier and provide its viscoelastic properties. The donation is comprised of white blood cells called leukocytes that fight viruses and bacteria in the body. Additionally, epithelial cells contribute to the barrier that includes mucus, epithelial cells, secretory immunoglobulin A (IgA), numerous antimicrobial peptides, and immune cells, which limit contact and translocation and impact commensal gene expression and prevent bacterial adhesion.

Epithelial cells also play a role in the immune response to helminths, with tuft cells in the intestine activating group 2 innate lymphoid cells (ILC2s) and contributing to anti-helminth immunity. The immune function in the gut begins with M cells in the follicle-associated epithelium, which sample and transport antigens from the intestinal lumen. This process is crucial for the recognition of helminths by host cells and the roles of various cell types, including macrophages, tuft cells, T and B lymphocytes, eosinophils, basophils, neutrophils, innate lymphoid cells type 2 (ILC2), and mast cells.

Tumor-associated macrophages (TAMs) are a heterogeneous population of immune cells that play dual roles in cancer biology. They can promote tumor progression by facilitating angiogenesis, metastasis, and immunosuppression, and they can also enhance antitumor immunity through the secretion of cytokines such as IL-12 and TNF- α . TAMs are involved in tumor proliferation, invasion, and metastasis, and they can secrete proteolytic enzymes such as cathepsins, matrix metalloproteinases (MMPs), and serine proteases, which are important components mediating ECM degradation and cell-ECM interactions. TAMs can also directly promote cytotoxic T-cell exhaustion, and many current immunotherapies aim to reactivate antitumoral cytotoxic T cells by inhibiting the PD-1/PD-L1 immune checkpoint pathway.

The inherent complexity of TAMs, including diverse cellular origins, post-differentiation heterogeneity, and context-dependent functional states, complicates the identification of optimal therapeutic windows and precise molecular targets. Targeting TAMs has become a major focus in cancer immunotherapy, with strategies such as disrupting their recruitment, depleting existing macrophage populations, inhibiting key signaling pathways to suppress pro-tumoral-like polarization, reprogramming TAMs toward antitumor phenotypes, and modulating metabolic processes. TAMs are also involved in the formation of pre-metastatic niches and can sustain IL-6 expression through a positive feedback circuit involving IL6, STAT3, and C/EBP β in LUAD. TAMs can also promote allogeneic CD10⁺ cancer stem cells through IL-6/STAT3/CD10 signaling and contribute to CSC maintenance by supporting stemness properties and establishing a specialized niche that promotes CSC survival and drug tolerance.

In summary, the roles of epithelial cells and TAMs in the immune response and cancer biology are multifaceted, with significant implications for therapeutic strategies.

Chapter: Location | Bone Marrow

Central tolerance mechanisms occur in the bone marrow, where clonal deletion eliminates B cells with self-reactive BCRs, ensuring immune system tolerance to self-antigens. Identification of

clonogenic common Flt3+M-CSFR+ plasmacytoid and conventional dendritic cell progenitors in mouse bone marrow highlights the diverse cellular origins within the bone marrow microenvironment. Macrophages, derived from bone marrow progenitors that circulate as monocytes before differentiating into tissue-resident macrophages, play key roles in pathogen clearance, antigen presentation, and tissue homeostasis. These cells mediate the clearance of neutrophil extracellular traps (NETs), which are involved in various inflammatory processes. Transfused stem cells become engrafted when they reach the bone marrow and start producing new blood cells, a process critical for hematopoietic reconstitution. Bone marrow transplantation is commonly performed for patients with certain cancers of the blood or bone marrow, such as multiple myeloma, leukemia, and some types of lymphoma, as well as for immune deficiencies. The T cell development process begins in the bone marrow, where hematopoietic stem cells (HSCs) give rise to common lymphoid progenitors (CLPs). The colony-forming unit-spleen (CFU-S) is used to assess the ability of infused bone marrow cells to give rise to clones of maturing hematopoietic cells in the spleens of irradiated mice, a phenomenon utilized in bone marrow transplantation. Bone marrow-residing T cells exhibit conserved behaviors in multiple myeloma patients undergoing BCMAxCD3 T cell engager (TCE) therapy. Bone marrow-derived hematopoietic stem cells (HSCs) fabricated into circulating blood monocytes are essential for macrophage development. Bone marrow transplants involve the patient receiving high doses of chemotherapy or radiotherapy, followed by the infusion of donor bone marrow or stem cells. Primary bone marrow cells are grown in specific growth media to support their expansion and maintenance. Bone marrow and stem cell transplants are also used to treat cancer, particularly leukemias, where donor T-cells have a graft-versus-tumor effect. Bone marrow transplantation from haploidentical donors may result in lower long-term survival rates.

Chapter: Location | Thymus

Upon leaving the thymus, T cells are referred to as recent thymic emigrants. These cells undergo a critical phase of development in the thymus, where they rearrange their T cell receptor (TCR) genes. All T cells originate from c-kit + Sca1 + haematopoietic stem cells (HSC) that reside in the bone marrow, and upon migration to the thymus, they differentiate into distinct subpopulations of mature T cells through proliferation and differentiation pathways. The earliest cells that arrive in the thymus are termed double-negative, as they express neither the CD4 nor CD8 co-receptor.

In the thymus, immature CD4 single-positive thymocytes that receive T cell antigen receptor (TCR) signals and have high affinity for self-peptides differentiate into FOXP3 + Treg cells, also known as thymic Tregs or tTregs. Central tolerance occurs in the thymus, where thymic dendritic cells (DCs) present self-antigens, leading to the apoptosis of high-affinity autoreactive T cells (negative selection) and guiding low-affinity CD4⁺ T cells to differentiate into Treg cells (positive selection). The presentation of self-peptides derived from endogenous proteins in the thymus is responsible for both the negative selection of self-reactive T cells and the positive selection of thymodependent regulatory T (tTreg) cells that can inhibit self-reactive T cells in the periphery.

The remaining cells exit the thymus as mature naive T cells, also known as recent thymic emigrants. In contrast, newly-generated FOXP3 + T reg cells in the thymus do not have a high level of Il2ra expression. The thymus shrinks by about 3% a year throughout middle age, leading to a corresponding fall in the thymic production of naive T cells, which leaves peripheral T cell expansion and regeneration to play a greater role in protecting older people. This novel knowledge

of normal and pathological functions of the thymus constitutes a solid basis for the development of a novel type of tolerogenic/negative 'self-vaccination' against type 1 diabetes.

Chapter: Location | Lymph Nodes

Once activated, leukocytes migrate to the lymph nodes, where they interact with T cells and B cells to initiate and shape the adaptive immune response. These interactions are crucial for the development of effective immune memory and the generation of a robust immune response against pathogens. Although observations suggest that the endothelial junction is the sole barrier a leukocyte needs to overcome *in vivo*, 'closing' the paracellular route did not affect lymphocyte entry to lymph nodes via high endothelial venules (HEVs), indicating a vessel- or tissue-specific preference of diapedesis route (Schulte et al., 2011). Leukocytes capture and process antigens, then migrate to lymph nodes to present these antigens to T cells, initiating and regulating T cell responses crucial for adaptive immunity.

Compared to an FDA-approved formulation, the redesigned lipid nanoparticles (LNPs) delivered more than three times as much mRNA to the lymph nodes relative to the liver. This enhanced delivery to the lymph nodes is particularly significant for vaccine development and gene therapy applications, as it can lead to more potent immune responses. Large complexes of intact antigen are presented in lymph nodes to B cells by follicular dendritic cells in the form of immune complexes. Interestingly, in studies on lymph nodes, a low pH acidic environment inhibited T cell monocarboxylate transporters (MCTs), causing negative feedback regulation of glycolysis and further inhibiting the effector function of CD8⁺ T cells; however, this did not prevent the initial activation of nascent T cells by dendritic cells.

Lymph node cortical sinus organization and relationship to lymphocyte egress dynamics and antigen exposure are critical for understanding immune responses. Modifying the size of LNPs has been shown to enhance their targeting to the lymph nodes, as demonstrated by Hideyoshi Harashima's research team. Nakamura et al. (2022) further explored the effect of size and charge of lipid nanoparticles prepared by microfluidic mixing on their lymph node transitivity and distribution. This memory subpopulation is commonly found in the lymph nodes and in the peripheral circulation, contributing to long-term immune memory.

In immunoperoxidase staining of sections of lymphoid organs, the HECA-452 antibody clearly stained all HEVs observed in tonsils, lymph nodes, and gut-associated lymphoid tissue (GALT). When an antigen is identified by the adaptive immune system, antigen-presenting cells (APCs) become active and migrate to the lymph nodes in order to activate B and T cells. The gut mucosal immune system, which consists of lymph nodes, lamina propria, and epithelial cells, constitutes a protective barrier for the integrity of the intestinal tract.

Chapter: Location | Skin

Tissue-resident memory T cells (T_{RM}) occupy the skin without recirculating, providing a rapid response to pathogens that breach the skin barrier. Interestingly, the vasculatures in the skin and colon selectively prefer the extravasation of PDGFR α -lineage Ly6C⁺ monocytes, which play a crucial role in immune responses at these sites. In the skin, monocytes mainly extravasate in the hypodermis, where laminin- α 2 isoforms colocalize with type-V collagen, facilitating their migration and activation.

Subcutaneous (s.c.) injection is commonly used for vaccination because resident and recruited antigen-presenting cells (APCs) are present in the skin and muscle, which can internalize and process mRNA-encoded antigens. Once microbes reach physical barriers such as the skin or intestinal tract mucosa, they are recognized by Toll-like receptors (TLRs), which activate immune cell responses. Tests may also be run on the skin to elicit an immune system reaction, providing a non-invasive method for assessing immune function.

Cell death triggers inflammation by exposing the components of dying cells to neighboring cells, as well as by compromising the barrier integrity in the skin and intestine, allowing microbes to infiltrate the tissue. This process is critical for initiating an immune response and preventing pathogen spread.

Chapter: Location | Tumor Microenvironment

Secreted by tumor cells and other cell types, including immune cells and stromal cells in the tumor microenvironment (TME), chemokines can modulate immune cell trafficking and lymphoid tissue growth, thus regulating anti-tumor immunological responses in a spatiotemporal manner. Within the TME and in chronic inflammatory settings, elevated TGF- β and IL-10 suppress effector T cell activation while expanding Tregs, fostering an immunosuppressive milieu. Regulation of the IL-23 and IL-12 balance by Stat3 signaling in the TME plays a crucial role in modulating immune responses. Jiang et al. (2023) highlight the role of the TME in PD-L1/PD-1-mediated tumor immune escape, emphasizing the complex interplay between tumor cells and the immune system.

Metabolic competition in the TME is a driver of cancer progression, as described by Chang et al. (2022). This metabolic shift leads to significant glucose depletion within the TME, while simultaneously producing lactic acid as a byproduct. Stromal cells in the TME, as discussed in Zhao et al. (2023), are accomplices of tumor progression, contributing to the complex ecosystem of the TME. On one hand, immune cells present in the TME can play the roles of immune surveillance and immune clearance by inducing tumor cell death; on the other hand, tumor cells can evade programmed cell death (PCD) by establishing immune tolerance.

Regulatory T cells (Tregs) in the TME have a complex role, as highlighted in Wang et al. (2023). These cells can suppress tumor immunity, hindering the body's innate ability to control the growth of cancerous cells. The dual role of tumor-associated macrophages (TAMs) in the TME is also significant, as they can both promote and inhibit tumor growth depending on their polarization state. High levels of cholesterol in the TME are positively correlated with CD8⁺ T cell failure, as noted in multiple studies.

The TME represents a highly complex ecosystem wherein dynamic interactions among malignant cells, stromal constituents, and immune cell populations critically influence disease progression and therapeutic efficacy. Hypoxia, arising primarily from the rapid proliferation of tumor cells that outpace the development of vascular supply, leads to regions of oxygen deprivation, further complicating the tumor microenvironment. The accumulation of adenosine in the TME promotes cancer immune evasion, and A2AR blockade rescues immune cell function. High numbers of Treg cells in the TME are indicative of a poor prognosis, as suggested by numerous studies.

CAR T cells are also not trafficked efficiently into the center of solid tumor masses, and the hostile TME suppresses T cell activity. Strategies have been developed to address this issue, including the use of exosomes secreted by TAMs in the TME, which highly express PD-L1 and inhibit CD8⁺ T

cell function through a Rab27a-dependent mechanism. Notably, knockdown of Rab27a in macrophages significantly enhances the efficacy of anti-PD-1 therapy. Proinflammatory cytokines in the TME inhibit antitumor immunity, further contributing to the immunosuppressive nature of the TME.

Chapter: Location | Germinal Center

Germinal center B cells undergo V-region somatic hypermutation, a process that introduces point mutations in immunoglobulin variable region genes to generate a diverse pool of B cell variants. This process, one of the four mechanisms that create immunoglobulin diversity, was described in detail in Chapter 4. Cells with mutations that improve affinity for antigen are selected, a critical step in the affinity maturation process. A temporal switch within the germinal center determines the differential output of memory B and plasma cells, ensuring that only the most antigen-specific B cells survive and differentiate.

The intrinsic properties of human germinal center B cells set antigen affinity thresholds, as highlighted in the study by Kwak et al. (2021). The mTORC1 kinase pathway is dynamically regulated in germinal centers, influencing the survival and differentiation of B cells. The microanatomic segregation of selection by apoptosis in the germinal center, as described by Mayer et al. (2018), ensures that only the most fit B cells survive. Germinal center B cells recognize antigen through a specialized immune synapse architecture, as noted by Nowosad et al. (2021).

Deleterious mutations are frequent in germinal centers, leading to the apoptosis of B cells that are quickly engulfed by macrophages, resulting in tingible body macrophages. Sustained antigen availability during germinal center initiation enhances antibody responses to vaccination, as observed by Tam et al. (2021). Germinal center centroblasts transition to a centrocyte phenotype according to a timed program and depend on the dark zone for effective selection, as discussed by Victora and Nussenzweig (2018).

Within germinal centers, isotype switching of immunoglobulin genes occurs after the onset of somatic mutation, as reported by Liu et al. (2001). Germinal center B cell dynamics are influenced by the control of inflammation, cytokine expression, and germinal center formation by BCL-6. Germinal center B cell maintenance and differentiation are controlled by distinct NF- κ B transcription factor subunits, as detailed in J. Exp. (2018). In the dark zone of germinal centers, centroblasts proliferate rapidly while undergoing somatic hypermutation, generating a diverse pool of B cell variants. DNA demethylation is observed at Ig promoters and enhancers following B cell activation, mediated in part by AID's deamination of 5-methylcytosine (5mC) to generate substrates for base excision repair and TET enzyme activity.

Germinal centers within the patches contain IgA⁺ B-cells, which differentiate into antibody-producing plasma cells, supporting antigen-specific and high-affinity IgA responses. Germinal centers found within Peyer's patches are areas where B lymphocytes proliferate, differentiate, and undergo somatic hypermutation to produce high-affinity antibodies.

Chapter: Location | Lung

Multiple immunohistochemical analyses of lung tissue from patients who died from COVID-19 confirm macrophage infiltration, a significant proportion of which have increased expression of the C5aR1 receptor with pro-inflammatory functions. Proteolytic cleavage of elastin, abundantly found

in elastic tissues such as skin and lung, by macrophage MMP-12 produces a hexapeptide VGVAPG to chemoattract monocytes via binding S-Gal receptor (Senior et al., 1984; Taddese et al., 2009). Tissue-resident NK cells have also been identified in sites like bone marrow, spleen and more recently, in lung, intestines and lymph nodes. Granulomas are most common in the lung, but they can appear anywhere in the body.

Increased viral infection in the lung and greater mortality were observed in *Ascaris suum* and vaccinia virus (VACV) co-infected mice. This may be because the host response to infection rather than direct injury of respiratory cells by viral infection accounts for the clinical and pathological changes in the lung. In a co-infection study of *H. polygyrus* and RSV, enteric helminth infection through pathways that remain undefined induced type I interferon (IFN) signaling in the lung to protect against viral infection, possibly through helminth interactions with the gut microbiota. Biochemical and morphological effects in the liver, kidneys and lungs of rats following intravaginal and intraperitoneal administration were noted. Limiting the infiltration of immune cells to the lungs or changing the quality of their response could lessen pulmonary inflammation. They focus on pathogens that infect the lungs, both because of the interesting immune dynamics at play in this organ system, and due to the global health relevance of understanding and treating lung infections.

In cancers such as breast and lung, high TAM infiltration correlates with increased metastatic potential and poor patient prognosis, as evidenced by associations with advanced stages and reduced survival rates. The present invention permits the targeted inhibition of lymphocyte and leukocyte entry into mucosal lymphoid and inflammatory tissues, into lymph nodes, into synovium and skin, and by extension into other discrete organs such as brain, heart, kidney, lung and liver. Regulatory T cells also perform tissue-specific roles in repair, tolerance, and modulation of inflammation across the central nervous system, gastrointestinal tract, joints, skin, and lungs. The upper portion of the lower lobe or the lower portion of the upper lobe are typically home to the Ghon focus, the predominant site of infection in the lungs.

Chapter: Location | Spleen

Then, in 1973, two Canadian scientists, Ralph Steinman and Zanvil Cohn, discovered an immune cell in the spleen and peripheral lymph node of a mouse with a unique "stellate" morphological appearance and named it a "dendritic cell" because of its myriad filopodial protrusions that extended from the cell body. Different animal models have demonstrated the importance of the spleen as a mediator in LPS-induced CS with regards to splenic macrophages or the activation of nonsplenic IL-6 production. In addition, an immediate precursor to myeloid and lymphoid dendritic cells of the spleen has been identified.

The spleen plays a crucial role in the pathogenesis of miliary tuberculosis, where the infection begins when a *Mycobacterium tuberculosis* bacterium, inhaled from the air, penetrates the lungs and reaches the alveoli. The spleen and mesenteric lymph nodes of germ-free (GF) mice have absent lymphocyte zones, indicating a far-reaching and systemic influence of the gut microbiome on the host and providing insight into the possible future pathologic development of numerous diseases. The spleen and mesenteric lymph nodes of GF mice have absent lymphocyte zones, indicating a far-reaching and systemic influence of the gut microbiome on the host and providing insight into the possible future pathologic development of numerous diseases.

The spleen is also involved in the expression of surface antigens on parasitized erythrocytes,

influencing the immune response. Improved T-cell responses and anticancer immunity were induced by the intravenous (IV) delivery of mRNA vaccines, which is due to its ability to mobilize the substantial antigen-presenting cell (APC) pools that are present in the spleen. The prolonged circulation time of lipid/pDNA particles in mice resulted in dose accumulation in distal tumors over 24 hours with minor accumulation in the liver and spleen. The replacement of helper lipids with charged alternatives in lipid nanoparticles facilitates targeted mRNA delivery to the spleen and lungs.

Chapter: Location | Endosomes

The cytosolic pathway involves the transport of antigens from the lumen of endosomes to the cytosol, where they are processed by proteasomes and then coupled to MHC class I molecules in the endoplasmic reticulum. When foreign substances, such as certain nanoparticles, are trapped in endosomes after cellular uptake, the lower pH environment causes ionizable lipids within the nanoparticles to become protonated. This protonation facilitates membrane destabilization, leading to the release of the payload into the intracellular compartment.

The foreign and pathogen-derived proteins are partially degraded in various cellular compartments, including endosomes, lysosomes, cytosol, and endoplasmic reticulum, by hydrolases and cathepsins. These degraded proteins are then presented as antigenic peptides on MHC class I and class II molecules on the surface of dendritic cells (DCs). The mechanism by which squalene-based cationic nanoemulsions escape from endosomes to deliver mRNA into the cytoplasm remains unclear.

Chapter: Location | Lamina Propria

The lamina propria is a critical component of the gastrointestinal tract, serving as a major mucosal effector site alongside the ileum's submucosa. It is also present in other mucosal areas such as the lacrimal, nasal, and salivary glands in the upper airways, and in mammary glands where it contributes to milk production. Specifically, the lamina propria resides within the lamina propria and extends into the submucosa of the ileum, playing a vital role in immune responses and mucosal defense. The origin of the lamina propria dendritic cell network has been studied, highlighting its importance in immune regulation. Additionally, the lamina propria, along with the intraepithelial layer, epithelial layer, and gut-associated lymphoid tissue (GALT), features structures such as Peyer's patches, isolated lymphoid follicles, and mesenteric lymph nodes (mLNs), which are primary locations for gut macrophages.

Chapter: Location | Kidneys

- **Article PubMed PubMed Central Google Scholar Harada, T. et al. Effects of recombinant human soluble thrombomodulin on neutrophil extracellular traps in the kidney of a mouse model of endotoxin shock.** This study investigates the impact of recombinant human soluble thrombomodulin on neutrophil extracellular traps (NETs) in the kidneys of mice subjected to endotoxin shock, highlighting the potential therapeutic benefits in managing kidney inflammation and injury.
- **In the kidney, CD206+macrophages with transcript expression characteristics of the efferocytic S1 macrophages have been shown to cause renal fibrosis (Cheung et al.,**

2022; Zimmerman et al., 2019). These macrophages play a critical role in the development of renal fibrosis by promoting the clearance of apoptotic cells, which can lead to chronic kidney damage.

- **Extra-pulmonary TB infection can affect the kidneys, spine, brain, lymph nodes, or bones - a sample cannot easily be obtained for culture.** Tuberculosis (TB) can disseminate beyond the lungs, affecting various organs including the kidneys, where obtaining a sample for culture can be challenging due to the difficulty in obtaining a representative sample.
- **PEG lipids reduce the tendency of particle aggregation by steric stabilization, thereby enhancing formulation stability and prolonging the blood circulation time by reducing clearance mediated by the kidneys and mononuclear phagocyte system (MPS) (67, 68).** Polyethylene glycol (PEG) modification of drug formulations can improve their pharmacokinetic properties by reducing interactions with the kidney and MPS, thus extending their circulation time in the bloodstream.
- **For slender lungfish, we confirmed the MHC-IIabSol sequence experimentally using RNA from kidney.** This finding indicates the presence of specific major histocompatibility complex (MHC) sequences in the kidney of slender lungfish, which could be relevant for understanding immune responses in these aquatic organisms.
- **Total RNA was isolated from kidney by use of “TRIZol” (Gibco) following the instructions of the manufacturer, except that the protocol was repeated for additional removal of DNA and proteins.** This method ensures the accurate isolation of RNA from the kidney, facilitating downstream applications such as gene expression analysis and transcriptomics.
- **Because of the increased molecular weight, AMV564 is more difficult to be cleared by the kidney compared with AMG330, which gives it a longer half-life.** The larger molecular size of AMV564 results in reduced renal clearance, leading to a prolonged duration of action and potentially higher therapeutic efficacy.
- **doi: 10.2337/db19-0287 CrossRef Google Scholar 85 Arazi A Rao DA Berthier CC Davidson A Liu Y Hoover PJ et al. The immune cell landscape in kidneys of patients with lupus nephritis.** This study provides insights into the immune cell composition in the kidneys of patients with lupus nephritis, highlighting the complex immune response and potential targets for therapeutic intervention.

Chapter: Location | Peripheral Blood

Peripheral blood plays a critical role in the immune response and tumor microenvironment (TME). Soluble tumor-derived signals orchestrate the recruitment of monocytes from peripheral blood, which then undergo polarization into pro-tumoral tumor-associated macrophages (TAMs) within the immunosuppressive TME. This process can be modulated by CCR2 antagonists, which have been shown to reduce the infiltration of M2-type TAMs and increase the number of CD8⁺ T cells and CD8⁺ tumor-infiltrating lymphocytes (TILs) with tumor-killing effects. However, the heterogeneity of responses between individuals and the limited access to tissue samples beyond the peripheral blood present significant challenges for human studies on helminths, yet they also offer

opportunities to enhance our understanding of human immunology. Additionally, quantitative DNA methylation analysis of FOXP3 has emerged as a new method for counting regulatory T cells in both peripheral blood and solid tissue, providing a valuable tool for studying immune cell populations.

Chapter: Location | Small Intestine

The small intestine is a critical segment of the gastrointestinal tract, playing a pivotal role in nutrient absorption and immune defense. The lamina propria of the small intestine contains dendritic cells that promote the de novo generation of Foxp3⁺ regulatory T (Treg) cells via retinoic acid, contributing to the maintenance of immune tolerance and preventing autoimmunity.

Peyer's patches, small, organized clusters of lymphoid tissue found in the mucosa and submucosa of the small intestine, particularly in the ileum, serve as key immune defense sites. These patches are essential for the induction of immune responses, including the development of IgA-producing cells, which are crucial for mucosal immunity. Isolated lymphoid follicles, single intestinal lymphoid aggregates with or without germinal centers, are also present along the small intestine, functioning as dynamic immune reservoirs for both T cell-independent and T cell-dependent induction of IgA.

The small intestine epithelia are equipped with Paneth cells located at the base of crypts, which secrete antimicrobial peptides (AMPs) in response to bacteria or pathogens in the gut lumen. This innate host defense mechanism is crucial for maintaining a balanced microbiome and preventing pathogen invasion. The presence of helminths supports initial *Salmonella* colonization, highlighting the complex interplay between different pathogens and the host immune system. The small intestine acts as the major site of immune priming, with a significant impact on the host's immune response to both pathogens and commensal bacteria.

Chapter: Location | Liver

Although lipid-based nanoparticles (NPs) are believed to passively target and accumulate in tumors through the enhanced permeability and retention (EPR) effect, a growing body of research has revealed that lipid-based NPs, without any ligand-modification for active targeting, mainly accumulate in the liver upon systemic administration (Kim M, et al., 2017). This phenomenon is particularly relevant in the context of organ transplantation, where the current standard of care requires patients to be on immunosuppressive drugs for many years, which can affect the skin, the liver, and the gastrointestinal (GI) tract. In the liver, which is both a source and a target of inflammation, hepatocyte injury and apoptosis induced by tumor necrosis factor- α (TNF- α) occur, with Kupffer cells amplifying cytokine release. Albumin (ALB), a negative acute-phase reactant synthesized in the liver, maintains oncotic pressure and exerts antioxidant and anti-apoptotic effects at the endothelial interface.

In the liver, assays have shown that memory CD8⁺ T cells generated by liver sinusoidal endothelial cells (LSECs) following Toll-like receptor 4 (TLR4) activation exhibit high levels of mitochondrial respiration and constitutively low levels of glycolysis to support their scavenger and sentinel functions. Some eggs from parasitic infections, such as those from *Plasmodium* species, are dragged by the blood flow into the liver where they are trapped in the small sinusoidal vessels, inducing a strong granulomatous response. This response is essential for compensating the damage caused by

the migration of the parasite in the host tissues, thereby allowing its establishment in the liver during the chronic phase of the disease. In the absence of interleukin-4 receptor α (IL4R- α) in macrophages (Mo), mice experience increased levels of Th1 cytokines, nitric oxide synthase-2 (NOS-2) activity, and immunopathology in the liver and gut, leading to increased egg expulsion.

Like delivery trucks taking the wrong exit off the highway, nanoparticles often end up accumulating preferentially in the liver instead of in lymphoid organs, where immune cells coordinate the body's response to a vaccine. This preferential accumulation in the liver can be a significant challenge for targeted drug delivery systems, as it can lead to off-target effects and reduced efficacy in other organs.

Chapter: Location | Endoplasmic Reticulum

The endoplasmic reticulum (ER) serves as a critical site for peptide loading onto MHC class I molecules. These peptides are translocated into the ER via the transporter associated with antigen processing (TAP), where they bind nascent MHC class I molecules to generate pMHCI complexes. The resulting peptides are then translocated to the ER via TAP transporters, loaded onto MHC-I, and trafficked through the Golgi apparatus to the cell surface. Supporting this process, the aminopeptidase complexes ERAP1 and ERAP2 play a role in peptide trimming within the ER, as detailed in the study by Saveanu, L. et al. Additionally, it is understood that upon activation of receptors, calcium is released from the ER, although the exact mechanism and its role in the overall process are still subjects of ongoing research.

Chapter: Location | Nucleus

Viral DNA is then imported into the nucleus and integrated into the cellular genome, where it reproduces. This process allows the virus to hijack the host cell's machinery for its own replication. Once translocated into the nucleus, activated p38 MAPK phosphorylates transcription factors such as ATF2 and NF- κ B or co-activators, thereby enhancing their binding to ISRE elements within ISG promoters (e.g., the Schlafen gene family) or to GAS elements, ultimately promoting ISG transcription. Upon translocation to the nucleus, these multiple transcription factors act to synergize or antagonize each other, more precisely tailoring the inflammatory response to the pathogen. The generation of oxidants by MPO liberates NE from the azurosome, allowing it to translocate to the nucleus where it aids in the decondensation of the chromatin by proteolyzing histones (Metzler et al., 2014). The dimer moves into the nucleus, binds to the DNA, and causes transcription of genes.

Chapter: Location | Vascular Basement Membrane

The vascular basement membrane plays a crucial role in the interaction between endothelial cells and leukocytes, facilitating the process of leukocyte extravasation. Gros et al. (2015) and Currie et al. (2022) have shown that the vascular basement membrane, in conjunction with endothelial focal adhesions, enables endothelial cells to generate forces stimulated by platelet-derived Ang1-Tie2 activation. These forces help in the formation of cortical actin bundles around passing leukocytes, which are essential for the stabilization of endothelial junctions. The vascular basement membrane also coordinates these processes through the transient exposure of collagens, which are integral to the structural support and force-dependent remodeling of endothelial junctions.

In general, the endothelium and the vascular basement membrane serve as the primary barriers for

leukocyte extravasation. The vascular basement membrane, through its interaction with the extracellular matrix (ECM) proteins, regulates various processes that are critical for the proper functioning of the vascular system. Faint yellow stars, which are processes regulated by the vascular basement membrane as a collective structure, highlight the dynamic nature of these interactions and the importance of the vascular basement membrane in maintaining the integrity of the vascular barriers.

Chapter: Location | Cytosol

After 2 days, *M. tuberculosis* progressively translocates from phagolysosomes into the cytosol in nonapoptotic cells. This translocation precedes the induction of apoptosis, suggesting a more complex mechanism than mere outgrowth within the phagolysosomal space. Cationic polymers can form structurally heterogeneous complexes with mRNA, known as polyplexes, which can be endocytosed by cells. Although the exact endosomal escape mechanism for polyplexes is unclear, it involves the delivery of mRNA into the cytosol. This result underscores that the translocation of *M. tuberculosis* to the cytosol is a distinct and critical step in the infection process, not merely a consequence of the bacteria outgrowing its phagolysosomal environment.

Chapter: Location | Peyer's Patches

Peyer's patches are observable as elongated thickenings of the intestinal mucosa measuring a few centimeters in length. These structures are characterized by the follicle-associated epithelium (FAE), which covers all lymphoid follicles and is a key component of the gut-associated lymphoid tissue (GALT). The function of Peyer's patches is to analyze and respond to pathogenic microbes in the ileum, playing a crucial role in the immune surveillance of the intestinal mucosa.

Microscopically, Peyer's patches appear as oval or round lymphoid follicles, similar to lymph nodes, located in the mucosa layer of the ileum and extending into the submucosa layer. The epithelium of Peyer's patches in the rat has been studied using morphological, immunohistochemical, and morphometrical techniques, highlighting its specialized structure and function. Although B-cell populations recovered after treatment ended, these findings may provide insight into the role that Peyer's patches may play in gastrointestinal diseases.

Peyer referred to Peyer's patches as plexus or agmina glandularum (clusters of glands), while Schenk thought that intestinal worms resided in Peyer's patches and that the orifices of the patches were the worms' mouths. Indeed, one could even be tempted to regard Peyer's patches as lymph glands embedded between the walls of the intestinal mucosa. The number of Peyer's patches peaks at age 15–25 and then declines during adulthood, reflecting changes in immune function over the lifespan.

Having too many or larger than normal Peyer's patches is associated with an increased risk of prion diseases and intussusception in children. As Peyer's patches are packed with immune cells and produce protective proteins such as secretory IgA to maintain gut balance, their dysfunction can trigger inappropriate immune responses, driving the inflammation and tissue damage.

Chapter: Location | Follicle-Associated Epithelium

The follicle-associated epithelium (FAE) differs from the epithelium of the villus mucosa in several

key aspects. The production of mucus in FAE is weak, and membrane-bound digestive enzymes are lightly expressed. Additionally, the enterocyte brush border glycocalyx in FAE exhibits different glycosylation patterns compared to other epithelial types. The basal lamina of FAE is more porous than that of intestinal villus, yet the paracellular pathway in FAE remains tightly closed to prevent the penetration of antigens and maintain continuous contact with immune cells. Claudin expression in FAE, particularly in rat Peyer's patches, defines a major restriction of the paracellular pathway. Finally, FAE is less permeable to ions and macromolecules, largely due to the higher expression of tight junction proteins.

Chapter: Process | Adaptive Immune Response

Although this defense is effective, a complete cure of the infection requires mounting an adaptive immune response that is controlled to prevent immune pathogenicity. The adaptive immune system, unlike the innate system, is characterized by a time lag from the pathogen invasion, as it involves the recognition of specific antigens, the distinction between self and non-self antigens, and the development of an antigen-specific response. The gut microbiota plays a significant role in the development of the major subtypes of CD4⁺ T cells: Th1, Th2, Th17, and Tregs. This mechanism enhances the breadth of the immune response, ensuring that mRNA vaccines can effectively stimulate both arms of the adaptive immune system. In Crohn's disease, the innate immune response exerts a greater impact compared to adaptive immunity due to the ability of macrophages to interact with T cells and initiate adaptive immune responses.

Rapid T cell activation induces aerobic glycolysis, as determined by extracellular flux analysis using a Seahorse XFe96 bioanalyzer. T cell differentiation involves several stages, each characterized by unique cellular and molecular changes, starting with hematopoietic stem cells (HSCs) in the bone marrow. Isotype switching occurs when a B cell changes which heavy chain constant domain it pairs with its variable chain domain. This process is crucial for the functional diversity of the humoral immune response, directed by cytokines, especially those released by armed effector CD4 T cells. Activation-induced cytidine deaminase (AID) plays a central role in isotype switching and somatic hypermutation, a programmed process of mutation affecting the variable regions of immunoglobulin genes. Somatic hypermutation introduces mutations in antibody genes, and clonal selection competitively favors B-cells that produce antibodies with higher antigen affinity.

The DNA rearrangements that underlie isotype switching and confer this functional diversity on the humoral immune response are directed by cytokines, especially those released by armed effector CD4 T cells. The C-terminal region of AID is responsible for a recombination function other than DNA cleavage in class switch recombination. Regulation of class switch recombination and somatic mutation by AID phosphorylation is also critical. The pathways used for processing the AID-generated U:G lesions determine the variety of base substitutions observed during somatic hypermutation. The CCCTC-Binding Factor locks premature IgH germline transcription and restrains class switch recombination. Enhanced intraswitch region recombination during immunoglobulin class switch recombination in 53BP1^{-/-} B cells suggests that 53BP1 plays a role in regulating this process.

In invertebrates without an adaptive immune system, ficolins are expanded and their binding specificities diversified to compensate for the lack of pathogen-specific recognition molecules. T cell differentiation following maturation involves naive T cells circulating through the circulatory

and lymphatic systems until presented with an antigen for which they bear the receptor. The specific adaptive immune response regulated by the TH cell depends on its subtype (i.e., TH1, TH2, TH17, and regulatory T-cell), which is distinguished by the types of cytokines they secrete. This process is a defining feature of the adaptive immune system.

Chapter: Process | T Cell Exhaustion

Defining 'T cell exhaustion' is a state where T cells lose their function and proliferative capacity, often in response to chronic antigenic stimulation. This condition is marked by the upregulation of inhibitory receptors such as PD-1, TIM-3, and LAG-3, which contribute to the impaired function of effector T cells. Disadvantaged effector response due to T cell exhaustion can lead to viral escape and further T cell exhaustion, creating a vicious cycle. Additionally, neutrophil extracellular traps (NETs) play a role in immune evasion by decorating tumor and immune cells with PD-L1, thereby impairing the cytotoxic activity of CD8⁺ T cells and natural killer (NK) cells.

Metabolic and epigenetic regulation are key factors in the development of T cell exhaustion. Chronic exposure models have shown that persistent antigen stimulation can induce T cell exhaustion, characterized by reduced proliferative capacity. This stage of sepsis also sees an increase in T cell exhaustion, highlighting its role in immune dysregulation. During transplantation, T cell exhaustion can develop following persistent antigen exposure, similar to the situation seen with alloantigen presence. These data suggest that T cell exhaustion plays an important role in the tolerance of a graft, primarily by depleting alloreactive CD8 T cells. Chronic infection has been shown to positively influence graft acceptance and long-term survival, mediated partly by T cell exhaustion. Recipient T cell exhaustion also provides conditions for NK cell transfer, further complicating the immune response.

Several studies have demonstrated the positive effect of chronic infection on graft acceptance and its long-term survival, mediated partly by T cell exhaustion. However, T cell exhaustion also carries disadvantages, including an increased risk of infections and tumor development. Cancer-associated cells and tumor cells themselves can actively induce T cell exhaustion at the site of the tumor, contributing to the persistence of the disease. Inhibitory receptors, such as PD-1, are vital in T cell exhaustion, and blocking them can reverse these changes, enhancing the efficacy of treatments like CAR-T therapies. This genetic modification not only reduces the risk of T-cell exhaustion and graft-versus-host disease but also enhances the overall efficacy of CAR-T therapies, offering a more durable and potent treatment option.

Chapter: Process | Innate Immunity

Innate immunity is the first line of defense against infectious pathogens and cancer, sensing and responding to the structure-conserved molecules of the pathogens (pathogen-associated molecular patterns, or PAMPs) as well as the endogenous ligands released from damaged cells (damage-associated molecular patterns, or DAMPs). This rapid and nonspecific response is mediated by various cell types and molecular pathways, including phagocytes, natural killer cells, and antigen-presenting cells. These cells utilize pattern recognition receptors (PRRs) to detect PAMPs, initiating an inflammatory response and activating the complement cascade.

The complement system, a crucial component of the innate immune response, is found in species earlier than vertebrates, with recent evidence suggesting its presence in protostome horseshoe crab

species. This system can be recruited and activated by antibodies generated by the adaptive immune system, highlighting the interplay between innate and adaptive immunity. The complement cascade involves a series of biochemical reactions that lead to the lysis of pathogens and the recruitment of immune cells to the site of infection.

Inflammasomes, multi-protein complexes found within immune cells, play a crucial role in detecting harmful microorganisms and stress signals, thereby triggering an inflammatory response. The inflammasome NLRP3, for instance, is activated by various stimuli, including microbial components and host-derived danger signals, leading to the processing and release of pro-inflammatory cytokines such as IL-1 β and IL-18.

NETs (neutrophil extracellular traps) are another important aspect of innate immunity, providing a new link between innate immunity and hypercoagulability. NETs are extracellular structures composed of DNA, histones, and antimicrobial proteins that can stimulate the coagulation process by activating platelets, the coagulation cascade, and the vascular endothelium. This mechanism is particularly relevant in conditions such as sepsis and acute respiratory distress syndrome.

The complement system, cytokines, and Toll-like receptors (TLRs) are key molecular regulators of innate immunity. TLRs recognize conserved molecular structures broadly expressed on or in microbial and viral pathogens, initiating a cascade of events that lead to the activation of immune cells and the production of pro-inflammatory cytokines. For example, the recognition of peptidoglycan from the microbiota by Nod1 enhances systemic innate immunity, while the gut microbiota and innate immune system interact through gut-associated lymphoid tissues (GALTs) and mucosa-associated lymphoid tissues (MALTs) to maintain homeostasis.

Innate immunity is also involved in the pathogenesis of various diseases. For instance, in multiple sclerosis, TLR9-mediated innate immunity modulates pathogenicity. Similarly, in Type 1 diabetes, the gut microbiota and innate immunity are implicated in the development of the disease. The innate immune response to pathogens such as SARS-CoV-2 elicits high levels of TNF- α and IFN- γ , which act synergistically to amplify panoptosis, a form of programmed cell death.

The innate immune system is not only involved in pathogen recognition but also in the regulation of adaptive immunity. For example, the innate immune response orchestrates the adaptive immune system, leading to a second, more potent wave of amplification. This interplay between innate and adaptive immunity is crucial for the effective defense against pathogens and the development of immunological memory.

In conclusion, the innate immune system is a complex and dynamic network of cell types and molecular pathways that work together to provide rapid and effective defense against pathogens and cancer. Understanding the mechanisms underlying innate immunity is essential for developing rational treatment strategies for various diseases, including infectious diseases, autoimmune disorders, and cancer.

Chapter: Process | Signaling Pathways

Induced by Notch signaling, transcription factors TCF-1, GATA-3, and Bcl11b play critical roles in promoting T cell lineage commitment by limiting other lineage differentiation. Notch signaling is an evolutionarily conserved pathway that plays an important regulatory role in cell fate determination, proliferation, differentiation, and dynamic homeostasis. The core components of the

JAK-STAT pathway include four non-receptor tyrosine kinases known as JAKs—JAK1, JAK2, JAK3, and TYK2—which associate with the cytoplasmic domains of over 50 types of transmembrane receptors, and seven transcription factors called STATs—STAT1 through STAT6, plus STAT5A and STAT5B. Pharmacologic inhibition of JAK-STAT signaling promotes hair growth, and JAK-STAT signaling as a target for inflammatory and autoimmune diseases has been explored. The JAK/STAT pathway is an important cascade of signal transduction for multiple growth factors and cytokines, which regulates gene expression and cell activation, proliferation, and differentiation. A second adapter that facilitates JAK/STAT pathway activation is STATIP (STAT-Interacting Protein), a WD40 protein. Below we describe the various cytokines and proteins that affect the JAK/STAT pathway and the various cellular activities to understand how the JAK/STAT pathway is involved in disease development and thus suggest more effective therapeutic approaches. Therapeutic attenuation of IFN signaling in these scenarios can restore immune balance, reduce inflammation, and protect against immune-mediated organ injury. Targeting IFN signaling in combination with immunotherapies has emerged as a promising strategy to enhance treatment efficacy. In mammals, the JAK/STAT pathway is the principal signaling mechanism for a wide array of cytokines and growth factors. Notch signaling and the bone marrow hematopoietic stem cell niche are also critical in the initial commitment of T cell lineage in the thymus. Notch signaling further cooperates with pre-TCR signals at the β -selection checkpoint to ensure metabolic fitness and survival of thymocytes expressing functional TCR β chains, eliminating those with nonproductive rearrangements. Notch signaling upregulation in TAMs enhances the immunosuppressive activity, as demonstrated in pancreatic cancer. The JAK-STAT pathway is subject to stringent negative regulation that fine-tunes both the amplitude and duration of signaling, thereby preventing aberrant activation that could trigger immunopathology or cellular dysfunction. Inhibitors of the JAK-STAT pathway, such as ruxolitinib and baricitinib, have demonstrated efficacy in autoimmune and autoinflammatory conditions by dampening chronic IFN signaling. The JAK-STAT pathway in keloid pathogenesis, proteomimetics of natural regulators of JAK-STAT pathway, and the effect of the JAK2 inhibitor TG101209 against T cell acute lymphoblastic leukemia (T-ALL) are also areas of ongoing research. The JAK/STAT pathway regulates the downstream signaling of numerous membrane proteins such as G-protein-associated receptors, integrins, and others. PIAS inhibits gene transcription by directly binding to STAT dimers and thereby blocks STAT binding to DNA. The JAK/STAT pathway is involved in multiple diseases and their physiological processes, including fibrotic diseases, cardiovascular disorders, and cancer. The role of the JAK/STAT pathway in immune regulation has attracted increasing attention, and biophysical forces occupy an important role in the induction of downstream signaling in the JAK/STAT pathway. The JAK/STAT pathway is activated in a variety of common solid tumors contributing to an aggressive phenotype. The JAK/STAT pathway mediates the role of biophysical forces, and current studies have shown that JAK/STAT-mediated mechanotransduction often has important effects on cell physiological processes. Dysregulated IFN signaling disrupts neurotransmitter homeostasis and accelerates the progression of various neuroinflammatory and neurodegenerative disorders through the excessive production of pleiotropic cytokines. Timing and dosing remain critical determinants of efficacy, as excessive IFN signaling risks triggering autoimmunity or resistance. Numerous single nucleotide polymorphisms (SNPs) have been identified in genes associated with IFN signaling, including IFN receptor subunits, JAK kinases, and STAT transcription factor. The JAK-STAT pathway plays a major role in many fundamental processes, such as apoptosis and inflammation, and dysfunctional proteins in the pathway may lead to a number of diseases. Computational models of JAK-STAT signaling based on the laws of

chemical kinetics have elucidated the importance of these different regulatory mechanisms on JAK-STAT signaling dynamics.

Chapter: Process | Cellular Processes

Cellular Processes

Regulation of proliferation, survival, and apoptosis by members of the TNF superfamily is a critical aspect of cellular homeostasis. The TNF superfamily includes various ligands and receptors that modulate these processes, influencing cell fate decisions. Apoptosis, a tightly regulated form of programmed cell death, is essential for eliminating unwanted or damaged cells, maintaining cellular integrity, and preventing inflammation. The intrinsic pathway of apoptosis is initiated by the oligomerization of BCL-2 family proteins BAK and BAX, while the exogenous pathway is triggered by membrane receptors such as TNFR1, death receptors, or Toll-like receptors (TLRs). These pathways are finely balanced by BCL-2 family proteins, which can either promote or inhibit apoptosis, depending on the cellular context.

Apoptosis and pyroptosis are two distinct forms of programmed cell death with different mechanisms and outcomes. Apoptosis is characterized by morphological changes such as membrane blistering, reduction in cell size, nuclear fragmentation, chromatin condensation, phosphatidylserine (PtdSer) exposure at the cell surface, and apoptotic body formation. In contrast, pyroptosis is a form of cell death mediated by the inflammasome complex, which mainly consists of a sensor, an adaptor, and a zymogen pro-caspase-1. Pyroptosis is closely related to nervous system diseases, infectious diseases, autoimmune diseases, cardiovascular diseases, and tumors. It is executed through gasdermin family pore formation, while necroptosis is driven by RIPK3-mediated oligomerization of MLKL, resulting in membrane rupture and subsequent DAMP release.

The PANoptosome, a multiprotein complex, drives pyroptosis, apoptosis, and necroptosis (PANoptosis). This unified death pathway integrates molecular components of these three pathways and is regulated by PANoptosome complexes. PANoptosis is defined as a unique innate immune inflammatory lytic cell death pathway driven by caspases and RIPKs, executed by gasdermins, MLKL, and potentially other unknown proteins cleaved by caspases. The interplay between these pathways can lead to panoptosis, a comprehensive death mechanism that is crucial in cancer progression and treatment.

Tumor cells can induce apoptosis of T lymphocytes through the expression of FasL on their surface, leading to FasL-Fas interaction. Additionally, IFN- γ can enhance antitumor immunity by acting on the tumor stroma and endothelial cells, as well as directly inducing apoptosis in tumor cells.

Autophagy, a process that metabolically suppresses CD8⁺ T cell antitumor immunity, is a promising target for eliminating intracellular pathogens. The role of autophagy in NET regulation has emerged as a pivotal modulator of NET formation through intricate crosstalk with key signaling networks.

In the context of cancer, the programmed death of tumor cells can be directly induced by certain immune cells or participate in a positive feedback loop of anti-tumor immunity by spontaneously triggering a switch from apoptosis to pyroptosis. This interplay culminates in panoptosis, a unified death pathway that integrates molecular components of pyroptosis, apoptosis, and necroptosis. The roles of apoptosis, pyroptosis, and necroptosis in regulating the immunosuppressive tumor

microenvironment (TME) and determining clinical outcomes of tumor therapy have been highlighted in recent studies, revealing some mutual crosstalk between the different cell death mechanisms involved in PANoptosis.

Chapter: Process | Inflammation and Immune Response

Inflammation and immune response are critical processes in the body's defense against pathogens and tissue damage. Central to these processes is the activation of the inflammasome, a multiprotein complex that plays a pivotal role in the production of pro-inflammatory cytokines and the induction of cell death. The transcription factor NF- κ B, when activated in the nucleus, promotes the transcription of genes such as NLRP3, Pro-IL-1 β , and Pro-IL-18, which are essential for inflammasome activation. This activation leads to the caspase 1-dependent secretion of proinflammatory cytokines, IL-1 β and IL-18, and the induction of a gasdermin D-mediated form of cell death known as pyroptosis. The inflammasome is a key player in host defense against intracellular pathogens, but some bacteria have developed strategies to dampen or evade inflammasome activation, as reviewed elsewhere.

The dysregulation of inflammasome activation can contribute to the pathology of several major diseases, including cancer, autoimmune disorders, inflammatory conditions, metabolic disorders, and neurodegenerative diseases. The second signal for inflammasome activation is provided by NLRP3 agonists, which activate NLRP3 to trigger inflammasome assembly and mature IL-1 β secretion. Caspase-6 is a key regulator of innate immunity, inflammasome activation, and host defense. The activation of the inflammasome and the production of IL-1 β induced by endoplasmic reticulum stress can produce a positive feedback loop, amplifying the inflammatory response and leading to hepatic steatosis and injury.

Inflammasome activation is a double-edged sword, facilitating host defense against pathogens while also causing self-destructive systemic inflammation. The mechanisms designed for host defense can trigger a cytokine storm, a paradox where the body eliminates pathogens ("winning the battle") but triggers multiple organ dysfunction syndrome (MODS) ("losing the war"). Persistent bacteremia/viremia can sustain a feed-forward loop, amplifying the cytokine storm and compromising epithelial/endothelial barriers, facilitating pathogen dissemination. This "pyrrhic victory" exemplifies the complex nature of the immune response.

Appropriate inflammasome activation is vital for the host to cope with foreign pathogens or tissue damage, while aberrant inflammasome activation can cause uncontrolled tissue responses that may contribute to various diseases. It is imperative to maintain a fine balance between inflammasome activation and inhibition, which requires fine-tuned regulation of inflammasome assembly and effector function. Negative regulatory molecules, such as BHB, TMEM176B, macrophage CGI-58, IL-10, and NO, inhibit inflammasome activation by targeting K⁺ efflux, cytosolic Ca²⁺, and mitochondrial function. The activation of inflammasome and the production of IL-1 β can lead to caspase-1-dependent mitochondrial damage and block of mitophagy, further contributing to the inflammatory response.

Chapter: Process | Angiogenesis

The tumor microenvironment (TME) is a complex ecosystem surrounding cancer cells, encompassing immune cells, stromal cells, blood vessels, and the extracellular matrix (ECM).

Tumor hypoxia triggers angiogenesis by upregulating the expression of vascular endothelial growth factor (VEGF), a key driver of new blood vessel formation. This process, known as vascularization, is crucial for tumor growth and metastasis, significantly impacting cancer progression. Disorganized angiogenesis further promotes the formation of an immunosuppressive TME, facilitating immune evasion and tumor growth.

Angiogenesis is not only a local phenomenon but also a systemic one, where cancer cells release substances that affect the surrounding TME as the tumor grows. These substances support immune evasion, angiogenesis, and ECM modification. The metabolic profiles and intracellular signaling pathways of TME cells play a critical role in solid tumor angiogenesis. Key contributors include vasculature that supports angiogenesis, fibroblasts responsible for ECM remodeling, neuroendocrine components, cytokine and chemokine networks, and immune populations such as T cells, B cells, neutrophils, mast cells, and tumor-associated macrophages (TAMs). TAMs, in particular, adapt functionally to promote angiogenesis and support tumor growth, contributing to angiogenesis, lymphangiogenesis, immunosuppression, hypoxia-induced tumorigenesis, metastasis, and drug resistance.

Initiation of the anti-tumor immune response is often triggered by the local tissue disruption associated with angiogenesis or tissue-invasive growth, alerting cells of the innate immune system to the presence of a growing tumor. The TME also promotes angiogenesis and the formation of new blood vessels, which are essential for tumor growth and serve as conduits for metastasis. TAMs play a pivotal role in tumor formation and progression by directly promoting angiogenesis and indirectly through the activation of pro-angiogenic programs in tumor cells via molecules like TNF- α , IL-1 α , and COX-2.

Chapter: Process | Metabolic Processes

Inhibiting glycolysis may be a potential therapeutic strategy, as it can favor the formation of regulatory T (Treg) cells at the expense of Th17 cells. In glycolysis, one glucose molecule is converted through a 10-step enzymatic process into two pyruvate molecules, either used for aerobic respiration and oxidative phosphorylation in the mitochondria or further converted into lactate during aerobic glycolysis. This metabolic pathway is crucial for the function of immune cells, including CD8⁺ T cells, by providing the energy required for their proliferation and differentiation. For instance, metabolites fuel cellular processes through pathways including glycolysis, the tricarboxylic acid (TCA) cycle, and oxidative phosphorylation, supporting the immune cells' energy needs.

The functional integrity of CD8⁺ T cells is closely linked to metabolic reprogramming, and metabolic imbalance of glycolysis and oxidative phosphorylation (OXPHOS) can cause dysfunction of these cells. Metabolic restoration through further glycolytic activity can ameliorate this dysfunction, providing new insights into restoring CD8⁺ T cell function. Inhibiting glycolysis through pharmacological approaches, such as the use of 2-deoxyglucose and rapamycin, or genetic approaches, such as Hif1a deficiency, can favor Treg cell formation at the expense of Th17 cells. However, the specific mechanisms underlying the metabolic shift from glycolysis to fatty acid oxidation (FAO) to support memory CD8⁺ T cell differentiation remain unclear.

The Warburg effect, first described by Otto Warburg in the 1920s, refers to the disproportionate use of anaerobic glycolysis by tumor cells even under aerobic conditions. This effect has been extended

to activated immune cells, where glycolysis and glutaminolysis are essential for TLR2-mediated T cell activation. Key obesity-associated inflammatory mediators induce macrophage PD-1 expression through mechanisms dependent on mTORC1 and glycolysis. The upregulation of glycolysis levels after CD8⁺ T cell activation provides metabolic support for their further proliferation and differentiation. TCR can rapidly induce glycolysis in the early stages of CD8⁺ T cell activation, independent of de novo transcription and translation, and acting independently of the joint action of CD28 and AKT.

Glycolysis is a metabolic pathway of cells, where glucose generates pyruvate and ATP under the action of key enzymes of glycolysis, including GLUT, hexokinase (HK), phosphofructokinase (PFK), and pyruvate kinase (PKM). The upregulation of glycolysis levels caused by the activation of naive CD8⁺ T cells correlates positively with their proliferative capacity and reduces the sensitivity of cell growth to metabolic inhibition, providing energy and biosynthetic material for their growth, proliferation, and differentiation. Strengthening glucose absorption to promote glycolysis does not affect cytotoxic T lymphocytes (CTLs) proliferation but greatly enhances their killing efficiency, indicating that changes in CD8⁺ T cell self-glycolysis levels are crucial for their function in eliminating tumor or pathogen-infected cells.

Glycolysis and fatty acid synthesis are key features of lipopolysaccharide (LPS)-activated macrophages, while interleukin-4 (IL-4)-activated macrophages mainly use oxidative phosphorylation and fatty acid oxidation to generate energy. The balance between glycolysis and FAO is associated with the long-term survival of memory CD8⁺ T cells. Increasingly, studies suggest that a rapid metabolic switch to glycolysis is essential during the reactivation process of CD8⁺ Tm cells, which also contradicts their resting metabolic mode. Under hypoxic pressure culture conditions, upregulation of the levels of glycolytic pathway-related proteins such as GLUT1 and GLUT3 in memory CD8⁺ T cells lead to an increase in glycolysis level, thereby promoting their effector functions of IFN- γ and granzyme B production, but the ability of producing TNF- α and IL-2 is limited.

The mechanism by which glycolysis affects IFN- γ production by memory CD8⁺ T cells is clear and provides a reference for targeting glycolysis to further enhance memory CD8⁺ T cell function; however, the detrimental effect of high glycolysis on the long-term residence of memory CD8⁺ T cells will increase the difficulty of affecting CD8⁺ T memory function by targeting glycolysis. The decrease of glycolysis and OXPHOS activity of CD8⁺ T cells caused by the deletion of nucleus accumbens-associated protein 1 (NAC1) contributes to the enhancement of memory formation of virus Ag-specific CD8⁺ T cells after virus infection. The imbalance of glycolysis and OXPHOS often leads to dysfunction of CD8⁺ T cells, and glycolysis might affect the partial failure of CD8

Chapter: Process | Macrophage Function

The modern framework of macrophage polarization began to take shape in 2000, when Charles D. Mills and colleagues coined the terms "classical" (M1) and "alternative" (M2) activation to describe two opposing macrophage phenotypes in mice, distinguished by their arginine metabolism — nitric oxide production in M1 versus ornithine in M2 — and their alignment with Th1 or Th2 cytokine responses. Although the M1/M2 dichotomy is a widely used framework for describing macrophage polarization in tumor immunity research, recent studies have increasingly suggested that this framework is overly simplistic and does not fully capture the complex role of macrophages within the tumor microenvironment. Metabolic regulation of macrophage polarization in cancer plays a

crucial role in tumor progression and immune evasion. Additionally, macrophage polarization has implications in metabolic diseases and the role of exercise, highlighting the importance of understanding these processes in various physiological contexts.

A plethora of advances have provided a framework for rationally purifying, describing, and manipulating macrophage polarization. Macrophage polarization requires dynamic and reversible epigenomic marks at enhancers and promoters of signal-responsive genes, indicating the complexity and plasticity of these cells. In the context of cancer, major pathways involved in macrophage polarization include the interplay between cytokines, growth factors, and metabolic cues. The tumor microenvironment, characterized by hypoxia, nutrient scarcity, and immunosuppression, significantly influences macrophage polarization, shaping the immune response and tumor growth.

From an evolutionary perspective, macrophage polarization represents a conserved mechanism across vertebrates, originating from ancient phagocytic cells that allow adaptation to varying microenvironments and immune challenges. This evolutionary conservation underscores the fundamental importance of macrophages in host defense and tissue repair. In the context of autoimmune diseases, macrophage polarization regulates the pathogenesis and progression, highlighting the dual role of these cells in both promoting and resolving inflammation. Targeting the temporal dynamics of macrophage polarization and restoring the M1-to-M2 transition represent promising strategies for treating diabetic wounds and aging-impaired healing, underscoring the therapeutic potential of modulating macrophage function.

Chapter: Process | Epithelial-Mesenchymal Transition

Epithelial-mesenchymal transition (EMT) is a predominant event in the morphological transformation that contributes to the acquisition of malignant biological properties, including invasion and metastasis. This process is crucial for the development of invasive cancer cells, enabling them to detach from the primary tumor, undergo EMT, degrade the extracellular matrix (ECM), and enhance their motility. The invasive process involves multiple steps, starting with the detachment from the surrounding tissue architecture, followed by the transition from an epithelial to a mesenchymal phenotype, the degradation of the ECM, and the increase in cellular motility. Exosome-Mediated Metastasis: From Epithelial-Mesenchymal Transition to Escape from Immunosurveillance further elucidates the role of EMT in the metastatic process, highlighting the involvement of exosomes in facilitating the transition and evading immunosurveillance mechanisms.

Chapter: Process | Complement System

The complement system is a crucial component of the immune system, serving as a functional bridge between innate and adaptive immune responses. This system enhances the ability of antibodies and phagocytic cells to clear microbes and damaged cells, promotes inflammation, and directly attacks pathogen cell membranes. Complement activation is an important feature of both IgM and IgG, and various studies have suggested that excessive complement activation is a hallmark of sepsis, ischemia-reperfusion injury, and HIV/AIDS. The complement system consists of over 50 plasma proteins and protein fragments, including the classical, lectin, and alternative pathways. Each pathway involves a series of cleavage reactions that culminate in the formation of an enzymatic activity called a C3 convertase, which is bound covalently to the pathogen surface.

Complement control proteins regulate all three pathways and protect the host from the destructive effects of uncontrolled complement activation. The complement system is also involved in synaptic pruning and neurodegeneration, where overactivation can lead to damage.

Complement activation leads to the production of anaphylatoxins, potent proinflammatory molecules, and can be manipulated during HIV/AIDS, further damaging the body. The complement system is manipulated during HIV/AIDS, leading to additional tissue damage. The complement system is implicated in a variety of clinical conditions, including autoimmune diseases and ischemia-reperfusion injury in transplantation. The complement system is also involved in the pathogenesis of sepsis, where excessive activation can lead to acute organ damage. The alternative pathway is continuously activated at a low level, analogous to a car engine at idle, due to spontaneous C3 hydrolysis. This pathway is particularly important in distinguishing self from non-self based on the surface expression of complement regulatory proteins. The lectin pathway is homologous to the classical pathway but uses mannose-binding lectin and ficolins instead of C1q. The complement system has the potential to be extremely damaging to host tissues, and its activation must be tightly regulated to prevent such damage.

Chapter: Process | V(D)J Recombination

V(D)J recombination is a critical process in the adaptive immune system that allows for the generation of immunoglobulins and T cell receptors (TCRs) to antigens that the organism has not previously encountered. This process involves the somatic rearrangement of variable (V), diversity (D), and joining (J) regions of the genes encoding antigen receptors, leading to a diverse repertoire of both T-cell and B-cell receptors. The process occurs in primary lymphoid organs, such as the bone marrow for B cells and the thymus for T cells, and rearranges V, J, and in some cases, D gene segments in a nearly random fashion.

The mechanism of V(D)J recombination is mediated by the enzymes called RAG1 and RAG2, which form the VDJ recombinase. Recombination Signal Sequences (RSSs) play a crucial role in this process, acting as recognition sites for the VDJ recombinase. The assembly of a 12/23 paired signal complex is a critical control point in V(DJ) recombination, ensuring the proper initiation and termination of the process. The VDJ recombinase is a diverse collection of enzymes that are essential for the successful recombination of V, D, and J gene segments.

Aberrant V(D)J recombination events can lead to life-threatening conditions, including common lymphoid neoplasms and severe combined immunodeficiency (SCID) due to RAG mutations. These mutations can result in immunodeficiencies and oncogenic translocations, highlighting the critical role of V(D)J recombination in both immune function and disease. The process is therefore a very costly one that must be strictly regulated and controlled to prevent such adverse outcomes.

In addition to generating a diverse repertoire of B cell receptors (BCRs) and T cell receptors (TCRs), V(D)J recombination also allows for the generation of memory B and T cells, which are essential for long-term immune memory and the ability to mount a rapid and robust response to previously encountered pathogens. The importance of V(D)J recombination in the adaptive immune system is underscored by its role in recognizing and responding to specific pathogens, as well as in maintaining the fidelity of the recombination process.

Future research directions in V(D)J recombination include understanding the role of long-range interactions in the process, which may provide insights into the broader mechanisms of gene

regulation. Studies focusing on these aspects have contributed to our understanding of gene regulation as a whole, highlighting the importance of V(D)J recombination in the context of adaptive immunity and beyond.

Chapter: Process | Critical Cellular Processes

Abstract Class-switch recombination (CSR) is a genetic process by which a B cell switches from the production of IgM to the production of IgG, IgE, or IgA. This process is crucial for the diversification of the immune response and the generation of effector functions tailored to different pathogens. CSR involves the deamination of cytosine residues by activation-induced cytidine deaminase (AID) in the S regions of immunoglobulin genes, leading to double-strand DNA breaks (DSBs) and subsequent intrachromosomal deletions. The events that occur in the equilibrium phase of cancer immunoediting are likely similar to those in tumor dormancy, where the immune system can control tumor growth.

Hypermutation, triggered by AID, is a key component of CSR. AID deaminates single-stranded DNA, generating mutations that contribute to the affinity maturation of B cells. Successful hypermutation enhances the binding affinity of antibodies, leading to the selection of B cells that produce superior antibodies. The free ends of the DNA are rejoined by non-homologous end joining (NHEJ), linking the variable domain exon to the desired downstream constant domain exon of the antibody heavy chain. In the absence of NHEJ, alternative pathways may be used, often involving microhomology joins.

The CSR process is tightly regulated to minimize off-target mutations. AID's nuclear export via CRM1 and cell-cycle restriction are critical for this regulation. However, mistargeting of hypermutation can result in mutations and translocations in oncogenes, contributing to the development of B-cell malignancies. The SHM machinery is specifically recruited by immunoglobulin enhancers to immunoglobulin loci, and the chromatin state plays a significant role in the transcriptional activation of the IGH locus.

Recent insights suggest that CSR in vivo depends on the activation of B cells via their BCR, along with secondary signals from CD40 and TLR signaling. The C terminus of AID may be required for interaction with a protein essential for CSR but not for SHM. The process of CSR involves changes in the chromatin state and the transcriptional activation of the IGH locus at the upstream and downstream switch (S) regions, enabling S-S DNA recombination.

Somatic hypermutation (SHM) is a diversity-generating mechanism that occurs primarily in the germinal centers of secondary lymphoid organs. SHM is tightly regulated to minimize off-target mutations and genomic instability. The key enzymes involved in CSR and SHM include recombination activating genes 1 and 2 (RAG), terminal deoxynucleotidyl transferase (TdT), and Artemis nuclease, a member of the NHEJ pathway. Unlike germline mutation, SHM affects only an organism's individual immune cells and is not transmitted to offspring.

The cancer immunoediting concept, which consists of elimination, equilibrium, and escape phases, provides a framework for understanding how the immune system can control tumor growth. In the equilibrium phase, the immune system can maintain a balance with the tumor, leading to a state of dormancy. Off-target SHM by AID can drive oncogenic transformation in B cell malignancies, particularly through mutations in proto-oncogenes. Research continues to explore the intricate balance of SHM, aiming to harness its benefits for immunity while mitigating its potential role in

disease.

Chapter: Receptor | T Cell Receptors

T-cell receptor (TCR) directed antibodies, such as Muromonab-CD3, are used to prevent T-cell activation and proliferation by binding to the T-cell receptor complex. Stimulation of the TCR can be achieved through various methods, including plate-bound anti-CD3 antibodies, anti-CD3/anti-CD28 antibodies bound to beads, or anti-CD3 antibody-coated antigen-presenting cells. Polymorphic residues on the top alpha helixes are crucial for the specificity of TCRs for antigen peptides and MHC molecules, a phenomenon known as MHC restriction. The activation of naive T cells is initiated upon engagement of the TCR by a specific antigen presented on antigen-presenting cells, mostly dendritic cells. The cytoplasmic domain of the T cell receptor zeta chain is sufficient to couple to receptor-associated signal transduction pathways. TCR contact residue hydrophobicity is a hallmark of immunogenic CD8⁺ T cell epitopes. The T cell receptor is composed of two separate peptide chains, TCR α and TCR β , produced from independent genes. The rearrangement of the alpha (α) chain follows β chain rearrangement, resembling V-to-J rearrangement described for Ig light chains. TCRs recognize antigens presented by MHC class I and trigger cytotoxic activity in Tc cells. The first signal for T cell activation is provided by binding of the T cell receptor to its cognate peptide presented on MHCII on an APC. TCR ligation strength and self-peptide engagement influence T cell activation. Concomitant with TCR cross-linking, PD-1 binds to PD-L2 and induces inhibitory signals by recruiting phosphatases to the ITSM in the cytoplasmic tail of PD-1, resulting in the dephosphorylation of effector molecules involved in downstream TCR signaling. TCR repertoire analysis reveals that tumor-recognizing T cells are limited to merely 10% of intratumoral CD8⁺ T cells. TCR activation triggers CD8⁺ T cell proliferation and mounts a specific cellular immune response. TCR-based platforms extend immune engagement to intracellular antigens, exemplified by the PRAME/HLA-A*02:01-directed ImmTAC IMC-P115C. TCR transgenic cells can adjust the OT-I CD8⁺ T cell activation threshold, causing metabolic reprogramming of cells with a marked increase in both glycolysis and OXPHOS, which promotes autocrine IL-4 production. TCR activation directly stimulates PYGB-dependent glycogenolysis to fuel the early recall response in CD8⁺ memory T cells. T cell activation requires three integrated signals: TCR recognition of peptide-MHC complexes, costimulatory receptor-ligand interactions, and cytokine signaling, which guides lineage fate.

Chapter: Receptor | Immune Checkpoint Receptors

CTLA-4 and PD-1 are key immune checkpoint receptors that play crucial roles in regulating T-cell activation and function. CTLA-4, a negative regulator of T-cell activation, is upregulated on all CD4⁺ and CD8⁺ T cells 2–3 days following activation. CTLA-4 exerts its anti-tumor effects by increasing the activity of effector CD4⁺ T cells and inhibiting Tregs-dependent immune suppression. CTLA-4 produces a strong inhibitory signal to terminate the proliferation and activation of T cells, which can be blocked by CTLA-4 inhibitors, thus restoring the activation of T cells. On the other hand, PD-1 is an upregulated receptor on the surface of T lymphocytes after activation and binds to PD-L1, leading to switching off or apoptosis. Anti-CTLA-4 and anti-PD-1 antibodies are called immune checkpoint inhibitors, which have shown significant efficacy in cancer immunotherapy.

Combining CTLA-4 and PD-1/PD-L1 blockade enhances antitumor immunity, as evidenced by

clinical trials showing superior antitumor activity in some types of cancers. The combination of CTLA-4 and PD-1 blockers for cancer treatment has been widely studied and has demonstrated synergistic effects. For instance, the combination of nivolumab (anti-PD-1) and ipilimumab (anti-CTLA-4) has shown improved outcomes in melanoma patients. However, the combination therapy is also associated with a higher risk of immune-related adverse events (irAEs) such as colitis, hypophysitis, pneumonitis, and thyroiditis.

Several additional markers have been described, such as high levels of CTLA-4 and GITR (glucocorticoid-induced TNF receptor) on regulatory T cells, though the functional significance of this expression remains to be defined. CTLA-4 and PD-1 receptors inhibit T-cell activation by distinct mechanisms. CTLA-4 could competitively suppress the binding of CD28 to CD80/CD86, halting the secondary signal of T cell activation. In contrast, PD-1 binding to PD-L1 tells the T cell to leave the other cell alone, leading to functional exhaustion and apoptosis.

The PD-1 pathway is involved in both tolerance and autoimmunity. In cancer, the PD-1 pathway is dysregulated, leading to immune evasion. Blocking PD-1/PD-L1 has shown promise in treating various cancers, including melanoma, renal cell cancer, non-squamous lung cancer, lymphoma, and gastrointestinal cancers. Anti-PD-1 Abs (nivolumab, pembrolizumab, and cemiplimab) and anti-PD-L1 antibodies (atezolizumab, avelumab, and durvalumab) have been approved by the FDA for some solid tumor and hematologic cancers.

While the combination of PD-1/PD-L1 and CTLA-4 blockade has shown significant benefits, it is also associated with a low response rate in certain cancers, lack of known biomarkers, immune-related toxicity, and innate and acquired drug resistance. Further clinical investigations are needed to improve the safety and strengthen the efficacy of dual PD-1/PD-L1 and CTLA-4 blockade. Additionally, combining LAG-3 blockade with PD-1/PD-L1 and CTLA-4 inhibition has shown promise in enhancing antitumor responses.

Chapter: Receptor | Chimeric Antigen Receptors

Chimeric antigen receptor (CAR) design is a sophisticated approach in cancer immunotherapy that involves the genetic modification of T cells to express a synthetic receptor. The basic principles of CAR design include the creation of a receptor composed of four distinct regions: an antigen recognition domain, an extracellular hinge region, a transmembrane domain, and an intracellular T cell signaling domain. This design allows the engineered T cells to specifically recognize and bind to cancer cells, thereby initiating an immune response against the tumor.

To enhance the functionality and efficacy of CAR-T cells, second-generation CARs were developed. These advanced CARs incorporate an additional costimulatory domain, such as CD28 or 4-1BB, positioned between the antigen-binding domain and the CD3 ζ signaling domain. This modification enables the CAR-T cells to receive both signal 1 (via CD3 ζ) and signal 2 (costimulation) for robust and sustained T cell responses. The inclusion of these costimulatory domains has been shown to significantly improve the persistence and effectiveness of CAR-T cell therapies.

Scientists have successfully harvested T cells from individuals, genetically modified them to express a chimeric antigen receptor (CAR) that specifically targets cancer cells, and then reinfused these modified cells into patients to combat their tumors. For instance, CAR-T cells targeting CD19 have been used to treat B-cell acute lymphoblastic leukemia (B-ALL) and diffuse large B-cell

lymphoma (DLBCL) with remarkable success. Notably, second- and third-generation CARs have been developed to further enhance the efficacy of these therapies by incorporating additional co-stimulatory signaling domains.

Several clinical trials have demonstrated the potential of CAR-T cell therapies. For example, the use of second-generation CAR-T cells targeting CD19 in heavily pre-treated patients at the National Cancer Institute, University of Pennsylvania, and Memorial Sloan Kettering Cancer Center resulted in complete remissions. Additionally, tandem CARs targeting both CD19 and CD22 have shown improved efficacy in preclinical models of B-ALL, preventing tumor escape through single-antigen downregulation and achieving sustained tumor clearance.

While CAR-T cell therapies have shown promise, challenges remain. Early clinical trials using first-generation CARs targeting solid tumor antigens such as MUC1 did not achieve long-term persistence of the transferred T cells or result in significant remissions. However, the development of second- and third-generation CARs has addressed some of these limitations, leading to more effective and durable immune responses against cancer.

Chapter: Receptor | Co-stimulatory Receptors

CD28-B7 interactions play a crucial role in T-cell activation, where the binding of CD28 on T cells to B7 molecules on antigen-presenting cells (APCs) enhances T-cell activation and proliferation. This interaction is pivotal for the development of regulatory T (T reg) cells, as T reg cells require CD28 co-stimulation, and B7.2 expression is largely restricted to the medulla, paralleling the development of Foxp3⁺ cells. The CD28 signaling pathway has been shown to regulate glucose metabolism, with glucose uptake being a limiting factor in T-cell activation and requiring both Akt-dependent and independent pathways. For instance, CD28-mediated signaling enhances the therapeutic performance of T-cell receptor (TCR) chimeric antigen receptor (CAR) T cells through FYN kinase signaling rather than LCK.

Costimulatory domains, including those of CD28 and 4-1BB, have been utilized to enhance the function of CAR T cells specific for tumor antigens. Targeting CD28, CTLA-4, and PD-L1 costimulation differentially controls immune synapses and the function of human regulatory and conventional T-cells. The CD28 superagonist antibody TGN1412, when tested in six healthy human volunteers, led to disastrous complications, including multiorgan failure due to a cytokine storm. This underscores the critical role of CD28 in T-cell activation and the potential risks associated with its manipulation. CD40, a TNF receptor family member expressed on tumor cells and antigen-presenting cells, including macrophages, induces proinflammatory cytokine release and upregulates CD80 and CD86 upon activation, thereby sustaining T-cell responses. CD40 strengthens B cell-T cell crosstalk through its interaction with CD40L on T helper cells, which is essential for the efficient recombination of a switch substrate retrovector in CD40-activated B lymphocytes.

Activated T cells express high levels of CD69 and CD25, which promote T-cell proliferation. The CD28 transmembrane domain is known to result in a highly expressed, stable receptor, and signaling domains from various co-stimulatory molecules, including CD28, CD27, CD134 (OX40), and CD137 (4-1BB), have been successfully tested. The addition of the CD28 signaling domain to chimeric T-cell receptors enhances chimeric T-cell resistance to T regulatory cells. CD40 agonists have shown efficacy against pancreatic carcinoma in both mice and humans, indicating their potential in cancer therapy. The development of CD28 and CD40 agonists and antagonists

continues to be an area of active research, with ongoing clinical trials evaluating their therapeutic potential.

Chapter: Receptor | Chemokine Receptors

Chemokine receptors are a family of G-protein coupled receptors that play crucial roles in cell signaling and immune responses. These receptors are involved in the migration of immune cells to sites of inflammation and infection. The primary chemokine receptors discussed in this section include CXCR3, CCR5, and CCR6, each with specific functions and interactions.

CXCR3: The CXCR3 receptor is known to bind to the chemokine ligands CXCL9, CXCL10, and CXCL11. These ligands are primarily expressed in inflamed tissues and are induced by interferon- γ . The interaction between CXCR3 and its ligands is critical for the recruitment of T-cells to sites of inflammation, where these cells secrete IFN- γ , further upregulating the expression of CXCL9, CXCL10, and CXCL11. This positive feedback loop enhances the recruitment of T-cells and other immune cells to the site of inflammation. Additionally, CXCR3 is involved in the recruitment of plasmacytoid dendritic cells to inflamed epithelia, as demonstrated by the study of Sisirak et al. (2013).

CCR5: CCR5 is a chemokine receptor that binds to the chemokine ligand CCL5 (RANTES). CCR5 antagonists, such as maraviroc, are used to block the virus's ability to engage the CCR5 co-receptor, thereby preventing cell entry. This mechanism is particularly important in the context of HIV infection, where the CCR5 receptor is a primary co-receptor for HIV-1 entry. The CCR5 Δ 32 mutation, which leads to the absence of the CCR5 receptor, has been associated with reduced susceptibility to HIV infection, providing a basis for curative HIV therapy (Allers & Schneider, 2011).

CCR6: CCR6 is a chemokine receptor that binds to CCL20. It is primarily expressed on memory B cells and is essential for their ability to mount a recall response to antigen. The positioning of memory B cells is mediated by CCR6, as shown by Elgueta et al. (2013). Furthermore, CCR6 is involved in the recruitment of plasmacytoid dendritic cells to inflamed epithelia, as demonstrated by the study of Sisirak et al. (2013). CCR6 also plays a role in the localization of dendritic cells and the maintenance of immune responses in mucosal tissues (Cook et al., 2013).

Additional Chemokine Receptors: Approximately 19 different chemokine receptors have been characterized to date, which are divided into four families depending on the type of chemokine they bind. CXCR receptors bind CXC chemokines, CCR receptors bind CC chemokines, CX3CR1 binds the sole CX3C chemokine (CX3CL1), and XCR1 binds the two XC chemokines (XCL1 and XCL2). Some of these receptors, such as CCR1, CCR2, CCR4, and CCR5, have been associated with tumor immunosuppression, as seen in the context of breast cancer (TNBC) where monocytes recruited via the CAF-driven CXCL12-CXCR4 axis acquire pro-tumor characteristics (Inoue et al., 2015).

In summary, chemokine receptors, particularly CXCR3, CCR5, and CCR6, play essential roles in immune responses and inflammation. Their interactions with specific chemokine ligands are critical for the recruitment and activation of immune cells, and they have significant implications in various diseases, including HIV infection and cancer.

Chapter: Receptor | Integrins and Adhesion Molecules

Since laminin carryovers are also bound via integrins, modulation of outside-in signaling might represent a common mechanism of how ECM carryovers modify leukocyte functions. This initial tethering event is followed by decelerative rolling on endothelium and gradual activation of integrins on leukocytes. The primary integrins involved in this process are the $\beta 2$ subfamily, which pair with the common α subunit (CD18).

Activated LFA-1 now binds more strongly to its Ig-like ligand, intracellular adhesion molecule 1 (ICAM-1), on the surface of the presenting cell. LFA-1 is found on circulating leukocytes and binds ICAM-1 and ICAM-2 on endothelial cells, while Mac-1 is also found on circulating leukocytes and binds ICAM-1 on endothelial cells. VLA-4, another integrin, is present on both leukocytes and endothelial cells and facilitates chemotaxis by binding to vascular cell adhesion molecule-1 (VCAM-1).

Cellular activation via extracellular chemokines causes pre-formed $\beta 2$ integrins to be released from cellular stores, and this is assisted through juxtacrine activation of integrins by chemokines and soluble factors released by endothelial cells. This results in the activation of leukocyte integrins, which bind to members of the immunoglobulin superfamily (Ig-SF) that mediate firm adhesion. In the activated state, integrins bind tightly to complementary receptors expressed on endothelial cells, with high affinity. Increased ECM stiffness, often reaching 4–10 kPa in tumors compared to <1 kPa in normal tissues, activates mechanotransducers such as integrins, focal adhesion kinase (FAK), and YAP/TAZ, promoting proliferation through Rho/ROCK-ERK pathways and cytoskeletal remodeling in hepatocellular and lung cancers.

Chapter: Receptor | B Cell Receptors

B cell receptors (BCRs) play a crucial role in the immune response by recognizing and binding to antigens. Biram A et al. (2024) have shown that BCR affinity differentially regulates the colonization of the subepithelial dome and the infiltration into germinal centers within Peyer's patches, highlighting the importance of BCR strength in B cell localization and function. Casola S et al. (2024) have demonstrated that the strength of BCR signaling determines the fate of B cells, influencing their differentiation and activation.

While some in vitro studies suggest that BCR signaling in germinal center (GC) B cells is attenuated compared to naïve B cells (Khalil et al., 2012; Kwak et al., 2018; Nowosad et al., 2016), receptor recoupling occurs such that the BCR strengthens phosphatidylinositol 3-kinase (PI3K) and weakens nuclear factor kappa-B (NF κ B) signaling, whereas CD40 signals via NF κ B but not PI3K (Luo et al., 2018; Nowosad et al., 2016). This differential signaling is crucial for the activation and differentiation of B cells.

BCR signaling is involved in both tolerance to self-derived antigens and activation in response to foreign antigens. It is well established that BCR-triggering events regulate these processes, ensuring that the immune system can respond appropriately to pathogens while avoiding autoimmunity. The variable region of the BCR, composed of heavy and light chains, is essential for antigen recognition and binding.

B cells produce antibodies or immunoglobulin molecules, which are critical for neutralizing pathogens and marking them for destruction. The B cell receptor (CD22), CD19, CD25, CD30,

CD21, CD81, and other surface markers are involved in B cell activation, differentiation, and development. Plasma cell markers such as CD138, IgG, CD27, CD38, and CD319 are also important for understanding B cell function and differentiation.

Negative selection of B cells occurs through the binding of self-antigen with the BCR. If the BCR can bind strongly to self-antigen, the B cell may undergo one of four fates: clonal deletion, receptor editing, anergy, or ignorance (B cell ignores the signal and continues development). Once the antigen is bound to the BCR, it forms an antigen-BCR complex, which initiates the internalization of the antigen into the B cell.

The BCR is closely related to other immune receptors such as the T cell receptor (TCR), and it interacts with molecules like Toll-like receptors (TLRs), major histocompatibility complex-II (MHC-II), B-cell lymphoma (BCL), inducible co-stimulator (ICOS), and its ligand (ICOSL). Additionally, B cells produce cytokines such as interleukin (IL) and B-cell activating factor (BAFF), which are involved in T–B cell interactions.

Understanding BCR signaling is crucial for developing therapeutic strategies to target pathological B cell receptor signaling in lymphoid malignancies. Further research in this area is essential for advancing our knowledge of B cell biology and improving treatments for various immune-related diseases.

Chapter: Receptor | Toll-Like Receptors

Toll-like receptors (TLRs) are pattern-recognition receptors (PRRs) that recognize conserved microbial products and activate the innate immune system. These receptors play a crucial role in the recognition of pathogen-associated molecular patterns (PAMPs) and are essential for the initiation of immune responses. TLRs are expressed on various cell types, including antigen-presenting cells (APCs) such as dendritic cells (DCs), macrophages, and B cells, as well as on non-immune cells like platelets and fibroblasts. TLRs function as transmembrane proteins characterized by leucine-rich-repeat (LRR) motifs in their extracellular domains and a Toll/interleukin-1 (IL-1) receptor (TIR) homology domain in their cytoplasmic regions. Upon ligand engagement, TLRs signal through several adaptor molecules, including myeloid differentiation primary response gene 88 (MyD88) and TIR-domain containing adaptor protein-inducing IFN β (TRIF), leading to the recruitment and activation of members of the tumor necrosis factor receptor-associated factor (TRAF) family of proteins and the TRAF family member-associated NF- κ B activator (TANK).

TLR signaling pathways are regulated by intracellular adaptor molecules, such as MyD88, TIRAP/Mal, and TRAM/TRIF, which provide specificity to individual TLR-mediated signaling pathways. TLR activation in APCs, especially in DCs, produces cytokines and upregulates costimulatory molecules essential for inducing T-cell responses. TLR signaling has been shown to promote both carcinogenesis and antitumorigenic activity. On one hand, TLR signaling can induce proinflammatory, antiapoptotic, proliferative, and profibrogenic signals within the tumor microenvironment or tumor cells themselves. On the other hand, TLR signaling can activate specific immune cells like DCs, natural killer cells (NKs), and cytotoxic T lymphocytes (CTLs), which have antitumorigenic activity. TLR4, a key TLR model linked to the MD-2 receptor, acts as the LPS (lipopolysaccharide) receptor, which is an outer wall component of gram-negative bacteria. TLR4 signaling activates transcription factors such as NF- κ B, MAPK, and IFN regulatory factors through MyD88 or TIR domain-containing pathways, promoting the synthesis of proinflammatory

cytokines (e.g., IL-6, IL-1 β , and TNF- α) and type I IFNs (IFN- α / β).

TLR4 is known to be expressed on immune cells but its expression on non-immune cells is not ubiquitous. TLR4 helps the body sense specific parts of bacteria called lipopolysaccharide (LPS). TLR5 recognizes the flagellin of bacteria, while TLR2 forms heterodimers with TLR1 or TLR6, each dimer having a different ligand specificity. TLRs recognize specific chemical signatures found on subsets of pathogens, and their activation can lead to the release of inflammatory cytokines and the polarization of macrophages to an M1 phenotype. TLR-mediated activation and maturation of DCs and macrophages are critical links between innate and adaptive immunity. TLR signaling pathways are complex and can be regulated by various factors, including endosomal TLR signaling, which is required for anti-nucleic acid and rheumatoid factor autoantibodies in lupus.

TLR agonists have been used in cancer vaccines and immunotherapy, and their use in mRNA vaccines has been supported by the detection of mRNA-LNPs by TLRs, MDA5, and NLRP3, leading to the secretion of IL-1 β , IFN- γ , and IL-6 through the innate immunity pathway. TLR7 and TLR9 recognize viral nucleic acids and stimulate IFN-1 expression, contributing to systemic lupus erythematosus. TLR7 produces proinflammatory cytokines, and TLR9 promotes the synthesis of IFN α and IFN β in plasmacytoid dendritic cells (pDCs). TLRs also play a role in the recognition of commensal microflora, which is required for intestinal homeostasis. The recognition of PAMPs by TLRs initiates the innate immune response, which then stimulates the adaptive immune response, with a significant effect on the differentiation, memory formation, and antibody formation of CD4⁺ T cells and CD8⁺ T cells.

Chapter: Receptor | Pattern Recognition Receptors

Pattern recognition receptors (PRRs) are a key component of the innate immune system, enabling the recognition of pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs). These receptors, including toll-like receptors (TLRs), Nod-like receptors (NLRs), and retinoic acid-inducible gene-like receptors (RLRs), initiate the innate immune response and drive inflammatory signaling cascades. For instance, TLRs and NLRs recognize various PAMPs and DAMPs, respectively, leading to the upregulation of proinflammatory cytokines and type I interferons. The expression of Nod2, a member of the NLR family, is dependent on the presence of commensal bacteria, highlighting the importance of the gut microbiota in modulating immune responses.

Membrane-bound and cytosolic PRRs, such as TLRs and NLRs, sense PAMPs and DAMPs, triggering proinflammatory pathways like the NF- κ B and MAPK pathways. These pathways facilitate the activation of transcription factors such as NF- κ B, AP-1, CREB, and NFAT, which induce the expression of proinflammatory genes. Natural ligands of PRRs, such as endo- and exotoxins, DNA, and lipids, activate these receptors, leading to the secretion of cytokines and chemokines. For example, the activation of TLRs and NLRs by their respective ligands can induce the production of key proinflammatory cytokines such as IL-6, IL-1 β , and TNF- α .

The recognition of PAMPs and DAMPs by PRRs triggers the formation of the inflammasome, a multiprotein complex that includes ASC and Caspase-1, which cleaves pro-IL-1 β and pro-IL-18. This process is crucial for the initiation of an inflammatory response. Additionally, PRRs such as TLRs and NLRs can also induce the production of type I interferons, which can affect adaptive immunity. The activation of PRRs by inflammatory factors initiates a signaling cascade, leading to

the recruitment of leukocytes.

Adjuvants, such as liposomal TLR agonists and NLR agonists, are used to boost the immunogenicity of vaccines by triggering PRRs. For instance, mifamurtide, a liposomal-based innate immune agonist, targets both TLR4 and NOD2, synergizing NF- κ B activation and pro-inflammatory tumoricidal activities. The recognition of PAMPs and DAMPs by NLRs, such as NLRP3, leads to the assembly and activation of the inflammasome, initiating an inflammatory response. The crosstalk between TLRs and NLRs is regulated by pathogen recognition receptors, especially Nod2, which modulates the number, size, and T-cell composition of Peyer's patches in response to the gut flora.

In summary, PRRs play a crucial role in the recognition and response to both exogenous and endogenous signals, initiating a cascade of events that lead to the activation of the immune system. The activation of PRRs by various ligands triggers the production of proinflammatory cytokines and type I interferons, which are essential for the defense against pathogens and the regulation of immune responses.

Chapter: Receptor | Cytokine Receptors

Cytokine receptors play a crucial role in cellular signaling pathways, linking to the STAT, Ras–MAPK, and phosphatidylinositol-3'-kinase (PI3K)–AKT pathways. These pathways converge at the nucleus, where they regulate gene expression. For instance, the IL-2 receptor, a key component in this chain, includes a γ -chain that is common to several other cytokines. Deficiency in this γ -chain is directly responsible for the x-linked form of Severe Combined Immunodeficiency (X-SCID).

Cytokine receptors typically function by interacting with specific cytokines on the surface of target cells. The IL-2 receptor, for example, consists of the α , β , and γ chains and is essential for the clone expansion and survival of activated T lymphocytes. The IL-2 receptor pathway is particularly significant, as it is mediated by the trimeric cell surface receptor IL-2a. However, this does not take into account autocrine signaling through cytokine receptors, where the release of these proteins leads to signaling within the same cell.

JAKs (Janus kinases) phosphorylate cytokine receptors, which can then bind to a protein called Grb2. Phosphorylated cytokine receptors can also be bound by PI3K, allowing for the activation of AKT. A classification of cytokine receptors based on their three-dimensional structure has been attempted, highlighting the diverse mechanisms by which these receptors can be activated and signal within the cell. Especially significant are the IL-2 receptor (CD25) and CD3 directed antibodies, which play critical roles in immune regulation and disease.

Chapter: Receptor | Death Receptors

Death receptors, including Fas (also known as DR2, APO-1, or CD95), tumor necrosis factor receptor 1 (TNFR1), and tumor necrosis factor-associated apoptosis-inducing ligand receptor (TRAILR), are members of the Tumor Necrosis Factor Receptor Superfamily (TNFRSF). By binding to TNFR1 and TNFR2, tumor necrosis factor (TNF) can induce either cell survival or cell death in a target cell, depending on the context and the formation of signaling complexes. When TNF binds to its receptors, a pathway of signals is triggered within the target cell, resulting in an inflammatory response. The extrinsic pathway, mediated mainly by death receptors and their

ligands, leads to the formation of the death-inducing signaling complex (DISC), which activates pro-caspase-8. TNFR1 contains a death domain in its cytoplasmic tail, enabling it to trigger cell death, whereas TNFR2, which lacks a death domain, is incapable of directly inducing cell death. TNFR1 exists in most cell types and binds to both transmembrane TNF (tmTNF) and soluble TNF (sTNF), whereas TNFR2 is expressed in limited cell types, including endothelial cells, fibroblasts, and subsets of neurons and immune cells. TNFR1 activation can trigger cell survival through complex I formation or cell death through complex II formation, while TNFR2 activation most often leads to cell survival. TNFR2 can also indirectly cause cell death by disrupting the cell death checkpoints of TNFR1. Through TNFR1 signaling, TNF can increase the surface expression of AMPA receptors and NMDA receptors in neurons, strengthening synaptic transmission. TNFR2 signaling promotes the proliferation and maturation of oligodendrocytes, which produce protective myelin sheaths around nerve cells, due to its homeostatic role in the central nervous system.

Chapter: Receptor | Fc Receptors

Inhibitory Fc receptors modulate in vivo cytotoxicity against tumor targets, allowing other Fc receptors to be engaged and increasing the signaling for phagocytosis. Many immune cells, including macrophages, neutrophils, and natural killer cells, contain Fc receptors on their surface that enable them to bind to antibodies attached to pathogens or infected cells, stimulating their phagocytosis or destruction. Receptors play a crucial role in phagocytosis, with complement receptors and Fc receptors being particularly important in this process.

Signal transduction in Fc receptor-mediated phagocytosis has been extensively studied, with early research indicating that Fc receptors can induce inflammatory responses and share characteristics with mast cells, such as the expression of the high-affinity Fc receptor for IgE (FcεRI) and Toll-like receptors (TLRs) like TLR2 and TLR4. The absence of an Fc region prevents engagement of Fc receptor-bearing cells, thereby limiting nonspecific immune activation by fragment-based bispecific TCEs. The neonatal Fc receptor (FcRn) regulates mucosal immune responses to luminal bacteria and mediates the transport of IgG into luminal secretions for delivery of antigens to mucosal dendritic cells. This response depends on the affinity of the Fc receptor expressed on NK cells, which can have high, intermediate, and low affinity for the Fc portion of the antibody. Fc receptors (FcR) and complement receptors (CR) are the best-characterized opsonic receptors in the immune system.

Chapter: Receptor | Natural Killer Cell Receptors

Natural Killer (NK) cell receptors play a crucial role in the immune system, particularly in the recognition and elimination of infected or abnormal cells. There are two main classes of NK cell receptors in mammals: the killer cell immunoglobulin-like receptors (KIR) and the structurally unrelated killer cell lectin-like receptors (KLR). KIRs are encoded in the leukocyte receptor complex (LRC), while KLRs are clustered in the natural killer complex (NKC).

KIRs are characterized by their ability to inhibit NK cell activation through the presence of immunoreceptor tyrosine-based inhibitory motifs (ITIM) in their cytoplasmic tails, which can decrease the state of activation. In contrast, NKG2D, a second important activating receptor, is unique in that it associates with the cytoplasmic adaptor protein DAP10 to initiate distinct signaling cascades. The engagement of NKG2A by HLA-E or Qa-1b inhibits the activity of NK cells,

preventing target cell lysis. In higher primates, both NKG2A and MHC-E are very well conserved, presenting a system for detection of “gross” MHC-I expression, which is highly conserved.

The evolution of NK cell receptors is complex and involves the expansion and diversification of gene families. For instance, in humans, the KIR gene complex is organized into four phylogenetic lineages (I, II, III, and V), each with specific binding affinities for MHC-I molecules. Lineage I KIRs bind HLA-G-peptide complexes, lineage II KIRs recognize epitopes A3/A11 on HLA-A and Bw4 on HLA-A and B, and lineage III KIRs bind to HLA-C epitopes C1 and C2. Single amino acid substitutions in key residues can significantly affect KIR binding, highlighting the importance of precise specificity.

In contrast to humans, rodents have expanded their Ly49 gene family, resulting in remarkable diversity across different inbred mouse strains. Lemurs, on the other hand, have only one non-functional copy of KIR, and mice do not encode any KIR in the LRC. The KIR gene complex in humans has undergone specific expansion in the telomeric part, resulting in a diverse repertoire of receptors. This expansion is not observed in other species, such as rodents, which have expanded their Ly49 genes instead.

The haplotypes encoding KIRs exhibit great differences in gene content and allelic polymorphism, with up to 17 genes encoded over approximately 150 kilo bases. This polymorphism and polygenicity are thought to improve hosts’ survival by allowing for a more diverse and adaptable immune response. However, the exact evolutionary selection pressure that led to the development of polygenic and polymorphic NK cell receptors remains unresolved.

NK cell receptors are crucial for the recognition of viral products and the detection of abnormal cells. The ligands for activating NK cell receptors, such as NKG2D, are upregulated in infected, stressed, and tumor cells, indicating their importance in immune defense. The expansion of NK cell receptors in different species, such as humans and cattle, suggests that these receptors play a significant role in host defense mechanisms. Further experimental elucidation of the KIR motifs in the MHC presented peptides is necessary to validate these predictions and understand the full extent of their functional significance.

Chapter: Receptor | Other Receptors

Receptor | Other Receptors

Blockade of EGFR improves responsiveness to PD-1 blockade in EGFR-mutated non-small cell lung cancer, highlighting the importance of receptor signaling in cancer immunotherapy (Sugiyama et al., 2018). FLT3 overexpression in acute leukemias provides new insights into molecular mechanisms, underscoring the role of receptor tyrosine kinases in hematological malignancies (Poubel et al., 2018). GPVI^{-/-} mice exhibit inflammatory bleeding, indicating the necessity of GPVI/Syk/Btk signaling for platelet function (Gros et al., 2015). Simultaneous blockade of GPVI and CLEC-2 results in more severe bleeding, suggesting a synergistic role in platelet function (Currie et al., 2022). Transfusion of wild-type platelets rescues bleeding in GPVI^{-/-} mice, confirming the requirement of GPVI/Syk/Btk signaling (Gros et al., 2015).

Tie2 signaling is crucial for endothelial junctional strengthening, involving actin filament remodeling and pericyte recruitment (Coffelt et al., 2016; Teichert et al., 2017). The Tie2 mechanism can be strengthened by inhibiting VE-PTP activity, which phosphorylates Tie2 and

downstream FGD5 (Frye et al., 2015; Braun et al., 2019). This mechanism is essential for preventing diapedesis-mediated bleeding, even when platelet plugging is present (Braun et al., 2020). CD19/CD22 and other dual CARs target multiple tumor antigens to enhance specificity and reduce off-target toxicity (Table 5). TandAb AFM11 efficiently recruits T cells for potent lysis of CD19+ tumor cells, demonstrating the effectiveness of bispecific antibodies (Reusch et al., 2018). CD33, a member of sialic acid-binding sialoadhesin receptors, is selectively expressed on AML cells, making it a promising target for therapy (Barnwell et al., 2005; Franke-Fayard et al., 2006).

CD36 functions as a receptor for *Plasmodium falciparum*-infected erythrocytes, playing a critical role in malaria pathogenesis (Barnwell et al., 2005; Franke-Fayard et al., 2006). CD94 is essential for NK cell-mediated resistance to viral diseases, highlighting its role in innate immunity (CD94, 2016). CSF-1R inhibition alters macrophage polarization and blocks glioma progression, indicating its importance in tumor microenvironment modulation (Pyonteck et al., 2015; Wen et al., 2018). CSF1R inhibitors, such as pexidartinib, enhance CD8+ T cell infiltration and improve the efficacy of anti-PD-1 immunotherapy (Lv et al., 2018; Quail et al., 2018). Dectin-1, a C-type lectin receptor, recognizes yeast polysaccharides and plays a role in antifungal immunity (Herre et al., 2003). TREM2 deficiency downregulates ATP-binding cassette transporter A1, leading to cholesterol accumulation and a pro-inflammatory macrophage phenotype (Molgora et al., 2015; Zhang et al., 2018).

IFNAR signaling in T cells promotes immunity when administered intravenously but inhibits it when administered subcutaneously, highlighting the importance of receptor context in immune responses (Brown et al., 2003). STAT-mediated EGFR signaling in cancer contributes to immune evasion, suggesting potential therapeutic targets (STAT-mediated EGFR, 2018). CD36 promotes macrophage infiltration via p110 γ signaling, enhancing metastasis (Qin et al., 2018). The NKC encodes genes from the CD94, NKG2, and Ly49 families, with exchange of subunits influencing binding specificity and functionality (Walter, 2011; Averdam et al., 2009). CSF-1R inhibitors, such as PXB17, reprogram pro-tumoral macrophages and induce CD8+ T cell infiltration, improving the immunosuppressive microenvironment (A novel potent CSF1R inhibitor, 2019). TREM2-mediated cholesterol efflux in macrophages inhibits anti-tumor immunity, linking macrophage function to cancer progression (Wang et al., 2018; Zhang et al., 2018). Targeting

Chapter: Treatment | Immune Checkpoint Inhibitors

Immune checkpoint inhibitors (ICIs) are a class of drugs that target specific molecules on immune cells, thereby enhancing the immune response against cancer. One of the hallmark features of exhausted T cells is the expression of inhibitory checkpoint molecules such as programmed cell death protein 1 (PD-1) and cytotoxic T lymphocyte antigen 4 (CTLA-4). By blocking these molecules, ICIs reinvigorate exhausted T cells that recognize tumor antigens, leading to durable responses in patients with melanoma and non-small cell lung cancer (NSCLC). For instance, anti-PD-1 antibodies, such as pembrolizumab (Keytruda) and nivolumab (Opdivo), have shown overall survival improvements exceeding 20% in responsive patients. Similarly, anti-CTLA-4 antibodies, such as ipilimumab (Yervoy), have also demonstrated significant efficacy, particularly when combined with anti-PD-1 antibodies.

Several immune checkpoint inhibitors have been approved by the FDA for the treatment of various cancers. Nivolumab, for example, is approved for melanoma, lung cancer, kidney cancer, bladder cancer, head and neck cancer, and Hodgkin's lymphoma. Atezolizumab (Tecentriq) is approved for

urothelial cancer, while durvalumab (Imfinzi) is approved for unresectable, locally advanced, or metastatic triple-negative breast cancer. The clinical combination of ipilimumab with nivolumab has shown superior outcomes compared to nivolumab alone in the treatment of metastatic melanoma. Additionally, pembrolizumab combined with chemotherapy has been approved for the treatment of advanced non-small cell lung cancer (NSCLC) and advanced triple-negative breast cancer.

Immune checkpoint inhibitors have also been associated with various immune-related adverse events (irAEs). These include endocrine effects such as hypophysitis, gastrointestinal toxicity, and skin reactions. The mechanism of antitumor activity of ipilimumab is still not fully understood, and it is often used in combination with other therapies to enhance its efficacy. For instance, combining ipilimumab with nivolumab has shown improved outcomes in patients with metastatic melanoma. Similarly, pembrolizumab combined with chemotherapy has been approved for the treatment of advanced NSCLC, regardless of PD-L1 levels.

Beyond the primary checkpoint molecules PD-1 and CTLA-4, other immune checkpoint inhibitors such as anti-LAG-3, anti-TIM-3, and anti-TIGIT are also being explored. These inhibitors target additional pathways that can suppress the immune response. For example, the combination of tiragolumab (α -TIGIT) plus atezolizumab has shown improved objective response rates and progression-free survival in PD-L1 positive NSCLC. The use of immune checkpoint inhibitors is also being explored in combination with other therapies, such as chemotherapy, radiation, and targeted therapies, to further enhance their antitumor activity.

In conclusion, immune checkpoint inhibitors have emerged as a frontline treatment for multiple cancers, including metastatic melanoma, NSCLC, renal cell carcinoma, and bladder cancer. The clinical success of these drugs has led to their widespread adoption in oncology. However, the development of resistance mechanisms to ICIs remains a challenge, and ongoing research is focused on understanding these mechanisms and developing strategies to overcome them.

Chapter: Treatment | Immunotherapy

Immunotherapy for cancer treatment has emerged as a promising approach, leveraging the body's immune system to combat malignancies. This method includes various strategies such as immune checkpoint inhibitors, adoptive cell therapies, and cancer vaccines. Immune checkpoint inhibitors, such as those targeting PD-1 and PD-L1, have shown significant efficacy in treating a variety of cancers, including non-small-cell lung cancer and melanoma. These inhibitors block the interaction between PD-L1 on tumor cells and PD-1 on T-cells, thereby enhancing T-cell function and allowing the immune system to attack cancer cells more effectively.

The gut microbiota plays a crucial role in the efficacy of immunotherapy, particularly in the context of PD-1/PD-L1 blockade therapy. Manipulating the gut microbiome could potentially enhance the response to these therapies, highlighting the importance of considering the microbiome in treatment strategies. Additionally, the cGAS-STING pathway, which bridges innate and adaptive immunity, is a promising target for cancer immunotherapy. Extracellular vesicles (EVs) derived from cancer cells can be harnessed to enhance immunotherapy by facilitating PANoptotic cell death pathways, thereby reinvigorating the immune response against tumors.

Several types of immunotherapy are currently in use, including immune checkpoint inhibitors, which block immune system checkpoints to allow immune cells to respond more strongly to cancer.

Other forms of immunotherapy include CAR T-cell therapy, which involves genetically modifying T-cells to target specific antigens on cancer cells, and tumor-infiltrating lymphocyte (TIL) therapy, which enhances the activity of T-cells already present in the tumor microenvironment. The use of nanoparticle-based systems for immunotherapy is also gaining traction, as these can simultaneously target multiple pathways to enhance antitumor responses.

In the context of type 1 diabetes mellitus, immunotherapy has shown potential in delaying the progression of the disease by revitalizing β -cells. This approach could pave the way for other immune-targeted interventions for type 1 diabetes. Furthermore, the recent regulatory approval of immunotherapy targeting T cells for delaying autoimmune destruction of pancreatic β -cells underscores the critical role of the immune system in disease pathogenesis.

Combination therapies, which integrate TAM-focused interventions with safer, multimodal immunotherapies tailored to the tumor-specific immune context, are key to advancing oncology care. These strategies aim to mitigate the risks associated with targeted strategies and maximize therapeutic benefits. The use of artificial antigen-presenting cells (aAPCs) and designer vaccine nanodiscs for personalized cancer immunotherapy represents a promising direction, as these can be tailored to individual patients based on their specific tumor characteristics.

Machine learning algorithms are increasingly being used to predict neoantigens with high accuracy, facilitating the development of personalized cancer vaccines. However, challenges remain in efficiently recruiting patients, optimizing sequencing technology, and developing off-the-shelf therapies against public neoantigens. The use of systemic immunostimulatory agents, such as Toll-like receptor (TLR) agonists, in combination with checkpoint inhibitors, has shown synergistic effects in enhancing antitumor responses.

In conclusion, immunotherapy offers a range of promising strategies for cancer treatment, from immune checkpoint inhibitors to adoptive cell therapies and cancer vaccines. Ongoing research is focused on refining these approaches and developing new technologies to improve their efficacy and safety.

Chapter: Treatment | Bispecific T Cell Engagers

Bispecific T-cell engagers (BiTEs) are engineered antibodies designed to bind simultaneously to a T-cell and a cancer cell, facilitating the direct destruction of the tumor by the immune system. These BiTEs have gained significant momentum in the treatment of hematologic malignancies, particularly in the context of acute lymphoblastic leukemia (ALL) and non-Hodgkin lymphoma (NHL). Blinatumomab, a canonical CD3/CD19 BiTE, was approved by the FDA in 2014 for the treatment of R/R Philadelphia chromosome-negative (Ph-) B-ALL and in 2017 for R/R Philadelphia chromosome-positive (Ph+) B-ALL. Clinical trials have shown that blinatumomab can achieve durable complete responses and long-term survival in patients, especially when administered in first complete remission (CR). However, the short half-life of blinatumomab necessitates continuous intravenous infusion to maintain therapeutic concentrations, which can be challenging in clinical practice.

Several other BiTEs have been developed and are currently in clinical trials. For instance, talquetamab, a GPRC5D-directed BiTE, is being evaluated for multiple myeloma (MM), and mosunetuzumab, the first FDA-approved CD20-directed T cell engager, has demonstrated durable responses in relapsed or refractory follicular lymphoma. Additionally, tarlatamab, a DLL3/CD3

BiTE, has been approved for small cell lung cancer (SCLC) but carries a black box warning for severe CRS and immune cytopenia. The development of next-generation BiTEs aims to enhance therapeutic efficacy while minimizing adverse events, such as cytokine release syndrome (CRS) and immune cytopenia (ICANS). Strategies include steric masking, prodrug formats, split-CD3 architectures, and dual-antigen logic gates to confine activation to tumor cells and enable higher, therapeutically effective dosing.

The pre-existing T cell landscape in patients with MM can influence the response to BiTEs, highlighting the importance of personalized treatment approaches. Clinical trials for BiTEs are advancing rapidly, with a focus on treating blood cancers and expanding research into solid tumors. The flexible structure of BiTEs, such as blinatumomab, allows for efficient binding to T cells and tumor cells, enhancing the tumor lysis effect. Furthermore, the use of step-up dosing schedules can mitigate CRS, with subcutaneous administration generally associated with slower absorption and lower peak cytokine levels compared to intravenous administration. Overall, the maturation of the field from early experimental formats to clinically optimized, disease-specific immunotherapies has been marked by the approval of several BiTEs, including blinatumomab, talquetamab, and mosunetuzumab.

Chapter: Treatment | Chemotherapy and Related Treatments

Antimicrobial Agents and Chemotherapy have been a cornerstone in the treatment of various infections, but their role in cancer treatment is multifaceted. Peng et al. (2020) demonstrated that chemotherapy induces the overexpression of programmed cell death-ligand 1 (PD-L1) via the nuclear factor- κ B pathway, which can foster an immunosuppressive tumor microenvironment in ovarian cancer. This finding highlights the complex interplay between chemotherapy and the tumor microenvironment.

Alizadeh et al. (2021) reported that doxorubicin, a common chemotherapeutic agent, can eliminate myeloid-derived suppressor cells (MDSCs) and enhance the efficacy of adoptive T-cell transfer in breast cancer. This suggests that doxorubicin can modulate the tumor microenvironment to improve immunotherapy outcomes. Rapamycin, an mTOR inhibitor, targets the mechanistic Target of Rapamycin (mTOR) pathway, which is activated by T cell receptor (TCR) stimulation and proliferation-inducing cytokines like IL-2 and IL-7. This mechanism can be harnessed to enhance T-cell function and potentially improve cancer treatment outcomes.

Metformin, a commonly used medication for diabetes, has been shown to promote antitumor immunity by endoplasmic-reticulum-associated degradation of PD-L1, as described by Cha et al. (2020). This finding underscores the potential of repurposing existing drugs to enhance cancer immunotherapy.

For many years, children with relapsed or refractory cancers faced limited treatment options, with chemotherapy being a primary modality. However, the advent of combination therapies, particularly those involving immune checkpoint inhibitors (ICIs) and chemotherapy, has significantly improved outcomes. Combination therapy with ICIs and chemotherapy has been highlighted in the medical field and extensively evaluated in clinical trials, especially for treating advanced non-small cell lung cancer (NSCLC). For instance, the IMpower150 trial demonstrated the efficacy of atezolizumab plus angiogenesis inhibitor and chemotherapy in advanced non-squamous NSCLC.

Chemotherapy, when combined with ICIs, can synergize to achieve both rapid and long-term cancer

control. This is supported by numerous studies, including the double-edged sword of neurotoxicity associated with chemotherapy, which can also enhance immune responses. Chemotherapy can induce the expression of chemokines in tumors, favoring T-cell infiltration and tumor control, as observed in cutaneous melanoma (Hong et al., 2020).

Doxorubicin and paclitaxel, two widely used chemotherapeutic agents, have been shown to deplete MDSCs and reprogram tumor-associated macrophages (TAMs) to an M1-like phenotype, respectively. These mechanisms can enhance antitumor immunity and improve treatment outcomes. For example, doxorubicin can trigger dendritic cells to release IL-1 β , which further activates IL-17 producing $\gamma\delta$ T cells and recruits anti-tumoral IFN- γ expressing CD8 $^{+}$ T cells into the tumor microenvironment (Alizadeh et al., 2021).

Metformin, as a glycolysis inhibitor, can inhibit macrophage activation and cytokine release, thereby limiting the induction of IL-1 β and boosting IL-10 production in LPS-activated macrophages (Hattori et al., 2008). This suggests that metformin can modulate the tumor microenvironment to enhance antitumor immunity.

In addition to these mechanisms, chemotherapy can also induce the processing and release of interleukin-1 β through the activation of the NLRP3 inflammasome, which can further enhance antitumor immunity (Noordam et al., 2016). Paclitaxel, another chemotherapeutic agent, can promote the differentiation of myeloid-derived suppressor cells into dendritic cells, thereby enhancing antitumor immunity (Wanderley et al., 2017).

Combining chemotherapy with PD-1 blockade in NSCLC has shown promising results, with studies demonstrating that chemotherapy can synergize with α -PD-1/PD-L1 to enhance antitumor immunity. This combination therapy has become a standard-of-care option for some cancer patients, with hundreds of ongoing clinical trials exploring its efficacy and safety (Table 2).

In conclusion, chemotherapy, when combined with ICIs, can modulate the tumor microenvironment to enhance antitumor immunity and improve treatment outcomes. These findings highlight the potential of combination therapies in cancer treatment and underscore the importance of understanding the mechanisms by which chemotherapy can enhance immune responses.

Chapter: Treatment | Corticosteroids and Related Therapies

Furthermore, their use is often combined with immunosuppressive agents such as corticosteroids to enhance efficacy in clinical settings. Corticosteroids are used in anaphylaxis primarily to decrease the incidence and severity of delayed or biphasic reactions. Common immunosuppressants include: Corticosteroids: Prednisone and methylprednisolone are often used initially to reduce inflammation and suppress immune activity. Corticosteroids block the release of cytokines by immune cells, and have been historically used as broad-spectrum immunomodulatory agents in patients with severe infections such as sepsis. Glucocorticoids have potent anti-inflammatory and immunosuppressive effects, and have been used to treat the acute phase of rheumatoid arthritis (RA).

In clinical settings, immunosuppression by some drugs, such as steroids, can either be an adverse effect or the intended purpose of the treatment. For instance, in the treatment of severe COVID-19 pneumonia, early anti-inflammatory treatment with high doses of intravenous anakinra with or without glucocorticoids has shown efficacy. In sepsis, the use of corticosteroids is a topic of ongoing research, with some studies suggesting certainties and others highlighting shadows. In

transplantation, corticosteroids are used for induction and maintenance immunosuppression, as well as for acute rejection. The most common corticosteroids used in transplantation are oral prednisolone and intravenous methylprednisolone.

Corticosteroids are also used in the treatment of anaphylaxis, with a Cochrane systematic review supporting their use. In the context of community-acquired pneumonia, hydrocortisone (200 mg/d) is advised for patients with shock requiring treatment with vasopressors. Combination therapy with hydrocortisone and fludrocortisone has shown additional benefit in some trials, but other studies have failed to consistently reproduce these results. The use of corticosteroids in sepsis, acute respiratory distress syndrome, and community-acquired pneumonia is detailed in the 2024 focused update guidelines by Chaudhuri et al.

Steroids are also used in the treatment of various autoimmune diseases, asthma, and urticaria. They help in reducing swelling and can be used in conjunction with other therapies such as plasma exchange (PLEX) to manage symptoms. Before using biologics to treat psoriasis, treatment with topical moisturizers or steroids, or light therapy may provide relief. The anti-inflammatory effects of glucocorticoids influence all types of inflammatory events, no matter their cause, by blocking the production of cytokines and depleting certain immune cells.

Chapter: Treatment | Janus Kinase Inhibitors

Janus kinase (JAK) inhibitors, commonly known as JAK inhibitors, are a class of drugs that inhibit the activities of one or more enzymes of the JAK family. These inhibitors have been approved for the treatment of various conditions, including rheumatoid arthritis, psoriasis, psoriatic arthritis, and ulcerative colitis. The introduction of JAK inhibitors has ushered in a new era in the treatment of these conditions, offering hope to patients who have often struggled with limited therapeutic options. Tofacitinib, the first JAK inhibitor to reach clinical trials, is indicated for rheumatoid arthritis, psoriatic arthritis, and ulcerative colitis. It is a specific inhibitor of JAK3, thereby blocking the activity of IL-2, IL-4, IL-15, and IL-21. Tofacitinib also inhibits JAK1 and JAK2 to a lesser extent, which in turn blocks IFN- γ and IL-6 signaling and consequently T_H1 cell differentiation.

Ruxolitinib, another JAK inhibitor, is a JAK1 and JAK2 inhibitor with efficacy for the treatment of polycythemia vera and myelofibrosis. It has been approved for the treatment of severe coronavirus disease 2019 (COVID-19) and is also used in the treatment of steroid-refractory acute graft-versus-host disease (cGVHD). Ruxolitinib cream has been used for the treatment of cutaneous lichen planus. Tofacitinib and baricitinib are the first orally available JAK inhibitors to be approved for treating rheumatoid arthritis and other autoimmune diseases.

JAK inhibitors have demonstrated significant anti-inflammatory potential across diverse dermatological disorders, including alopecia areata, psoriasis, vitiligo, and atopic dermatitis. They are administered orally and include tofacitinib, ruxolitinib, and baricitinib. These medications are immunomodulators and disease-modifying antirheumatic drugs (DMARDs), and they are a subclass of tyrosine kinase inhibitors. However, the use of JAK inhibitors has been hampered by adverse events, which could be linked to the JAKs that are blocked. The US Food and Drug Administration (FDA) requires a boxed warning for the JAK inhibitors tofacitinib, baricitinib, and upadacitinib to warn about the risks of serious heart-related events, cancer, blood clots, and death.

Topical tofacitinib and ruxolitinib have been used for the treatment of alopecia. JAK inhibitors may cause the following side effects: they weaken the immune system and make the body susceptible to

infections. Patients of all ages treated with JAK inhibitors are at higher risk of Varicella zoster virus (VZV) infection. The drugs under investigation for JAK inhibitors include remdesivir, IFN γ -1b, emapalumab, anakinra, canakinumab, glucocorticoids, and ruxolitinib.

In some cases, medications to treat other conditions may interact with JAK inhibitors, and it is important for doctors to provide proper treatments. JAK inhibitors are being tested in models of multiple myeloma, and they can be classified according to their selectivity for specific JAK isoforms: pan-JAK inhibitors, JAK1 inhibitors, JAK1/2 inhibitors, JAK1/3 inhibitors, and JAK3 inhibitors. The use of JAK inhibitors in the treatment of immune and hematologic disorders has advanced significantly, but more clinical studies are needed to further explore the mechanisms of action of JAK inhibitors in different infectious diseases.

Chapter: Treatment | Biologic and Targeted Therapies

Safety of biologic therapy in rheumatoid arthritis has been well-documented, with a focus on the use of disease-modifying antirheumatic drugs (DMARDs) such as hydroxychloroquine and methotrexate, which aim to slow disease progression. Tocilizumab, a humanized monoclonal antibody against the IL-6 receptor, has been approved by the FDA for moderate-to-severe rheumatoid arthritis in adults who have not responded adequately to DMARDs. Rituximab, a human–mouse chimeric mAb against CD20 receptors, is used to induce apoptosis of B cells, inhibit B cell proliferation, and remove abnormal B cell proliferation, making it effective in treating autoimmune diseases and certain cancers.

TNF inhibitors, such as infliximab, etanercept, adalimumab, certolizumab, and golimumab, are standard treatments for rheumatoid arthritis and inflammatory bowel diseases. However, they are associated with adverse events including injection site reactions, infusion reactions, demyelinating diseases, granulomatous diseases, cytopenias, cutaneous vasculitis, psoriasis, and rarely antineutrophilic cytoplasmic associated vasculitis. The use of TNF blockers has been linked to increased risks of infections such as Varicella-zoster virus, Epstein–Barr virus, and Cytomegalovirus. Despite their efficacy, TNF blockers like etanercept have shown partial efficacy in reducing symptoms, while others like adalimumab and infliximab have been shown to worsen symptoms in some cases.

Biologic response modifiers (BRMs) are a class of medications that target the disease-causing mechanism, often used in treating immune-mediated conditions. They modulate the immune response by interfering with cytokine effects, inhibiting T cell activation costimulation, and depleting or inhibiting B cells. Prominent players in the global biologic response modifiers market include Swedish Orphan Biovitrum, Bristol-Myers Squibb, Janssen Inc., Merck & Co., Biogen, AbbVie Inc., Amgen, Novartis, Roche Holding AG, and Eli Lilly & Co. The market is segmented into various product types, including Interferon, Tyrosine Kinase Inhibitors, Angiogenic Inhibitors, Tumor Vaccines, Interleukins, Colony Stimulating Factors, TNF- α , and Monoclonal Antibodies.

The global biologic response modifiers market is expected to grow at a CAGR of XX% from 2023 to 2030, with North America dominating the market due to advanced healthcare infrastructure and high adoption rates of biologic therapies. However, the high costs of biologics limit their availability to all patients. Adverse effects and contraindications of biologics include bone marrow suppression, hepatotoxicity, cardiopulmonary and renal dysfunction, disseminated intravascular coagulation, and often death. Contraindications for biologics include active infections,

demyelinating diseases, immunodeficiency syndromes, cancer, certain neurologic disorders, pregnancy or breastfeeding, and a history of heart failure or tuberculosis.

Chapter: Treatment | Radiotherapy

Radiotherapy plays a crucial role in cancer treatment by directly damaging tumor cells and inducing immunogenic cell death. This process enhances the antitumor immune response, promoting T cell infiltration into the tumor microenvironment (TME) and expanding the T-cell receptor (TCR) repertoire. Additionally, radiotherapy upregulates the expression of programmed death-ligand 1 (PD-L1) on tumor cells, making them more susceptible to immune checkpoint inhibitors such as α -PD-1/PD-L1. The combination of radiotherapy with α -PD-1/PD-L1 has shown synergistic effects, enhancing the overall antitumor response.

In the context of immune-desert tumors, radiotherapy can overcome low immunogenicity-mediated immune tolerance by inducing immunogenic cell death, increasing the release of cancer antigens, and promoting the function of antigen-presenting cells (APCs). This synergistic effect is further supported by the induction of type I interferon-dependent innate and adaptive immunity, as highlighted in studies by Burnette et al. and others. However, the optimal fractionation, dose, schedule of radiotherapy, and the irradiated tumor volume and regional lymph nodes remain areas of ongoing research.

Integrating radiotherapy with other treatment modalities, such as immunotherapy, can enhance the overall efficacy of cancer treatment. For instance, combining radiotherapy with α -PD-1/PD-L1 inhibitors has shown promising results in clinical trials, as evidenced by the Durvalumab and tremelimumab combination in metastatic non-small-cell lung cancer. Moreover, the use of nanomodulators can further enhance the effectiveness of radiotherapy by promoting the antigen-presenting function of tumor-associated macrophages, as demonstrated by Zhao et al.

While the integration of radiotherapy with immunotherapy offers significant benefits, challenges remain. These include the need for optimal treatment schedules and the identification of the most effective irradiated tumor volumes. Nonetheless, the potential of radiotherapy to enhance the role of neoantigen vaccines and to benefit from other immunotherapeutic strategies makes it a valuable component in the comprehensive treatment of cancer.

Chapter: Treatment | Interferon Therapy

Interferon therapy involves the administration of human leukocyte interferon to treat various conditions. Three vials of human leukocyte interferon are commonly used in this therapy. Interferon beta-1a and interferon beta-1b are specifically utilized to treat and control multiple sclerosis, an autoimmune disorder. Interferon therapy is also employed in the treatment of some cancers, often in combination with chemotherapy and radiation. Additionally, it has been noted that "Secondary prevention of recurrence by interferon therapy after ablation therapy for hepatocellular carcinoma in chronic hepatitis C patients" is a recognized application. Furthermore, the frequency of certain single nucleotide polymorphisms (SNPs) differs significantly by race, which partly explains the observed differences in response to interferon therapy between European-Americans and African-Americans.

Chapter: Treatment | Other Treatments

Baricitinib and Other Treatments

Baricitinib, a selective oral JAK1 and JAK2 inhibitor, has shown promise in various clinical settings. It has been demonstrated to reverse HIV-associated neurocognitive disorders in a SCID mouse model and reservoir seeding in vitro (Gavegnano et al., 2021). Additionally, baricitinib has been found to restrain immune dysregulation in patients with severe COVID-19 (Bronte et al., 2021). The FDA has issued an emergency use authorization for the combination of baricitinib and remdesivir to treat hospitalized patients with COVID-19 (FDA, 2021). Baricitinib is also effective in treating moderate to severe atopic dermatitis (Guttman-Yassky et al., 2021).

JAK inhibitors, such as tofacitinib and ruxolitinib, selectively inhibit different JAK subtypes and effectively reduce the expression of various cytokines, including IL-6, which plays a crucial role in sepsis and severe infections (Smolen et al., 2021). The anti-inflammatory agents aspirin and salicylate inhibit the activity of I(kappa)B kinase (IKK) and prevent the phosphorylation of I κ B α (Hoang et al., 2021). Aspirin and salicylate also inhibit the production of prostaglandins in the body (Vane, 1971).

Immune checkpoint inhibitors (ICIs), adoptive cell therapy (ACT), and cancer vaccines (CVs) are representative techniques within the landscape of immunotherapy. ICIs, such as monoclonal antibodies targeting PD-1, enhance the cytotoxic capacity of T-cells against malignant cells (Vafaei et al., 2022). However, only a limited proportion of patients exhibit a favorable response to ICIs (Vafaei et al., 2022). Research is ongoing to develop new ICIs that target different immune checkpoints, such as LAG-3, TIM-3, and VISTA (Vafaei et al., 2022).

Angiogenesis inhibitors reshape the tumor microenvironment (TME) by promoting T cell infiltration and dendritic cell (DC) maturation, enhancing the differentiation towards M1-like macrophages, decreasing the ratio of regulatory T cells (Tregs) and myeloid-derived suppressor cells (MDSCs), and alleviating hypoxia-induced PD-L1 expression (Xu et al., 2021).

Chemotherapy, when administered at doses below the maximum-tolerated dose, can directly eliminate immune suppressor cells and enhance the functions of effector cells (Xu et al., 2021).

Combining baricitinib with direct-acting antivirals has been suggested to reduce infectivity, viral replication, and inflammation associated with COVID-19 (Vafaei et al., 2022). CDK4/6 inhibitors, such as palbociclib and ribociclib, promote NF- κ B activation, increase T cell chemoattractant and PD-L1 levels, and prevent PD-L1 degradation in cancer cells (Xu et al., 2021). CDK4/6 inhibitors improve the efficacy of α -PD-1 in murine models by enhancing lymphocyte infiltration and tumor-infiltrating lymphocyte (TIL) activities (Xu et al., 2021).

PARP inhibitors (PARPis) activate the cGAS-STING pathway in cancer cells, increasing T cell recruitment and upregulating PD-L1 expression by inactivating GSK3 β signaling, which attenuates antitumor immunity (Xu et al., 2021). The combination of PARPis with α -PD-1/PD-L1 has shown promising results in multiple types of cancers, including metastatic castrate-resistant prostate cancer and metastatic triple-negative breast cancer (Ramalingam et al., 2021; Vinayak et al., 2021).

In conclusion, various treatments such as baricitinib, JAK inhibitors, ICIs, angiogenesis inhibitors, CDK4/6 inhibitors, and PARPis offer promising therapeutic opportunities in different clinical settings.

Chapter: Condition | Sepsis and Septic Shock

Sepsis is a critical condition characterized by a systemic inflammatory response to infection, often leading to multiple organ dysfunction syndrome (MODS) and high mortality rates. The late phase of sepsis is marked by an increased microbiological burden and a higher death rate, underscoring the need for effective therapeutic interventions. Targeting cytokines, pathogen-associated molecular patterns (PAMPs), and damage-associated molecular patterns (DAMPs) via blood purification offers a promising approach to modulate the immune response. Early lymphocyte depletion, a hallmark of sepsis, is associated with increased mortality, highlighting the importance of preserving immune function. Tocilizumab, an IL-6 receptor monoclonal antibody, has shown efficacy in modulating cytokine dysregulation during sepsis, particularly in the context of COVID-19–related sepsis, suggesting its broader potential in inflammatory syndromes.

Sepsis is more dangerous in elderly, immunocompromised, and critically ill patients, with a significant risk of secondary bacterial or fungal infections contributing to long-term mortality. The core pathological mechanism of sepsis is the cytokine storm, where the continuous activation of the immune system leads to severe tissue and organ damage. This process can result in systemic inflammatory response syndrome (SIRS) and MODS, posing a substantial threat to patient health. The clinical manifestation of sepsis includes high fever, fatigue, myalgia, nausea, capillary leakages, tachycardia, cardiac dysfunction, liver failure, and kidney impairment. Excessive amounts of tumor necrosis factor (TNF) can cause septic shock, a severe form of sepsis that can be fatal without emergency medical treatment.

Subsequent secondary infections often act as the leading cause of death in patients with severe sepsis or septic shock. In murine models of sepsis, marked platelet aggregation, thrombin activation, and fibrin clot formation within neutrophil extracellular traps (NETs) have been observed, highlighting the role of NETs in tissue damage and organ failure. The use of immune checkpoint inhibitors (ICIs) in sepsis requires caution due to the potential for immune-related adverse events. The neutrophil-to-lymphocyte ratio (NLR) is a reliable immunological marker for sepsis, with elevated values consistently associated with microbiologically confirmed sepsis. In elderly diabetic patients, an $NLR \geq 3.482$ is associated with adverse outcomes, underscoring the need for dynamic monitoring of this biomarker.

The adage "the road to hell is paved with good intentions" aptly describes the paradoxical nature of inflammatory responses in sepsis, where the immune system's response can transition from localized to systemic, leading to MODS and high mortality. The use of red cell distribution width (MDW) as a predictor of mortality in sepsis is increasingly recognized, with elevated MDW values consistently associated with poor outcomes. Serum albumin trend is a significant predictor of mortality in ICU patients with sepsis, while ambulation ability, albumin, and C-reactive protein can predict 28-day mortality in elderly sepsis patients. Advances in multiomics and targeted therapies provide new hope for managing infections and sepsis, emphasizing the need for personalized medicine and combination therapies.

The immune response to sepsis evolves dynamically, often progressing from an early hyperinflammatory phase to a state of immunosuppression dominated by a chronic antigen-driven response (CARS). This shift enhances endothelial barrier function while minimizing vasodilation, leading to improved hemodynamics, reduced organ damage, and increased survival in animal models of sepsis. The role of triggering receptor expressed on myeloid cells-1 (TREM-1) inhibitors

as "inflammatory amplifiers" is crucial in understanding the pathophysiology of sepsis. The use of Janus kinase (JAK) inhibitors can reduce proinflammatory cytokine levels, alleviate organ damage, and increase survival rates in sepsis models. The dysregulation of essential fatty acid (EFA) metabolism may contribute to the pathogenesis of sepsis, highlighting the importance of nutritional interventions.

The sequential hypothesis is dominant in most severe infections and sepsis, where nonselective inhibitors may impair protective functions, leading to variable or disappointing clinical results. Future research should focus on combination therapies and personalized medicine based on the heterogeneity of infections and sepsis. The immune system's role in sepsis progression, resolution, and long-term outcome is complex and evolving, necessitating a multidisciplinary approach to improve patient outcomes.

Chapter: Condition | Cancer

Cancer is a complex and multifaceted disease characterized by uncontrolled cell growth and division. The classical theory posits that oncogenic mutations in malignant cells initiate cancer, but recent research has expanded our understanding to include the role of chronic inflammation and immune evasion mechanisms. Chronic inflammation, as suggested by Rudolf Virchow in 1863, is now recognized as a key driver of cancer, with leukocytes often infiltrating neoplastic tissues and contributing to tumor progression.

Cancer cells employ various strategies to evade immune surveillance. One such mechanism involves downregulation of MHC class I molecules, which allows tumor cells to avoid detection by the immune system. Additionally, the PD-1/PD-L1 pathway plays a crucial role in tumor immune escape, as it protects cells from T cell attack. Therapeutic monoclonal antibodies targeting PD-1 or PD-L1 have shown remarkable efficacy in treating several types of cancers, including melanoma, non-small cell lung cancer, and renal cell carcinoma.

Immune modulation strategies are increasingly being explored as a means to combat cancer. For instance, mRNA vaccines have been developed to target both infectious pathogens and cancer. These vaccines have demonstrated potent and versatile responses, with recent reports highlighting their efficacy in protecting against a wide range of infectious diseases. In the context of cancer, mRNA vaccines have been designed to target neoantigens, which are unique to individual tumors, thereby enhancing the specificity and effectiveness of the immune response.

Tumor regression in cancer patients has been achieved through the use of T cell-engaging antibodies, such as those that target CD19 in B-cell malignancies. However, challenges remain in expanding these therapies to solid tumors, where immunosuppressive microenvironments and the lack of tumor-specific antigens limit efficacy. To address these challenges, researchers are developing armored CAR T cells, which secrete cytokines or incorporate logic-gated signaling to enhance safety and potency.

Interleukin-6 (IL-6) has been implicated in both chronic inflammation and cancer. Anti-IL-6 therapy, initially developed for autoimmune diseases, has also been evaluated for cancer treatment due to its role in chronic inflammation. Similarly, IL-11, used to treat thrombocytopenia in cancer patients, has shown promise in modulating the immune response.

The JAK-STAT signaling pathway is another target for cancer therapy. Excessive JAK-STAT

signaling is responsible for some cancers and immune disorders, and JAK inhibitors have been proposed as potential therapeutic agents. However, the dual role of TNF in cancer, acting as both a promoter and inhibitor, complicates the development of TNF-targeted therapies.

In addition to these targeted therapies, natural immunity plays a critical role in fighting cancer. T-helper-1-cell cytokines can drive cancer into senescence, while IFNs enhance tumor immunogenicity through MHC upregulation, antigen presentation, and CTL recruitment. Autophagy modulators and cGAS-STING inhibitors offer promising strategies to indirectly modulate interferon activity in chronic inflammation and cancer.

Cancer stem cells have been proposed to explain the development of cancer and resistance to treatment. Activation of the JAK/STAT signaling pathway or induction of other signals that interact with the JAK/STAT pathway can promote the production and acquisition of drug resistance by cancer stem cells. Understanding the molecular mechanisms underlying PANoptosis, a form of cell death, may provide deeper insights into the occurrence and treatment of cancer.

In conclusion, the field of cancer research is rapidly advancing, with new therapeutic strategies emerging from a deeper understanding of immune responses and metabolic shifts. These developments hold promise for improving outcomes and reducing the burden of cancer worldwide.

Chapter: Condition | Leukemia

Leukemia is a type of cancer that affects the blood and bone marrow, characterized by the uncontrolled growth of abnormal white blood cells. These cells, which are part of the immune system, can accumulate in the bone marrow and interfere with the production of normal blood cells. Cytokine release syndrome (CRS) is a critical side effect observed in patients undergoing treatment with novel therapeutics, such as chimeric antigen receptor (CAR) T-cell therapies, particularly for acute lymphoblastic leukemia (ALL). This syndrome is a result of the massive release of cytokines by activated T cells, leading to systemic inflammation and potentially life-threatening complications.

Multiple myeloma, another hematological malignancy, arises from the transformation of a B-1 B cell into a malignant plasma cell. Understanding the pathogenesis of multiple myeloma within the bone marrow environment is crucial for identifying new therapeutic targets. For instance, the STAT3 antisense oligonucleotide AZD9150 has shown promise in a subset of patients with heavily pretreated lymphoma, as evidenced by a phase 1b trial. This treatment modality aims to inhibit the STAT3 pathway, which is often hyperactivated in lymphoma cells.

In the context of hematological malignancies, the clinical relevance of neutrophil extracellular traps (NETs) is evident, especially in chronic granulomatous disease (CGD) patients. CGD is a genetic disorder characterized by a deficiency in NADPH oxidase, leading to impaired reactive oxygen species (ROS) production and compromised NETs release. This condition predisposes patients to recurrent infections, highlighting the importance of NETs in antifungal defense.

Several biologics, including FLT3-targeted tyrosine kinase inhibitors and bispecific T cell-engaging (BiTE) antibodies, have shown cytotoxic effects on leukemia cells in vitro and in vivo. BiTEs, such as blinatumomab, have demonstrated significant efficacy in treating relapsed or refractory ALL, although some cases may experience antigen escape, where leukemia cells do not express CD19 and thus evade recognition by CD19-targeted CAR T cells. The development of additional BiTE

molecules targeting other hematologic malignancies, such as multiple myeloma and acute myeloid leukemia, continues to advance the field.

In multiple myeloma, the proliferation of malignant plasma cells is a hallmark of the disease. The use of monoclonal antibodies, such as daratumumab, has shown clinical efficacy in heavily pretreated patients. The targeting of signaling pathways, including aberrant nuclear factor-kappa B (NF- κ B) activity, is also a promising therapeutic approach. The Food and Drug Administration (FDA) and the European Medicines Agency have approved several CAR T-cell therapies for the treatment of blood cancers, including non-Hodgkin lymphoma and multiple myeloma.

Prophylactic therapy with trimethoprim-sulfamethoxazole (Bactrim, Septra) is effective in reducing the incidence of severe infections in patients with chronic granulomatous disease (CGD), a condition characterized by defective NADPH oxidase activity. Bone marrow transplantation from HLA-identical donors can be curative in patients with cellular immune deficiencies, including CGD, and may also benefit patients with other hematological disorders.

In summary, the treatment landscape for hematological malignancies, including leukemia and multiple myeloma, is rapidly evolving with the development of novel therapeutics such as BiTEs and CAR T-cell therapies. Understanding the pathogenesis and identifying new therapeutic targets are critical for improving patient outcomes.

Chapter: Condition | Autoimmunity

Follicular B helper T cells play a crucial role in the antibody responses and autoimmunity, facilitating the differentiation of B cells into plasma cells and memory B cells. These cells are involved in the mediation of inflammatory responses, infection, autoimmunity, transplant rejection, and graft-versus-host disease (GVHD). The involvement of cytokines in these processes underscores their importance in the immune system's regulation.

Vaccines and adjuvants are closely related to autoimmunity. For instance, mRNA-LNP-based drugs can cause autoimmunity through multiple mechanisms: mRNA acts as an autoantigen, triggering the autoimmune process via Toll-like receptor 7 (TLR7), while lipids in the nanoparticles (LNPs) act as adjuvants, enhancing the innate immune response. This can lead to an exacerbated autoimmune response, especially in the context of mRNA-LNP vaccines.

Autoimmune disease susceptibility in humans is highly associated with HLA class I (human MHC I) polymorphisms, which are prone to autoantigen presentation to CD8⁺ T cells. This genetic predisposition highlights the importance of HLA in the development of autoimmune responses. CAR T cells can be used to target autoimmune diseases by eradicating auto-reactive populations of memory B cells, offering a promising therapeutic approach.

Th1 patterns of cytokine production are associated with inflammation and autoimmune disease, while Th2 patterns are characteristic of allergic responses and asthma. The breakdown of a single mechanism of self-tolerance can lead to various autoimmune diseases, underscoring the complexity of these conditions. Adoptive cell therapy using regulatory T cells (Tregs) is currently being explored in clinical trials for various autoimmune diseases, aiming to restore immune homeostasis.

NETs (neutrophil extracellular traps) play a significant role in immune homeostasis, inflammation, autoimmune diseases, metabolic dysregulation, and cancer. Their formation and degradation are critical in maintaining immune balance. In systemic lupus erythematosus (SLE), two subsets of

dendritic cells contribute to the onset and severity of the disease, with plasmacytoid dendritic cells (pDCs) overproducing interferon- α , which interferes with the ability of conventional dendritic cells (cDCs) to maintain peripheral tolerance.

Autoimmune diseases such as SLE and vasculitis show a strong NET phenotype, with autoantibodies against proteins readily released from neutrophils in the process of NETosis, such as anti-dsDNA and anti-neutrophil cytoplasmic autoantibodies. This highlights the importance of NETs in the pathogenesis of these conditions. The immunopathogenesis of SLE involves excessive production of type I interferons, which can drive abnormal proliferation of immune cells and uncontrolled cytokine secretion, leading to systemic inflammation and tissue pathology.

In rheumatoid arthritis (RA), autoantibodies known as rheumatoid factors (RF) and anti-citrullinated peptide antibodies (ACPA) are present, and NETs are a source of citrullinated autoantigens that stimulate inflammatory responses. Patients with RA are at increased risk for early coronary artery disease, metabolic bone disease, and various cancers, which may be related to underlying, uncontrolled systemic inflammatory processes and medications.

Hashimoto thyroiditis, also known as Hashimoto's disease, is an autoimmune condition characterized by the presence of autoantibodies against thyroid peroxidase and thyroglobulin. It can cause the thyroid to produce insufficient thyroid hormone, leading to hypothyroidism. Individuals with Hashimoto's thyroiditis are at increased risk for other autoimmune illnesses, including celiac disease, type 1 diabetes, vitiligo, alopecia areata, Addison disease, Sjögren's disease, and rheumatoid arthritis.

Autoimmune diseases such as RA and SLE are associated with dysregulation of various signaling pathways, including those involving type I and type III interferons. These pathways are under intensive study for their possible role in the pathogenesis of autoimmune disorders. The gut microbiome plays a significant role in the pathogenesis of autoimmune disorders, with alterations in the microbiota contributing to the development of conditions such as SLE.

In conclusion, the interplay between genetic predisposition, environmental factors, and immune system dysregulation is critical in the development of autoimmune diseases. Understanding these mechanisms is essential for developing effective therapeutic strategies.

Chapter: Condition | COVID-19

Since the onset of the COVID-19 pandemic, the virus has caused significant public health and economic challenges worldwide. The virus, Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), has led to over 153 million infections and over 3.2 million deaths, as reported by the Johns Hopkins University Coronavirus COVID-19 Dashboard. The development of COVID-19 vaccines and therapeutic antibodies has been a critical strategy to combat the pandemic. Since 2020, Elsevier has provided free access to a comprehensive resource center on the novel coronavirus, including research available in PubMed Central and other publicly funded repositories. This resource highlights the importance of understanding the pathophysiology of COVID-19, including cytokine storms, immunosuppression, and the role of neutrophil extracellular traps (NETs).

Elevated NET release has been documented in numerous patients with COVID-19, contributing to detrimental coagulopathy, immunothrombosis, and pulmonary endothelium damage within the alveoli. The excessive formation of NETs is linked to the onset of acute lung injury (ALI), acute

respiratory distress syndrome (ARDS), and an increased risk of immunothrombosis. The relationship between coronavirus disease 2019 (COVID-19) and autoimmunity is still being studied, but there is evidence suggesting that COVID-19 may trigger autoimmune reactions in some individuals. The Janus kinase (JAK)/signal transducer and the activator of the transcription (STAT) pathway plays a central role in driving hyperinflammation in COVID-19, leading to the cytokine storm.

Modulation of the body by complement with infection, excessive complement activity, and the delayed interferon-I (IFN-I) response contribute to severe COVID-19 symptoms and disease. The ongoing pandemic has also raised concerns about the effectiveness of monoclonal antibody (mAb) therapies and the potential evasion from vaccine-induced immunity due to the continued evolution of SARS-CoV-2 variants. The development of broadly neutralizing antibodies (NAbs) for prophylactic and therapeutic use has been a focus, with mRNA vaccines playing a vital role in global response. The Pfizer-BioNTech and Moderna mRNA vaccines have shown high efficacy and authentic safety profiles, helping to save millions of lives.

In the context of vaccine efficacy, studies have shown that mRNA vaccines can broadly induce anti-spike antibodies and neutralizing antibodies (nAbs). However, the success of these vaccines highlights the need for ongoing research into optimizing formulations, expanding the range of target pathogens, and improving the stability of mRNA constructs. The benefits of helminth co-infection and the role of natural killer cell-based immunotherapies in managing COVID-19 are also areas of interest. The development of mRNA vaccines has demonstrated the potential of the lipid nanoparticle (LNP) platform, generating substantial revenue and providing hope for future treatments.

Chapter: Condition | Specific Cancers

In specific cancers, the tumor microenvironment (TME) plays a critical role in both tumor progression and therapeutic response. This section highlights key findings from various cancers, focusing on lung, breast, prostate, and melanoma, and discusses the implications for treatment strategies.

Lung Cancer

In lung cancer, the combination of radiotherapy with anti-PD-1/PD-L1 therapies has shown promising results, building better translational research platforms (Kordbacheh et al., 2020). Additionally, elevated tumor mutational burden and increased neoantigen expression in cancers such as melanoma, head and neck, non-small cell lung cancer (NSCLC), bladder, and microsatellite unstable cancers are generally more responsive to anti-PD-1/PD-L1 therapy (Rosenthal et al., 2020). TAM reprogramming via the STAT3/CD47-SIRP α axis facilitates acquired resistance to EGFR-tyrosine kinase inhibitors (TKIs) in lung cancer (Liu et al., 2020). Furthermore, inhibition of JAK2 signaling by TG101209 enhances radiotherapy in lung cancer models (Pham et al., 2014).

Breast Cancer

In breast cancer, TAM levels are significantly associated with poor survival outcomes. High TAM infiltration is linked to increased tumor recurrence rates and shorter disease-free survival (Hahne, 2020). Macrophages define the invasive microenvironment in breast cancer, and neutrophil-lymphocyte ratio predicts the prognosis of patients with hepatocellular carcinoma after liver

transplantation (Hilf et al., 2020). TAMs also play a role in the pre-metastatic niche, where they facilitate tumor metastasis (Li et al., 2020). The immune tumor microenvironment in breast cancer is influenced by estrogen and its receptors, with estrogen receptor α (ER α) signaling differentially modulating HLA-class II expression (Mostafa et al., 2020).

Prostate Cancer

In prostate cancer, TAMs are involved in tumor relapse and systemic disease dissemination. IFN- α has been shown to activate dormant hematopoietic stem cells in vivo, potentially contributing to tumor relapse (Weide et al., 2020). TAMs also play a role in the tumor microenvironment, where they can be targeted by bisphosphonates to inhibit tumor growth (Zhang et al., 2020).

Apolipoprotein E (APOE) in the TME binds to TREM2 on macrophages, promoting androgen receptor (AR) expression and upregulating immune mediators such as IL-10, TGF- β 1, IL-23A, and CCL2 (Lyu et al., 2020).

Melanoma

In melanoma, elevated TAM numbers are associated with increased tumor recurrence rates. The presence of Fas ligand in melanoma cells contributes to tumor immune escape (Hahne, 2020). Additionally, the use of mRNA vaccines has shown promise in activating neoepitope-specific T cells, reducing recurrence rates, and improving survival (Ribas et al., 2014). The melanoma cell expression of Fas ligand and the use of mRNA vaccines highlight the importance of targeting TAMs and neoantigens in melanoma treatment (Weide et al., 2020).

Conclusion

The tumor microenvironment is a complex network that influences cancer progression and therapeutic response. Targeting TAMs and neoantigens in specific cancers such as lung, breast, prostate, and melanoma offers promising avenues for improving treatment outcomes. Further research is needed to develop more effective and personalized treatment strategies.

Chapter: Condition | Chronic Diseases

In chronic diseases, the role of inflammation and immune cell polarization is paramount. In osteoarthritis (OA), the polarization spectrum of synovial macrophages extends beyond the conventional M1/M2 classification, making targeted modulation of these cells a promising therapeutic strategy (Perez-Sanchez, C. et al.). Obesity and type 2 diabetes involve adipose tissue macrophage polarization that drives metabolic dysfunction, with prolonged interferon exposure suppressing endothelial nitric oxide synthase (NOS3) expression and nitric oxide (NO) production, leading to endothelial dysfunction and promoting atherosclerosis (Mouton AJ, Li X, Hall ME et al.). Various age-related diseases such as Alzheimer's disease, hypertension, arthritis, atherosclerosis, type 2 diabetes, and cancer are attributed to inflammaging, a chronic low-grade inflammation associated with immune cell infiltration in adipose tissue (34).

Inflammatory bowel diseases (IBD), including Crohn's disease and ulcerative colitis, are chronic inflammatory disorders of the gastrointestinal tract, thought to result from inappropriate inflammatory responses to intestinal microbes (Wojcik, G. L. et al.). Asthma, a common chronic disorder of the airways, is characterized by variable and recurring symptoms, airflow obstruction, bronchial hyperresponsiveness, and underlying inflammation (Green RH, Brightling CE, McKenna

S, et al.). Airway remodeling, a key feature of asthma, includes subepithelial fibrosis, smooth muscle hypertrophy, goblet cell hyperplasia, and angiogenesis, contributing to the accelerated loss of lung function over time (34).

The development of asthma involves a complex interplay of genetic and environmental factors, with common symptoms including a persistent cough, wheezing, shortness of breath, and chest tightness (34). Asthma is more common in prepubertal boys but boys are more likely to grow out of their asthma during adolescence than girls (34). The incidence of asthma is reduced in association with certain infections, exposure to other children, and less frequent use of antibiotics (34). By using spirometry, physicians can diagnose asthma and assess the severity of obstruction (34).

Inflammatory diseases such as asthma, autoimmune diseases (type I diabetes, multiple sclerosis), and IBD are linked to the activation of various cytokines and immune cell populations. For instance, in psoriasis, lesional skin is populated with an increased aggregation of natural killer cells, type-3 innate lymphoid cells, TH17 cells, and $\gamma\delta$ T-cells, leading to the production of inflammatory cytokines such as IL-17 and IL-22 (34). In asthma and allergic disorders, the cytokine thymic stromal lymphopoietin (TSLP) activates JAK1 and STAT5 signaling in airway epithelial cells and dendritic cells, fostering type 2 immune responses that recruit eosinophils to the lung tissue and perpetuate airway hyperresponsiveness (34).

The role of inflammation in chronic diseases is further highlighted by the involvement of neutrophil extracellular traps (NETs) in cancer-related pathologies like thrombosis, organ failure, and metastasis formation (Fuchs TA, Brill A, Duerschmied D, Schatzberg D, Monestier M, Myers DD Jr., et al.). The hygiene hypothesis suggests that a loss of normal balance between two opposing populations of T helper (Th) lymphocytes may explain some of the dramatic increases in asthma prevalence in westernized countries (34).

In summary, chronic diseases such as OA, IBD, and asthma are characterized by persistent inflammation and immune cell polarization, with a significant impact on disease progression and management. Targeted modulation of immune cell phenotypes and the development of novel therapeutic strategies are critical for improving patient outcomes.

Chapter: Condition | Infectious Diseases

Infectious diseases, such as malaria and tuberculosis, pose significant global health challenges. Malaria, caused by Plasmodium parasites, primarily affects individuals in tropical and subtropical regions. Simulations of malaria infections in naïve hosts reveal that parasitaemia levels of various antigenic variants under different switching assumptions (random and preferential) do not show major qualitative differences. This suggests that the immune response to malaria is robust and adaptable, regardless of the switching mechanism of the parasite. Turner et al. (1998) conducted an immunohistochemical study on the pathology of fatal malaria, highlighting the severe inflammatory response and tissue damage associated with the disease. The biology and disease mechanisms of malaria are complex, involving antigenic variation and rapid switching of virulence genes, as evidenced by Frank et al. (2007) and Horrocks et al. (2004).

Tuberculosis, on the other hand, is a chronic infectious disease caused by Mycobacterium tuberculosis. The pathology of tuberculosis is characterized by the inflammatory reaction of the host to the eggs deposited in tissues, leading to granuloma formation. The granuloma contains necrotic tissue at its center, forming small white nodules known as tubercles. The disease can affect various

organs, but it most commonly involves the lungs, known as pulmonary tuberculosis. Active tuberculosis can develop in individuals without HIV infection in 5–15% of cases, while the rate is higher in those co-infected with HIV, reaching 30%. The disease has a long history, with Robert Koch identifying the causative agent in 1882, and J. L. Schönlein first using the term "tuberculosis" in 1832. The development of treatments for tuberculosis has been a long journey, with the first genuine success in immunization achieved by Albert Calmette and Camille Guérin in 1906 using an attenuated bovine-strain tuberculosis. The World Health Organization declared tuberculosis a global health emergency in 1993, and numerous strategies have been implemented to combat the disease, including the use of the bacillus Calmette-Guérin (BCG) vaccine.

In the context of malaria, the pathogenesis involves the switching of surface antigens by the parasite, which can evade the host immune response. This switching is associated with chromosomal position, as noted by Frank et al. (2007). Similarly, in schistosomiasis, an acute and chronic disease caused by trematode worms, the pathology is driven by the host's inflammatory response to the eggs deposited in tissues. The immune response in schistosomiasis is controlled by antigen-specific regulatory T cells primed in the presence of Toll-like receptor 2 (TLR2), as highlighted by Herbert et al. (2004).

Tuberculosis remains a significant public health issue, with global trends showing a decline in rates due to improvements in sanitation, vaccination, and public health measures. However, the emergence of drug-resistant strains, such as multi-drug resistant tuberculosis (MDR-TB), continues to pose a challenge. The World Health Organization's Global Plan to Stop Tuberculosis aimed to save 14 million lives between 2006 and 2015. The diagnosis of tuberculosis is often difficult, and the use of diagnostic tools such as the tuberculin skin test and interferon-gamma release assays is crucial. The treatment of active tuberculosis involves a combination of antibiotics, and prompt and appropriate treatment is essential for cure.

Chapter: Condition | Primary Immunodeficiency

Primary immunodeficiency (PI) is a heterogeneous group of more than 550 genetically determined disorders that impair the immune system's ability to protect against infections, malignancies, and other threats. PI is subdivided into types that cause T-cell deficiency, B-cell deficiency, both T-cell and B-cell deficiency, complement deficiency, phagocyte deficiency, and immunoglobulin A (IgA) deficiency. The 10 warning signs of primary immunodeficiency, developed by the Jeffrey Modell Foundation Medical Advisory Board, include persistent thrush or fungal infection on skin or elsewhere, infection with normally harmless tuberculosis-like bacteria, and family history of a primary immunodeficiency. These signs are critical for early diagnosis and intervention.

Severe combined immunodeficiency (SCID) is associated with profound deficiencies of T-cell and B-cell function (and sometimes natural killer cell function). The SCID but not the RAG-2 gene product is required for S mu-S epsilon heavy chain class switching. Flow cytometry is essential for quantifying lymphocyte subsets, including CD4+ and CD8+ T cells, B cells (via CD19 or CD20 markers), and natural killer cells, aiding in the identification of disorders like SCID or X-linked agammaglobulinemia. The clinical presentation of PIDs shows significant variability, with some individuals remaining asymptomatic carriers, such as in selective IgA deficiency, while others experience late-onset disease in adulthood, as seen in common variable immunodeficiency (CVID), where diagnosis often occurs after years of symptoms.

Primary immunodeficiency diseases are distinct from secondary immunodeficiencies that may result from other causes, such as viral or bacterial infections, malnutrition, immunoglobulin (Ig) loss, malignancy, or treatment with drugs that induce immunosuppression. It is important to note that PIDs are broadly classified as disorders of adaptive immunity (i.e., T cell, B-cell, or combined immunodeficiencies) or of innate immunity (e.g., phagocyte and complement disorders). The disorders affecting the activity of the T-cell arm of the adaptive system, such as SCID, require immune reconstitution as soon as possible. Reduced switching in SCID B cells is associated with altered somatic mutation of recombined S regions. Severe disorders such as SCID require definitive therapy for immune reconstitution (e.g., bone marrow transplantation, hematopoietic stem cell transplantation, gene therapy) as soon as possible, which has led to the application of newborn screening to this population.

The term “common variable immunodeficiency” encompasses a heterogeneous group of disorders that cause hypogammaglobulinemia (serum IgA levels below 5 mg per dL [0.05 g per L]). Patients with common variable immunodeficiency have a poor response to vaccines (decreased IgG antibody response) and an increased risk of developing autoimmune disorders and malignancy. In contrast, patients with IgA deficiency have a normal IgG response to vaccinations. It is essential to determine whether a patient's infections are caused by PID or by something else. PID can also lead to other conditions and issues, which, depending on the type of PID and the areas of the immune system that are weakened, may include developmental delays and slow growth in children, inflammation and damage to internal organs, autoimmune diseases, anemia or other problems of the blood, and digestive difficulties. PID may also lead to complications including damage to the heart, lungs, nervous system, or digestive tract, as well as an increased risk of certain cancers.

Chapter: Condition | Graft-Versus-Host Disease

Graft-versus-host disease (GVHD) is a complication that can occur after a stem cell or bone marrow transplant (BMT) that uses donor cells. What is graft-versus-host disease? GVHD is an immune condition that occurs after transplant procedures when immune cells from the donor (known as the graft or graft cells) attack the recipient patient host's tissues; the disease is a side effect that is common after an allogeneic bone marrow transplant (stem cell transplant). This procedure, however, carries a substantial risk of graft-versus-host disease (GVHD), where donor T cells recognize the recipient's tissues as foreign and mount an immune response.

Laboratory and histopathological confirmation of GVHD begins with routine blood tests, particularly for hepatic involvement, where elevations in liver enzymes such as alanine aminotransferase (ALT) and aspartate aminotransferase (AST), often exceeding twice the upper limit of normal, alongside increased alkaline phosphatase and bilirubin, support the diagnosis. Diagnosis of acute GVHD relies on standardized clinical criteria that evaluate organ involvement in the skin, upper and lower gastrointestinal tract, and liver. Chronic GVHD, on the other hand, typically develops more than 100 days after transplant and is characterized by progressive tissue fibrosis and scarring across various organs.

To prevent GVHD, strategies such as using umbilical cord blood as a source of donor cells, removing some of the T cells from the donor stem cells before infusion, and better HLA matching can be employed. GVHD is broadly classified into acute (typically within the first 100 days post-transplant) and chronic (after day 100 or as progression from acute) forms, though overlap syndromes exist. Treatment for those with grade 2-4 acute GVHD usually consists of continuing the

original immunosuppressive prevention and adding a corticosteroid such as methylprednisolone or prednisone taken orally (by mouth).

Acute GVHD classically manifests as a triad of organ involvement affecting the skin, gastrointestinal (GI) tract, and liver, occurring in approximately 70%, 74%, and 44% of cases, respectively. Patients who have some GVHD may have a lower risk of the cancer returning after transplant. The risk of GVHD is significantly amplified by human leukocyte antigen (HLA) disparity between donor and recipient; for instance, transplants with a full 10/10 HLA match exhibit a significantly lower incidence of severe acute GVHD compared to those with an 8/8 match or greater mismatches.

GVHD can affect many parts of the body, including the eyes, skin, and mouth, as well as the gastrointestinal tract (gut), liver, and lungs. The risk of GVHD usually increases with the recipient's age and with the degree of HLA differences between donor and recipient unless fully T-cell depleted. Prevention of GVHD is challenging, but measures such as better HLA matching and the use of immunosuppressive drugs can help reduce the risk. Treatment for GVHD can depend on a number of factors, including the type of GVHD and where it is located. Steroids remain the first line of treatment for acute GVHD, but only about 50% of patients respond to treatment, otherwise having steroid-refractory GVHD (SR-GVHD).

In the clinical setting, a retrospective analysis of 567 patients who received allogeneic transplantation from HLA-identical sibling donors noted that statin use by the donor (but not by the recipient) was associated with a decreased risk of grade 3-4 acute GVHD. Because chronic GVHD usually develops more than 100 days after transplant, it's important that recipients examine their body regularly for signs of GVHD. Management sometimes involves a skin biopsy to confirm a diagnosis of GvHD of the skin.

Chapter: Condition | Cytokine Release Syndrome

Cytokine release syndrome (CRS) is a condition characterized by the excessive release of proinflammatory cytokines such as IL-6 and TNF- α following rapid immune engagement, often observed after treatment with CAR T-cells. This immune activation can precipitate CRS, leading to symptoms such as fever, hypotension, and organ dysfunction. The term "cytokine storm" was first referenced in 1993 by James Ferrara in the context of graft vs. host disease, highlighting the long-standing recognition of this phenomenon. The management of CRS is complex and requires a tailored approach, as therapeutic strategies are etiology-specific and a "one-size-fits-all" protocol does not exist.

CRS is the most common issue encountered after CAR T-cell therapy, and its severity can be life-threatening. In 2018, a systematic review highlighted that CRS might be a fatal complication of CAR-T cell immunotherapy. To address CRS, various therapeutic strategies have been explored, including the use of corticosteroids and tocilizumab, as well as plasma exchange. The future of CRS management involves precision treatment, immune phenotyping, and the use of cutting-edge technologies such as multiomics and artificial intelligence to discover new biomarkers and therapeutic targets. Additionally, the development of next-generation targeted therapies aimed at specific cytokines, cell death pathways, and immune cell functions is being pursued.

The pathogenesis of CRS involves the activation of immune signaling cascades triggered by pathogen-associated molecular patterns (PAMPs), damage-associated molecular patterns (DAMPs),

and PANoptosis. Organ injury during CRS is driven not only by proinflammatory responses but also by reciprocal interactions between pathogens and dysregulated immunity. Metabolic disturbances play a significant role in the occurrence and maintenance of CRS, with infection and inflammation leading to dramatic remodeling of energy metabolism. The clinical manifestations of CRS include a wide range of adverse reactions such as fever, hypotension, and organ dysfunction, among others.

Chapter: Condition | Genetic Disorders

DiGeorge syndrome results in abnormal migration of the third and fourth branchial pouches during embryogenesis, leading to hypoplasia to aplasia of the thymus and parathyroid glands. This condition is caused by a chromosomal 22q11.2 deletion affecting TBX1, which can result in thymic hypoplasia, conotruncal heart defects, and hypocalcemia. Examples of other genetic disorders affecting T cell function include Wiskott-Aldrich syndrome, where mutations lead to thrombocytopenia, eczema, and recurrent infections due to cytoskeletal defects in hematopoietic cells.

Causes of partial insufficiencies of T cell function include acquired immune deficiency syndrome (AIDS) and hereditary conditions such as DiGeorge syndrome, chromosomal breakage syndromes, and B cell and T cell combined disorders like ataxia-telangiectasia and Wiskott–Aldrich syndrome. Patients with chronic graft-versus-host disease (GVHD) will typically require prolonged courses of steroids, generally 2 to 3 years, and some may require lifelong treatment. The development of chronic GVHD is closely related to a helpful graft-versus-tumor/leukemia effect. The National Institutes of Health (NIH) consensus criteria, first established in 2005 and revised in 2014, provide a comprehensive framework for assessing multiorgan involvement beyond the time from transplant.

Belumosudil (Rezurock), a Rho-associated coiled-coil kinase 2 (ROCK2) inhibitor approved by the FDA in 2021 for chronic GVHD after at least two prior lines of systemic therapy, demonstrated an overall response rate of 73% (with 6% complete responses) in the phase 2 ROCKstar trial, particularly benefiting lung and skin manifestations by reducing fibrotic signaling. Lemzoparlimab, another anti-CD47 antibody, is in Phase III studies for myelodysplastic syndromes.

Chapter: Condition | Miscellaneous Conditions

Mast cell activation syndrome (MCAS) is a condition characterized by the overactivation of mast cells, leading to the release of excessive amounts of mediators such as histamine, leukotrienes, and prostaglandins (Theoharides TC, Valent P, Akin C. Mast cells, mastocytosis, and related disorders). MCAS has been frequently associated with postural orthostatic tachycardia syndrome (POTS) and Ehlers-Danlos syndrome, although the exact nature of the connection remains unclear (Theoharides TC, Valent P, Akin C. Mast cells, mastocytosis, and related disorders). Mastocytosis, a rare disease, involves the accumulation of mast cells in organs such as the spleen, liver, gut, and skin (Theoharides TC, Valent P, Akin C. Mast cells, mastocytosis, and related disorders).

Anaphylaxis is a life-threatening allergic reaction that can cause shock and affect various organ systems, including the skin, respiratory, cardiovascular, and gastrointestinal systems (Simons FER, Arduso LRF, Bilo MB, El-Gamal YM, Ledford DK, Ring J, et al. World Allergy Organization anaphylaxis guidelines: Summary). It is primarily a clinical diagnosis, and laboratory studies are not usually required (Simons FER, Arduso LRF, Bilo MB, El-Gamal YM, Ledford DK, Ring J, et al.

World Allergy Organization anaphylaxis guidelines: Summary). Anaphylaxis can occur following immunization with a COVID-19 mRNA vaccine, among other triggers (Moneret-Vautrin DA, Morisset M, Flabbee J, Beaudouin E, Kanny G. Epidemiology of life-threatening and lethal anaphylaxis: A review).

Signs and symptoms of anaphylaxis include urticaria, angioedema, respiratory compromise, hypotension, and gastrointestinal upset (Simons FER, Arduzzo LRF, Bilo MB, El-Gamal YM, Ledford DK, Ring J, et al. World Allergy Organization anaphylaxis guidelines: Summary). Up to 500–1000 fatal cases of anaphylaxis per year are estimated to occur in the United States (Moneret-Vautrin DA, Morisset M, Flabbee J, Beaudouin E, Kanny G. Epidemiology of life-threatening and lethal anaphylaxis: A review). Treatment for MCAS is directed at preventing mediator release and blocking mediator effects, often using cromolyn and/or ketotifen, along with H1 and H2 blockers, aspirin, and montelukast (Theoharides TC, Valent P, Akin C. Mast cells, mastocytosis, and related disorders).

Immunodeficiency, a hallmark of acquired immunodeficiency syndrome (AIDS), caused by the human immunodeficiency virus (HIV), results from a failure or absence of elements of the immune system, including lymphocytes, phagocytes, and the complement system (Introduction Immunodeficiency results from a failure or absence of elements of the immune system, including lymphocytes, phagocytes, and the complement system). Mast cell activation syndrome is commonly misdiagnosed due to the wide range of symptoms it can present (Jennings S, Russell N, Jennings B, Slee V, Sterling L, Castells M, et al. The mastocytosis society survey on mast cell disorders: patient experiences and perceptions). The pathophysiology of anaphylaxis involves the interaction of antigen with IgE on basophils and mast cells, triggering the release of histamine, leukotrienes, and other mediators (Pathophysiology of Anaphylaxis Interaction of antigen with IgE on basophils and mast cells triggers release of histamine, leukotrienes, and other mediators that cause diffuse smooth muscle contraction (eg, resulting in bronchoconstriction, vomiting, or diarrhea) and vasodilation with plasma leakage (eg, resulting in urticaria or angioedema)).

In multiple sclerosis (MS), the immune system attacks the myelin sheath, leading to demyelination and axonal injury, causing various neurological symptoms (10 There is debate about whether the inflammation leading to MS is initiated within or outside the central nervous system; however, T cells, B cells, macrophages (including central nervous system microglia), astrocytes, inflammatory mediators, and blood-brain barrier permeability are all involved in a response that is associated with myelin sheath destruction, axonal injury, and clinical symptoms). The diagnosis of MS depends on a combination of clinical findings, imaging, and laboratory results (A patient history, neurologic examination, and application of the 2017 McDonald Criteria are needed to

Chapter: Cell_Function | Antigen Presentation

Antigen presentation is a critical process in the immune system, enabling the recognition of foreign pathogens and the activation of T lymphocytes. This process involves the binding of peptide fragments derived from antigens to major histocompatibility complex (MHC) molecules and the subsequent display of these complexes on the surface of antigen-presenting cells (APCs) for T cell recognition. The nature of antigen presentation is such that MHC class I glycoproteins present endogenous antigens that originate from the cytoplasm, while MHC class II molecules present exogenous antigens derived from the endoplasmic reticulum (ER) via the transporter associated with antigen processing (TAP) (A).

The process of antigen presentation begins with the proteasome cleavage of proteins into peptide fragments, which are then transported into the ER by TAP. These peptides bind to MHC molecules in the ER and are subsequently transported to the cell surface for T cell recognition. This process is influenced by cytokine signal transduction and lymphocyte differentiation, ensuring that early pathogen sensing translates into precise and sustained adaptive immune responses (Muller et al., 2018; Gfeller et al., 2022).

Antigen presentation profiling has revealed the recognition of lymphoma immunoglobulin neoantigens, highlighting the importance of this process in identifying and responding to abnormal proteins (Khodadoust et al., 2020). The "who, how, and where" of antigen presentation to B cells also underscores the complexity and specificity of this process. The combined understanding of antigen presentation by MHC molecules allows for the exploitation of this knowledge to improve the responses of the cellular arm of the immune system to vaccinations and immunotherapies (Wieczorek et al., 2017).

Supporting the primary facts, mRNA translation plays a role in direct MHC class I antigen presentation, and peptide diffusion, protection, and degradation occur in nuclear and cytoplasmic compartments before antigen presentation by MHC class I (Apcher et al., 2018; Reits et al., 2005). Type I interferon upregulates surface molecules such as CD40, CD69, CD86, and MHC-II, which promote T-B cell interactions and antigen presentation (Rosalia et al., 2019). Additionally, deficiencies in the antigen presentation machinery can influence the heterogeneous tumor antigenic landscape, as demonstrated by the application of NeoDisc for personalized antigen discovery (Rosalia et al., 2019).

Certain interferon-stimulated genes (ISGs) encode a wide array of effector proteins that collectively restrict viral replication, modulate immune cell activity, and regulate processes such as antigen presentation, cell-cycle arrest, and programmed cell death. Suppressor of cytokine signaling 1 (SOCS1) negatively regulates the antigen presentation process, while B. malayi cystatins can impede antigen presentation in APCs via interference with host cysteine protease function, reducing T cell priming (136, 137). The role of SOCS1 in antigen presentation and the function of macrophages and dendritic cells is also significant (Suppressor of cytokine signaling 1, 2023).

Antigen presentation by mass spectrometry should be embedded in a broader, integrated characterization of host immunity, providing robust, reliable, and quantitative measurements. This approach can help in evaluating the wider host antigen presentation, monitoring the status of the host microenvironment, and assessing host immune exhaustion before and after antigen dosing to better understand the capacity to respond to therapy.

Chapter: Cell_Function | Phagocytosis

Phagocytosis is a critical cellular process involving the ingestion of extracellular particulate material such as invading pathogens or dead/dying cells by phagocytic cells. This mechanism is essential for the innate immune response, helping to eliminate harmful substances and maintain homeostasis. Oxidized phosphatidylserine-CD36 interactions play an essential role in macrophage-dependent phagocytosis of apoptotic cells, facilitating the recognition and clearance of dying cells. Phagocytosis is also observed in protists like amoebae, where it serves as a means to acquire nutrients. Non-Opsonin Phagocytosis involves the direct recognition of pathogens by phagocytes using pattern-recognition receptors (PRRs) that detect microbial molecules, enhancing the immune

response against invading pathogens.

During phagocytosis, a respiratory burst occurs in macrophages, where membrane-bound oxidase is activated, leading to the reduction of oxygen to superoxide anion, a reactive oxygen species (ROS). This ROS is further converted to hydroxyl radicals and hydrogen peroxide, which are potent oxidizing agents. In the phagolysosome, the enzyme myeloperoxidase catalyzes the reaction between hydroperoxide and chloride ions, producing hypochlorite, a toxic substance that can destroy ingested microorganisms. This process is similar to the mechanism used in household bleach, highlighting the importance of ROS in phagocytic defense.

CD47, a molecule that negatively regulates phagocytosis, interacts with SIRP α on macrophages, leading to the phosphorylation of the immunoreceptor tyrosine-based inhibition (ITIM) motif in the cytoplasmic tail of SIRP α . This interaction recruits the phosphatase SHP-1, which inhibits phagocytosis. Integrins and Fc γ receptors also cooperate during phagocytosis, promoting adhesion to opsonized particles. The membrane-associated cortical cytoskeleton undergoes alterations during phagocytosis, involving the action of coronins, cofilin, and gelsolin, which regulate F-actin dynamics.

Phagocytosis can be opsonin-dependent, where pathogens are coated with opsonins that enhance recognition by phagocytes, or non-opsonin-dependent, where phagocytes recognize pathogens directly using PRRs. An improved understanding of phagocytosis is crucial for developing future therapeutics related to infections and inflammation. For instance, targeting the CD47-SIRP α axis may be effective in preventing the spread of certain cancers. Defects in phagocytosis can lead to increased susceptibility to infections and inflammatory diseases, underscoring its importance in immune function.

Phagocytosis is a form of endocytosis, involving the transport of substances from the extracellular matrix into the cell. It is one of several types of endocytosis, including receptor-mediated endocytosis and pinocytosis. Phagocytosis is performed by phagocytes such as neutrophils and macrophages, which are very efficient at internalizing antigens and processing them into peptide fragments for presentation on their membrane. This process is essential for immune surveillance and maintaining homeostasis.

Chapter: Cell_Function | Efferocytosis

Efferocytosis is the process of macrophages (or phagocytes) engulfing apoptotic cells and thereafter synthesizing pro-resolving mediators. This process plays a crucial role in determining the fate of tissue inflammation, tipping the balance towards resolution. Chakraborty et al., 2021 have shown that efferocytosis can be promoted by Piezo1 activation, highlighting the importance of mechanical signaling in this process. Furthermore, soluble factors produced by other cell types can facilitate efferocytosis, suggesting a complex interplay between different cell types in the inflammatory response. The hardening and fibrotic renal tissue may also promote efferocytosis via Piezo-1 activation, indicating a potential role in tissue repair mechanisms.

Efferocytosis not only prevents secondary necrosis and inflammation caused by the leakage of apoptotic cell contents but also actively induces macrophages to secrete anti-inflammatory factors like IL-10, thereby promoting tissue repair. Efficient efferocytosis is crucial for maintaining tissue homeostasis and preventing chronic inflammation. Abnormal neutrophil traps and impaired efferocytosis have been linked to liver injury and sepsis severity after binge alcohol use,

underscoring the importance of this process in various pathologies.

Taken together, the central role of macrophage efferocytosis in determining tissue inflammation fate and the beneficial effects presented by efferocytosis effectors in numerous pathologies emphasize the vast therapeutic potential of this process. While efferocytosis modulation holds immense promise, caution must be taken in a pathology-specific manner to evaluate its impacts on the disease.

Chapter: Cell_Function | Cross-Presentation

Cross-presentation is a specialized process that allows exogenous antigens to be presented on MHC class I molecules, primarily by dendritic cells, to prime CD8⁺ T cells against threats that do not directly infect the antigen-presenting cells (APCs), such as viruses in dying cells or tumor antigens. This mechanism is essential for the effective activation of CD8⁺ T cells, which are crucial in clearing infected or transformed cells. Regardless of the exact mechanism, cross-presentation enables MHC I molecules to present peptides from antigens that are typically handled by MHC II molecules. This process is critical for the immune system to recognize and eliminate a wide range of pathogens and tumors that do not directly interact with the APCs.

Chapter: Cell_Function | Cytotoxicity

Cytotoxicity is a critical mechanism by which the immune system eliminates harmful cells, including cancer cells and virus-infected cells. Studies have shown that M1 macrophages are associated with the classical response, inflammation, and cytotoxicity, while M2 macrophages are linked to the alternative response and anti-inflammation (Mills et al., 2000). Targeting Multiple Myeloma with AMG 424, a novel anti-CD38/CD3 bispecific T-cell-recruiting antibody, has been optimized for enhanced cytotoxicity and cytokine release (Zuch de Zafra et al., 2018). BiTEs (bispecific T-cell engagers) have been demonstrated to induce significant cytotoxicity by T cells in the absence of costimulation, such as through anti-CD28 antibodies and IL-2 (Zuch de Zafra et al., 2018).

Besides neutralizing viruses through Fab-mediated mechanisms, neutralizing antibodies (NAbs) can also inhibit viral infection through Fc-mediated effector functions such as antibody-dependent cell-mediated cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), and complement-dependent cytotoxicity (CDC) (Zuch de Zafra et al., 2018). ADCC is considered a primary therapeutic mechanism of monoclonal antibody (mAb) therapy, and clinical resistance often involves impaired cytotoxic immune effector cell responses (Zuch de Zafra et al., 2018).

The cytotoxicity of dendritic cells (DCs) against tumor cell killing has been elucidated in recent studies (Zuch de Zafra et al., 2018). Cytotoxicity is typically measured and calculated using an LDH release kit according to the protocol (Zuch de Zafra et al., 2018). Twenty-five years ago, immunologists began to explore how key enzymatic reactions of cellular metabolism could affect the immune response, leading to the field of "immunometabolism" (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018).

Cationic additives in nanosystems can activate the cytotoxicity and inflammatory response of human neutrophils, with lipid nanoparticles versus polymeric nanoparticles showing different effects (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018). Antibody-dependent cell-mediated cytotoxicity (ADCC) of regulatory T cells (Tregs) has been confirmed to substantially contribute to

the antitumor activity of ipilimumab (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018). Fc domains of bispecific antibodies (BsAbs) can recruit natural killer (NK) cells and macrophages to induce ADCC and complement-dependent cytotoxicity (CDC) (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018).

The neutralization activity of non-nAbs is usually undetectable, but they can exert their protective effects through Fc-mediated effector functions such as ADCC, ADCP, and CDC (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018). To improve safety while preserving extended serum half-life, Fc regions are frequently silenced through the use of IgG4 backbones or Fc-engineering mutations that abrogate Fc γ receptor and complement binding, thereby preventing ADCC and CDC (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018).

ADCC is also referred to as antibody-dependent cell-mediated cytotoxicity (ADCC) and is a mechanism by which infected cells are routinely opsonized with antibodies for detection by immune cells (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018). The contribution of ADCC to tumor cell killing can be measured using a specific test that uses NK-92, an immortal line of NK-like cells licensed to NantKwest, Inc. (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018). A mouse model where extreme inflammation was triggered by viral infection revealed worsened disease progression associated with depletion of NK cells or presence of NK cells lacking perforin-mediated cytotoxicity (A., Lin, C. F., Chang, Y. T., & Fang, J. Y., 2018).

Human T-lymphocyte cytotoxicity and proliferation directed by a single chimeric TCRzeta/CD28 receptor have been studied (A., Lin,

Chapter: Cell_Function | Neutrophil Extracellular Traps

Neutrophil extracellular traps (NETs) are web-like structures composed of decondensed chromatin and antimicrobial proteins that neutrophils deploy to immobilize and eliminate invading pathogens. Unlike lytic mechanisms, vital NETosis allows neutrophils to survive and continue effector functions such as phagocytosis and migration post-release. NETs have been shown to form within blood vessels during sepsis, particularly in lung capillaries and liver sinusoids, indicating their role in systemic inflammatory responses. The formation of NETs is a dynamic process involving neutrophil multitasking in vivo, as highlighted by Yipp et al. (2013).

NETs are networks of extracellular fibers, primarily composed of DNA from neutrophils, which bind pathogens. These structures are essential under physiological and pathological conditions, giving them both cytotoxic and anti-microbial features. Excessive generation of NETs is widely acknowledged as a mediator of further pathophysiological abnormalities following SARS-CoV-2 infection, as well as in other conditions such as sepsis, where they contribute to tissue damage and inflammation. NETs components, including extracellular DNA, histones, neutrophil elastase, and matrix metalloproteinases, damage the blood–brain barrier by degrading tight junction proteins and activating endothelial cells through β 2 integrins.

Histone citrullination, catalyzed by PAD4, is a central driver of chromatin decondensation during NETosis. Three mechanistically distinct forms of NET release, collectively referred to as NETosis, have been delineated: suicidal NETosis, vital NETosis, and mitochondrial NETosis. The process of NETosis involves an unusual type of programmed, purposeful cell death, resulting in the extracellular release of a web of chromatin heavily impregnated with antimicrobial proteins. This process is critical for neutrophil function in host defense but can also contribute to disease

pathogenesis, exacerbating inflammation and host tissue damage.

NETs play a role in various autoimmune conditions, such as systemic lupus erythematosus (SLE), where excessive NETs release combined with impaired clearance due to DNase deficiency, protective factors (C1q, anti-NETs antibodies, anti-dsDNA), or inhibitors promotes NETs persistence. Neutrophil extracellular traps (NETs) have also been implicated in the development and progression of liver metastases after surgical stress, as well as in promoting deep vein thrombosis in mice. Additionally, NETs have been shown to enhance the early inflammatory response in Sendai virus-induced asthma phenotype and to promote the growth and metastasis of hepatocellular carcinoma via the activation of cancer-associated fibroblasts.

Targeting strategies for modulating NETs involve inhibiting peptidylarginine deiminase 4 (PAD4), a critical enzyme in NETosis. Physical exercise promotes DNase activity, enhancing the capacity to degrade NETs. Neutrophil extracellular traps (NETs) are also involved in the pathogenesis of various diseases, including cancer, where they can mediate the transfer of cytoplasmic neutrophil antigens to myeloid dendritic cells, contributing to ANCA induction and associated autoimmunity. In sepsis, NETs can induce intestinal damage and thrombotic tendency in inflammatory bowel disease, and in chronic granulomatous disease, restoration of anti-*Aspergillus* defense by NETs in human patients is calprotectin-dependent.

In summary, NETs are a fundamental component of the innate immune system, enabling neutrophils to deploy a web-like structure to immobilize and eliminate invading pathogens. However, if this process is uncontrolled, NETs can contribute to disease pathogenesis, exacerbating inflammation and host tissue damage.

Chapter: Cell_Function | Pyroptosis

Pyroptosis, an inflammatory form of cell death, is a critical process in the immune response against pathogens and cellular stress. It is characterized by caspase-1-mediated cell membrane perforation, leading to the release of pro-inflammatory factors such as IL-1 β and IL-18. This process is dependent on the activation of inflammasomes, large multiprotein assemblies that respond to pathogen-associated and stress-associated cellular insults. The nonclassical pathway of pyroptosis involves LPS in the cytoplasm, activating caspase-4/5/11 and cleaving GSDMD to initiate the process. Inflammasome activation, particularly the NLRP3 inflammasome, is a key step in triggering pyroptosis.

In the context of cancer, pyroptosis can be harnessed to inhibit the proliferation of cancer cells. For instance, ZnNPs-induced pyroptosis has been shown to enhance DCs maturity and increase the frequency of effector-memory T-cells, thereby boosting the immune response against tumors. However, unchecked pyroptosis can lead to cytokine storms, which not only promote multiorgan failure but also reshape the immune landscape through the interplay of pyroptosis, necroptosis, and PANoptosis, creating a vicious cycle of damage.

Recent studies have identified pyroptosis as a promising target for treating severe COVID-19-associated diseases. Caspase-1 inhibitors, such as AC-YVAD-CMK, have been shown to inhibit pyroptosis and improve survival in murine models of sepsis. Additionally, pyroptosis is not limited to the canonical pathway; it can also occur through the non-canonical pathway, which involves TLR1/2, TLR2/6, and TLR7/8, bypassing the need for RIPK1 and K⁺ efflux.

Pyroptosis plays a crucial role in various diseases, including non-alcoholic steatohepatitis (NASH), where it activates the GSDMD pathway and the canonical pathway. In colon cancer cells, pyroptosis is driven by the activation of caspases 1 and GSDME through Liver X receptor and lobaplatin signaling. The morphology of pyroptosis is distinct from MLKL channel-mediated necroptosis, characterized by non-selective gasdermin-D pore formation.

In vivo experiments have demonstrated that pyroptosis can enhance anti-tumor immune activity by promoting the maturation of dendritic cells and increasing the frequency of effector-memory T-cells. However, the unchecked release of cytokines can lead to inflammatory diseases and organ damage. Pyroptosis is also a key signaling node for cell death, acting as a co-primary regulator of pyroptosis, apoptosis, and necroptosis, and can be part of larger cell death-inducing complexes called PANoptosomes, which induce PANoptosis.

In summary, pyroptosis is a double-edged sword, providing both protective and detrimental effects. Its role in immune responses and its potential as a therapeutic target make it a critical area of study in both basic and clinical research.

Chapter: Pathogen | Viral Pathogens

Viral Pathogens: SARS-CoV-2 and Its Interactions with the Immune System

T-cell responses and therapies against SARS-CoV-2 infection have been extensively studied, highlighting the importance of these cells in combating the virus. Zhou et al. (2020) reported that acute SARS-CoV-2 infection impairs dendritic cell and T cell responses, which can lead to a weakened immune response and increased susceptibility to viral replication. Neutrophil extracellular traps (NETs) also play a significant role in SARS-CoV-2 pathogenesis, as Veras et al. (2020) demonstrated that NETs mediate tissue damage and contribute to the severity of COVID-19.

The interplay between SARS-CoV-2 and the type I interferon (IFN-I) response is complex. Infection with SARS-CoV-2 leads to the suppression of the IFN-I pathway, primarily through the nonstructural protein 1 (NSP1), which inhibits STAT1 phosphorylation and global host translation, thereby blocking interferon-stimulated gene (ISG) expression (Walls et al., 2020). This suppression is a critical mechanism by which the virus evades the host's immune response, contributing to the severity of the disease.

SARS-CoV-2 can trigger an over-exuberant immune response, characterized by high levels of circulating pro-inflammatory cytokines. This cytokine storm can lead to acute respiratory distress syndrome (ARDS) and systemic inflammatory response syndromes (SIRS), which are major contributors to the morbidity and mortality associated with severe COVID-19 (Veras et al., 2020).

Antibody neutralization is a critical aspect of the immune response against SARS-CoV-2. Wang et al. (2020) reported that naturally enhanced neutralizing breadth against SARS-CoV-2 one year after infection is possible, indicating that the immune system can develop potent neutralizing antibodies over time. However, the emergence of immune escape variants, such as the P.1 variant, has challenged the effectiveness of humoral immunity (Wang et al., 2021). Studies have shown that increased resistance to antibody neutralization is a hallmark of these variants, necessitating a detailed understanding of neutralizing antibodies to guide vaccine design strategies (Pinto et al., 2021).

The spike (S) glycoprotein is the primary target for neutralizing antibodies, with most antibodies

binding to the receptor-binding domain (RBD) to block virus binding to the host cell (Walls et al., 2020). However, the D614G variant of SARS-CoV-2 exhibits efficient replication and transmission, highlighting the importance of monitoring and understanding viral mutations (Korber et al., 2020). Structural studies have revealed that ultra-potent SARS-CoV-2 neutralizing antibodies can be elicited by sequential vaccination, providing a basis for the development of effective therapeutic strategies (Tegally et al., 2021).

The immune response to SARS-CoV-2 is multifaceted, involving both humoral and cellular components. CD4⁺ T cells indirectly contribute to pathogen clearance by providing help to macrophages, CD8⁺ cytotoxic T lymphocytes (CTLs), and B cells (Geers et al., 2021). Additionally, the Th17 response, characterized by massive pro-inflammatory cytokine production, is often elicited in infections by bacterial and viral microorganisms, including SARS-CoV-2 (Chen et al., 2021).

The effectiveness of SARS-CoV-2 mRNA vaccines has been demonstrated in clinical trials, with mRNA vaccines offering several advantages over other types of vaccines. These vaccines are effective at inducing SARS-CoV-2 IgG responses, even at very low dosages (Jackson et al., 2020). The mRNA vaccines foster potent antigen-specific germinal center responses associated with neutralizing antibody generation (Lederer et al., 2021).

In conclusion, the immune response to SARS-CoV-2 is complex and involves multiple mechanisms. Understanding these mechanisms is crucial for developing effective therapeutic and preventive strategies against the virus.

Chapter: Pathogen | Bacterial Pathogens

Bacterial pathogens pose significant challenges to human health, with various species contributing to a wide range of infections and diseases. **Listeria monocytogenes** is a major human pathogen and the causative agent of listeriosis, a severe illness characterized by fever, gastrointestinal symptoms, and meningitis. **Escherichia coli** is another significant bacterial pathogen, particularly in the context of antibody expression for diagnostic protein production. However, formula-fed infants are more susceptible to pathogenic bacteria such as **Escherichia coli** and **Clostridioides difficile** compared to breast-fed infants, who have a more favorable gut microbiome dominated by beneficial **Bifidobacterium**.

Staphylococcus aureus is known for its ability to degrade neutrophil extracellular traps (NETs), leading to immune cell death, and for inducing necroptosis in the lungs, contributing to lung damage. **Mycobacterium tuberculosis** is the causative agent of tuberculosis (TB), a disease that affects the lungs and can be exacerbated in individuals coinfecting with HIV, increasing the risk of reactivation to 10% per year. The bacterium is slow-growing and can survive within macrophages, utilizing mechanisms such as the inhibition of phagolysosome biogenesis and autophagy to evade the host immune response. **Mycobacterium tuberculosis** can also induce neutrophil extracellular traps (NETs) that activate human macrophages, highlighting the complex interplay between the bacterium and the host immune system.

The immune response to bacterial infections is fundamentally different from that to viruses, involving the recognition and destruction of extracellular bacteria by the innate immune system, including the classical and alternative complement pathways and phagocytosis. Reactive oxygen species (ROS), such as nitric oxide, play a crucial role in killing intracellular bacteria within

activated macrophages. The bacterium can survive and replicate within the phagosome, eventually destroying its host macrophage and spreading the infection. The slow growth rate of **Mycobacterium tuberculosis** necessitates prolonged culture periods, making it a challenging pathogen to study and treat.

Some bacteria, such as **Faecalibacterium prausnitzii**, **Roseburia intestinalis**, and **Anaerostipes butyraticus**, produce short-chain fatty acids (SCFAs) by fermenting complex carbohydrates, which modulate the host's immune system and provide colonocytes with carbon. These bacteria enhance the functions of dendritic cells (DCs) and T cells, contributing to better responses to immunotherapy. Conversely, reductions in beneficial bacteria like **Faecalibacterium prausnitzii** increase the risk of post-operative recurrence in Crohn's disease (CD) patients. The mechanisms by which these bacteria protect from disease are thought to involve modulation of T regulatory (T Reg) cell responses.

In the context of co-infections, helminth infection can either be beneficial or detrimental to bacterial and viral infections, with enteric or cutaneous helminth infection reducing the severity of respiratory bacterial or viral infections in helminth-endemic regions. The presence of bacteria in cancer or immune cells further confirms the role of microbiomes in tumor microenvironments (TME), although the specific roles remain under investigation. The immune system's ability to differentiate between beneficial and pathogenic bacteria is crucial, with Peyer's patch dendritic cells internalizing pathogenic bacteria and dead cells, highlighting the importance of these immune sensors in the intestine.

In summary, bacterial pathogens like **Listeria monocytogenes**, **Escherichia coli**, **Staphylococcus aureus**, and **Mycobacterium tuberculosis** pose significant health risks through various mechanisms, including immune evasion, pathogen localization, and the induction of cell death pathways. Understanding these mechanisms is essential for developing effective prophylaxis and therapeutic strategies.

Chapter: Pathogen | Parasitic Pathogens

During the blood-stage infection with *Plasmodium falciparum*, members of the *var* gene-encoded Erythrocyte Membrane Protein 1 (PfEMP1) family are exposed on the surface of infected red blood cells (RBCs). This exposure is crucial for the parasite's ability to evade the host immune system. Freitas-Junior et al. (2002) highlighted the frequent ectopic recombination of virulence factor genes in telomeric chromosome clusters of *P. falciparum*, which contributes to the parasite's antigenic variation. Ralph et al. (2005) noted that antigenic variation in *P. falciparum* is associated with the movement of *var* loci between subnuclear locations, further complicating immune recognition. The absence of erythrocyte sequestration and lack of multicopy gene family expression in *P. falciparum* from a splenectomized malaria patient underscores the importance of these mechanisms in immune evasion (Smith et al., 1995).

The malaria parasite *P. falciparum* avoids recognition and clearance by the immune system through sequential switching between members of the *var* multi-gene family, which encode the immunodominant surface proteins PfEMP1 (Smith et al., 1995; Bull et al., 1998; Smith et al., 2000; Warimwe et al., 2013). Proteomic studies of *Plasmodium knowlesi* SICA variant antigens demonstrate their relationship with PfEMP1, highlighting the conserved mechanisms of antigenic variation across different malaria species (Rodríguez et al., 2010). The expression of *var* genes is

regulated by two regions with separate promoters, one upstream of the coding region and a second within the intron (Smith et al., 1995). The differential expression of *var* gene groups is associated with morbidity caused by *P. falciparum* infection in Tanzanian children (Rottmann et al., 2005).

Antisense long noncoding RNAs regulate *var* gene activation in the malaria parasite *P. falciparum* (Smith et al., 2013), and active transcription is required for the maintenance of epigenetic memory in the parasite (Smith et al., 2013). PfSET10, a *P. falciparum* methyltransferase, maintains the active *var* gene in a poised state during parasite division (Smith et al., 2013). Kaestli et al. (2004) reported a longitudinal assessment of *P. falciparum* *var* gene transcription in naturally infected asymptomatic children in Papua New Guinea, highlighting the dynamic nature of *var* gene expression. The PCR cycling conditions optimized for *P. falciparum* cDNA were 95°C for 15 min followed by 40 cycles of 94°C for 30 s, 58°C for 25 s, and 68°C for 30 s, followed by a final extension step at 68°C for 10 minutes (Smith et al., 1995).

In *P. falciparum*, the PfEMP1 protein is associated with pathogenicity mediated through the adherence of infected RBCs via CRDs to the endothelium of post-capillary venules (Smith et al., 1995). Accumulated immunity to antigenic variants has been associated with the reduction of clinical severity in adults and older children (Smith et al., 1995; Bull et al., 1998; Smith et al., 2000; Warimwe et al., 2013). The *pir* multigene family of *P. falciparum* also plays a role in antigenic variation and beyond (Smith et al., 1995). The *P. knowlesi* model provides insights into antigenic variation, with dramatic transcriptomic differences observed in *M. mulatta* and *M. fascicularis* with *P. knowlesi* infections (Smith et al., 1995).

Chapter: Pathogen | General Pathogen Concepts

Pathogens, including viruses and bacteria, are a significant challenge to the human immune system. The immune response is finely tuned to recognize and eliminate these threats, but pathogens have evolved various strategies to evade detection and destruction. T cell activation is a critical component of the adaptive immune response, where approximately (2×10^{11}) naïve T cells in the human body express about (10^{10}) different T cell receptors (TCRs) capable of recognizing various antigenic peptides. Peptides derived from pathogens and transformed cells are presented by antigen presenting cells (APCs) in the context of MHC-II to activate CD4⁺ T cells. This process is crucial for initiating a robust immune response.

Neutrophils, another key player in the innate immune response, utilize a mechanism called neutrophil extracellular traps (NETs) to entangle and destroy pathogens. While NETs are effective in concentrating protease attack, they can also degrade extracellular matrix (ECM) components, potentially causing collateral damage. The process of antigen presentation by APCs involves capturing antigens, processing them, and presenting them to naïve T lymphocytes as peptides bound to class II major histocompatibility molecules (MHC) located on the surface of dendritic cells (DCs). Unlike traditional vaccines, mRNA vaccines use synthetic mRNA encoding the antigen, allowing host cells to produce the antigen and elicit strong immune responses. This approach has the advantage of rapid development, as it does not require the cultivation of live pathogens.

Pathogens can manipulate the host immune system in various ways. For instance, the virus can block the TAK1 kinase, leading to caspase-8-dependent cleavage of gasdermin D and subsequent cell death. This mechanism is critical for the pathogen's survival and evasion of the host's immune response. The entry and spread of pathogens within the host tissue must be controlled to prevent the

persistence of infection. In viral infections, such as influenza, HIV, and respiratory syncytial virus, there is often an excessive recruitment of neutrophils, which can exacerbate the inflammatory response and contribute to tissue damage.

Viruses have evolved mechanisms to evade the immune system, including the downregulation of MHC-I molecules, which are crucial for presenting viral peptides to CD8⁺ T cells. The Epstein-Barr virus (EBV), which causes infectious mononucleosis, is one of the viruses most consistently linked to the development of multiple sclerosis (MS). The immune system must be capable of recognizing and responding to a vast array of pathogens, including viruses and bacteria, as well as unhealthy or infected cells. Experimental evidence from mouse models suggests that some pathogens may manipulate regulatory T cells to suppress the host's immune response, thereby enhancing their own survival.

In viral infections, the virus can exploit various mechanisms to inhibit the activation of pattern recognition receptors (PRRs) and their downstream signaling cascades. For example, the virus can conceal its pathogen-associated molecular patterns (PAMPs) or directly interact with PRRs, transcription factors, or adaptor proteins involved in innate immune signaling. The virus can also interfere with interferon (IFN) signaling, degrade Janus kinase (JAK)/signal transducer and activator of transcription (STAT) components, and trigger pyroptosis, a form of programmed cell death that can amplify the inflammatory response.

The immune system's ability to detect and respond to pathogens is essential for protecting against a wide range of diseases. Vaccines, such as mRNA vaccines, can be developed rapidly using the pathogen's genetic code, providing a significant advantage in responding to emerging infectious diseases. The development of universal influenza vaccines remains a challenge due to the rapid mutation rates of viruses like influenza and HIV. However, recent studies have identified potential targets for vaccines against influenza, HIV, and SARS-CoV-2, which could help in developing more effective and broadly protective vaccines.

In summary, the immune system faces a complex and dynamic challenge from pathogens, which have evolved sophisticated mechanisms to evade detection and destruction. Understanding these mechanisms is crucial for developing effective vaccines and therapies to combat infectious diseases.

Chapter: None | PD-1 and PD-L1 Pathway

Immunostimulatory cytokine treatment or immunoinhibitory cytokine blockade plus α -PD-1/PD-L1: Some cytokines, including GM-CSF, IFN- α , IL-2, IL-7, IL-12, IL-15, and IL-18, exhibit antitumor activity by stimulating immunity, inhibiting proliferation, or inducing apoptosis in cancer cells. Additionally, the engagement of adenosine 2A receptor (A2AR) with adenosine elicits immunoinhibitory effects, suppressing the activities of tumor-infiltrating CD8⁺ T cells and hampering the function and differentiation of dendritic cells (DCs).

Metabolic modulators plus α -PD-1/PD-L1: The interaction between adenosine and A2AR suppresses the activities of tumor-infiltrating CD8⁺ T cells and impairs the function and differentiation of DCs. A series of preclinical studies have demonstrated that co-stimulatory molecule agonists can improve the efficacy of α -PD-1/PD-L1. Epigenetic modifiers, such as histone acetylation, regulate PD-L1 expression, further enhancing the therapeutic potential of α -PD-1/PD-L1. Moreover, α -PD-1/PD-L1 has been shown to enhance chimeric antigen receptor (CAR)-T cell therapy by rescuing CAR-T cell exhaustion.

In this review, we summarized the synergistic antitumor efficacies and mechanisms of α -PD-1/PD-L1 in combination with other therapies. Bispecific antibodies targeting PD-1/PD-L1 and co-stimulatory molecules have been shown to synergize with α -PD-1/PD-L1. The safety and efficacy of α -PD-1/PD-L1 combined with IL-2 pathway agonists (such as NKTR-214/BEMPEG), PEGylated IL-10 (Pegilodecakin), IL-12 plasmids (Tavo), IL-15 agonists (ALT-803), or PEGylated IFN- α have been validated in cancer patients. Logically, removing these negative factors could enhance the therapeutic effect of α -PD-1/PD-L1 and alleviate drug resistance.